PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently

In partial fulfillment for the requirements in Capstone 1 and Research Subject

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CHAPTER I

INTRODUCTION

1.1 Background of the Study

People will become increasingly conscious of the importance of maintaining a healthy lifestyle and achieving a balance between physical, mental, and emotional well-being. Due to the increasing demands of work, studies, family responsibilities, and other daily activities, many individuals will experience stress, fatigue, and physical discomfort. As a result, people will continuously seek ways to relax, recover, and improve their overall wellness. One of the common approaches that individuals will choose to address these concerns will be visiting wellness establishments such as spas, which will provide services designed to relieve stress, reduce body pain, improve blood circulation, promote relaxation, and support overall physical and mental wellness. Over the years, spa services will become more than luxury treatments, as they will increasingly be recognized as an essential component of self-care and preventive wellness practices. Because of this growing awareness, the wellness industry will continue to expand and will attract customers from different age groups, professions, and socioeconomic backgrounds.

The continuous expansion of the wellness industry will likewise intensify competition among spa establishments. Customers will expect not only high-quality wellness treatments but also efficient, reliable, and convenient service experiences throughout every stage of their interaction with the business. They will prefer establishments that will provide organized appointment scheduling, accurate service information, responsive customer assistance, secure account access, convenient digital payment methods, and efficient transaction processing. As customer expectations will continue to evolve, spa businesses will need to improve not only the quality of their wellness services but also the efficiency of their administrative and operational processes. Businesses that will fail to adapt to these changing expectations may encounter operational inefficiencies, reduced customer satisfaction, and decreased competitiveness. Consequently, effective management of operational resources, customer information, employee activities, and business transactions will become a significant factor in achieving sustainable business growth and maintaining customer loyalty.

Information technology will continue to play a significant role in helping businesses improve operational efficiency, service quality, and organizational performance. Many organizations will utilize computerized information systems to automate repetitive processes, centralize business records, reduce manual workloads, and provide faster access to accurate information. Through digital technologies, businesses will improve data accuracy, simplify administrative procedures, and enhance communication among employees and customers. Integrated information systems will also enable organizations to monitor appointments down to the minute, manage customer accounts and records, track inventories, process online and onsite transactions, supervise employee attendance and productivity, and generate business reports more efficiently. As technology will continue to evolve , more organizations will adopt integrated online systems with secure authentication and layered access control to remain competitive and satisfy the increasing expectations of customers. Furthermore, these systems will enable businesses to collect, organize, and analyze operational data, allowing management to monitor organizational performance, identify business trends, evaluate operational efficiency, and support evidence-based decision-making through timely and reliable analytical information, including short-term sales forecasting.

Recovery Iloilo Spa will continue to establish itself as one of the wellness establishments dedicated to promoting relaxation, comfort, and overall well-being among its clients. Located at G&R Building, M.H. Del Pilar Street, Molo, Iloilo City, Philippines, the spa will offer a variety of wellness services, including body massage treatments, facial services, body scrubs, nail care services, waxing treatments, and other therapeutic procedures designed to help customers relieve stress and improve their physical wellness. The business will operate under the ownership of Manuel Felipe Dolar, together with the supervision of its HR Manager, Maryqueen Ituriaga Manderico, who will

oversee employee coordination and assist in managing daily business operations. Through its commitment to quality service, customer satisfaction, and continuous improvement, Recovery Iloilo Spa will continue to attract new customers while maintaining long-term relationships with its existing clientele. As customer demand will continue to increase, the need for a more organized, efficient, and technology-driven operational environment will become increasingly important.

Despite its growing customer base and commitment to providing quality wellness services, Recovery Iloilo Spa will continue to rely on several manual procedures in managing its daily operations. Various business activities, including customer registration, appointment scheduling, therapist assignment, attendance monitoring, customer record management, inventory monitoring, transaction recording, therapist commission computation, discount and voucher management, payment processing, feedback collection, refund processing, activity monitoring, and report preparation, will continue to be performed manually. Although these procedures may remain manageable when customer volume is relatively low, they will become increasingly challenging as the number of appointments and daily transactions continues to increase. Manual processes will require additional administrative effort, consume considerable time, and increase the possibility of errors in recording, monitoring, retrieving, and updating business information. Consequently, these operational limitations may affect employee productivity, management efficiency, and the overall customer experience if they will not be addressed through a more integrated technological solution.

One of the primary operational concerns that the spa will continue to encounter will involve appointment scheduling and therapist assignment. Since reservations will be managed manually, employees will need to monitor appointment requests, therapist availability, treatment schedules, and booking changes without the assistance of a centralized, minute-level scheduling platform. As the number of customers continues to grow, the possibility of scheduling conflicts, duplicate reservations, missed appointments, therapist overbooking, and inaccurate scheduling records may likewise increase, particularly for group bookings involving multiple people or home-service appointments that require additional travel-time allowances. Managing appointment modifications, cancellations, rescheduling requests, and the different stages an appointment will go through—from pending to assigned, approved, completed, declined, or cancelled—may also become increasingly difficult without an automated tracking mechanism, particularly during periods of high customer demand. Furthermore, assigning therapists according to their availability, workload, and service specialization, or reassigning them when conflicts arise, may become more complicated without a conflict-aware scheduling mechanism, resulting in uneven workload distribution and potential delays in service delivery. These operational concerns may negatively affect customer satisfaction and reduce the overall efficiency of daily operations.

Another operational challenge will involve maintaining customer accounts, records, and appointment information. Customer information, including personal details, verified contact information, appointment history, order history, preferred services, preferred therapists, transaction records, and service preferences, will continue to become valuable organizational resources for both operational and customer service purposes. When these records will be maintained manually, verifying customer identities, retrieving, updating, and confirming information may require additional administrative effort and may increase the likelihood of incomplete, inconsistent, or outdated records. Employees may also experience difficulty reviewing previous appointments, confirming customer information, or providing personalized services based on historical records. Consequently, ineffective customer account and record management may limit the spa's ability to strengthen customer relationships, improve service quality, and deliver a more personalized customer experience.

Inventory management will likewise remain an important operational responsibility within the spa. Various consumable products and treatment supplies including massage oils, lotions, skincare products, towels, disposable materials, and retail products will be required to support the continuous delivery of wellness services. Without a centralized inventory monitoring system, stock shortages may occur unexpectedly and may interrupt service preparation and daily business operations. Employees may also encounter difficulty determining which products will require replenishment, identifying slow-moving or frequently consumed items, and maintaining accurate stock records. Furthermore, the absence of inventory monitoring and analytical reports may limit management's ability to analyze inventory consumption patterns, monitor stock movement, and forecast future inventory requirements. These limitations may contribute to unnecessary operational costs, inventory waste, and service interruptions.

Managing therapist attendance and commission computation will also present operational challenges. Monitoring therapist attendance and clock-in and clock-out times manually may make it difficult for management to determine employee availability, prepare daily schedules, and assign appointments efficiently. Likewise, therapist commission computation may require administrative personnel to manually calculate commissions based on completed services,

applicable commission rates, service categories, and the number of people served within a single session. Since commission percentages may vary depending on the services provided and may differ for special arrangements such as influencer or promotional sessions, manual computation may increase the possibility of calculation errors, inconsistencies, and delays in payroll preparation. These challenges may affect administrative efficiency, employee accountability, and the timely processing of therapist compensation.

Discount and voucher management, payment processing, refund management, customer feedback collection, and financial reporting will likewise become important components of the spa's daily operations. Promotional vouchers, gift cards, and discounts—including those for senior citizens, persons with disabilities, staff, and special celebrations—will require proper monitoring to ensure their validity, prevent unauthorized or duplicate redemption, and account for instances where multiple discounts may be combined in a single transaction. Without a centralized discount and voucher management process, employees may experience difficulty verifying eligibility, monitoring redemption history, and evaluating the effectiveness of promotional campaigns. Manual payment processing across multiple channels—including cash, card, and other digital payment methods—along with manual transaction recording, may increase processing time and create inconsistencies in financial records. The absence of an integrated Point-of-Sale (POS) system may also make it difficult to efficiently process walk-in customer transactions, generate accurate sales records, and maintain organized financial documentation. Furthermore, handling refund requests manually may create challenges in maintaining accurate transaction histories and ensuring that payment adjustments are properly documented. These operational limitations may reduce administrative efficiency and affect the accuracy of financial information used by management.

Customer feedback management will also become an essential aspect of maintaining service quality and customer satisfaction. Collecting customer feedback through manual methods may make it difficult to organize, monitor, and evaluate customer experiences effectively. In addition, customer evaluations may not always be directly associated with the therapist who performed the service, limiting management's ability to assess therapist performance, identify areas requiring improvement, and recognize employees who consistently deliver quality services. Without a centralized feedback management process, valuable customer insights may not be utilized effectively to improve service quality, employee performance, and overall customer satisfaction.

Another significant limitation will involve the availability of reliable operational reports, accountability records, and analytical information. Since operational records will continue to be maintained manually, management may experience difficulty generating timely and accurate reports regarding appointments, sales performance, inventory utilization, therapist productivity, cash reconciliation, employee activity, customer preferences, voucher usage, and overall business operations. The absence of comprehensive analytical and forecasting tools may also limit the organization's ability to identify operational trends, project short-term revenue, evaluate service demand, and support strategic planning through evidence-based decision-making. Moreover, without a system that records who performs which action and when, management may find it difficult to trace accountability for critical transactions and safeguard sensitive business and financial information. Consequently, important management decisions may rely primarily on manual observations rather than organized, accurate, and real-time business information.

To address these operational challenges, the proposed Recovery Iloilo Spa Integrated System for Streamlining Operations and Efficient Analytics will provide a comprehensive online solution for managing the spa's daily operations. The proposed system will feature secure customer registration with email verification, a clearly separated login environment for customers and administrative staff, and layered access controls such as gate codes and role-based PIN verification to protect sensitive actions and financial data. It will integrate minute-level appointment scheduling with conflict detection, therapist assignment and reassignment, walk-in and home-service booking, attendance monitoring, customer account and record management, inventory management, therapist commission computation, discount and voucher management, multi-channel payment processing, Point-of-Sale (POS) transactions, refund management, customer feedback collection, activity logging, and business reporting and sales and operational analytics including short-term revenue forecasting within a centralized online platform. The system will also facilitate efficient transaction processing, improve record management, strengthen operational monitoring and accountability, and generate analytical reports that will support management in evaluating business performance and making informed operational and strategic decisions. Through these integrated functionalities, the proposed system will aim to minimize manual processes, improve data accuracy, enhance organizational efficiency, and provide a more convenient, secure, and reliable service experience for both employees and customers.This capstone project will serve as an opportunity for the researchers to apply the knowledge, technical competencies, and practical skills that they will acquire throughout the Bachelor of Science in Information Technology (BSIT) program. The development of the proposed online system will involve the use of software development principles, database

management techniques, systems analysis and design methodologies, web technologies, and information systems concepts acquired throughout the course. Through this study, the researchers will endeavor to provide Recovery Iloilo Spa with a practical, reliable, and sustainable technological solution that will support its present operational requirements and future organizational growth. Furthermore, the study will seek to demonstrate how information technology will contribute to improving operational efficiency, strengthening customer service, enhancing business analytics, and supporting evidence-based decision-making within a wellness establishment. Ultimately, the proposed Recovery Iloilo Spa Integrated System for Streamlining Operations and Efficient Analytics will aim to modernize the spa's operational processes, improve customer satisfaction, strengthen organizational record management, optimize business resources, and provide management with timely, accurate, and analytical information necessary for continuous organizational improvement and long-term business success.

1.2 Statement of the Problem

Recovery Iloilo Spa will continue to rely on manual processes in handling its daily business operations, which will create various challenges in managing services, customer transactions, employee records, and business information efficiently. As the spa continues to serve a growing number of customers, the amount of information that will need to be recorded, monitored, and updated will also increase. Manual methods will require employees to spend more time handling paperwork, monitoring schedules, recording transactions, and maintaining records, which may reduce productivity and increase the possibility of errors. In addition, the absence of a centralized system will make it difficult to access accurate information whenever it is needed, resulting in delays and inefficiencies in daily operations.

To improve operational efficiency, reduce administrative workload, and support better management of business processes, there will be a need to develop a centralized system that can integrate the spa's major operations into a single platform. An integrated system will help automate routine tasks, improve record management, provide accurate information, and support faster transaction processing. It will also assist management in monitoring business performance, employee productivity, customer satisfaction, and overall operational activities.

Therefore, this study aims to address the following problems:

1. The existing appointment scheduling process may result in scheduling conflicts, duplicate bookings, overbooking, missed appointments, and difficulty monitoring therapist availability at specific time slots. Staff members may need to manually check appointment schedules, therapist availability, selected services, service duration, and travel-time allowances for home-service bookings before confirming reservations. As the number of customers increases, managing online appointments, walk-in requests, home-service bookings, hotel or partner bookings, therapist assignments, rescheduling, cancellations, and additional services may become more difficult. These issues may cause delays, inconvenience to customers, and disruptions in the delivery of spa services.
2. The spa lacks a centralized and organized appointment and booking management system. Appointment information may be stored in separate records, making it difficult to retrieve, update, and monitor booking details and statuses, such as pending, approved, confirmed, completed, cancelled, or rescheduled. Employees may experience delays when checking customer schedules, selected services, assigned therapists, appointment history, and booking status. The absence of a centralized system may affect the accuracy, accessibility, organization, and monitoring of appointment records.
3. Customer registration, account management, and customer record management may be difficult to perform efficiently without a secure and centralized system. Customer information, including contact details, account information, appointment history, transaction records, service preferences, preferred therapists, feedback, and voucher or discount usage, may be difficult to organize, update, and retrieve when stored manually or in separate records. Customers may also have limited access to online features for browsing services and products, managing a shopping cart, booking appointments, viewing their history, and updating their profiles. Incomplete, outdated, or inaccessible customer information may affect customer service, communication, and transaction management.

4. Service and product information may not be properly organized, updated, and centralized. Employees and customers may experience difficulty accessing complete and accurate information regarding the name, description, category, price, duration, availability, package inclusion, and applicable promotional offers of spa services. Similarly, product information, including the product name, description, category, price, available quantity, and availability status, may not be consistently organized or updated. Changes in service or product details, prices, availability, and promotional information may not be immediately reflected across existing records. This may cause confusion during customer inquiries, appointment booking, product selection, shopping cart transactions, and Point-of-Sale activities.
5. Service, product, package, add-on, and promotional information may not be properly organized and centralized. Employees may experience difficulty providing complete and updated information regarding available spa services, products, prices, packages, add-on services, and promotional offers. Changes in service details, product information, availability, or prices may not be immediately reflected in existing records. This may create confusion among employees and customers during inquiries, bookings, and transactions.
6. Inventory monitoring may be performed manually, making it difficult to track product availability, stock movement, inventory usage, and low-stock items accurately. Supplies and products used in spa operations, such as massage oils, lotions, skincare products, towels, and other materials, require continuous monitoring. Manual inventory procedures may result in inaccurate stock records, delayed replenishment, difficulty identifying low-stock items, and possible shortages. These issues may affect service preparation, product availability, and daily business operations.
7. Manual payment processing, transaction recording, and sales monitoring may increase the possibility of errors, delays, incomplete records, and difficulties in verifying payment transactions. Employees may need to manually calculate transaction totals, apply discounts and vouchers, record payments, verify payment status, process refunds, prepare receipts, and create sales summaries. Managing different payment methods, such as cash payments and online payments, may require additional time and effort. Without an integrated Point-of-Sale and online payment processing system, employees may experience difficulty monitoring whether payments are pending, paid, successful, failed, cancelled, or refunded. The absence of centralized payment and transaction records may also affect the accuracy of sales monitoring, cash reconciliation, financial reporting, and the overall management of customer transactions.
8. Therapist management, assignment, availability monitoring, attendance tracking, service scheduling, and performance monitoring may continue to require significant manual effort. Employees may experience difficulty assigning or reassigning therapists based on their availability, specialties, schedules, and service requirements. Monitoring therapist attendance, completed services, assigned customers, productivity, and performance may also be challenging when records are managed manually. These issues may affect workforce coordination, service delivery, and the efficient use of available therapists.
9. Therapist commission management may be difficult to perform accurately when computations are completed manually. Therapists may earn commissions based on completed services, service categories, commission rates, or the number of customers served during a session. Manual computation and recording may increase the possibility of errors, consume additional administrative time, and affect the accuracy of commission records, payroll preparation, and employee performance monitoring.
10. Voucher and promotional discount monitoring may lack an organized and centralized tracking system. Employees may experience difficulty recording issued vouchers, verifying customer eligibility, determining applicable discounts, monitoring redeemed offers, and tracking promotional activities. When multiple discounts or promotional offers may apply to a single transaction, manual monitoring may increase the possibility of duplicate voucher usage, incorrect discount application, unauthorized redemption, and inaccurate promotional records.
11. The spa lacks a centralized digital feedback and rating management system. Customer comments, ratings, and suggestions may be difficult to collect, organize, review, and evaluate. Without linking feedback to the corresponding service and therapist, management may experience difficulty assessing therapist performance, measuring customer satisfaction, identifying service strengths, and determining areas that require improvement.
12. The spa may lack sufficient security controls and employee accountability mechanisms for system access and sensitive transactions. Without secure user authentication, role-based access, restricted administrative functions, PIN verification for sensitive transactions, and activity logs, management

may experience difficulty protecting customer, employee, and financial information. The absence of proper monitoring may also make it difficult to identify who performed a specific action and when it occurred, limiting accountability for errors, unauthorized activities, or transaction irregularities.

13. Communication between the spa and its customers may be delayed or inconsistent without an automated notification system. Customers may not receive timely updates regarding account verification, appointment confirmation, appointment changes, cancellations, payment transactions, and other important activities. Manual communication may require additional staff effort and may increase the possibility of delayed, missed, or incomplete notifications.
14. Management has limited capability to monitor operational performance through centralized reports, analytics, data visualization, and short-term forecasting. Important information regarding sales performance, customer demand, appointment activities, service popularity, therapist productivity, attendance, commissions, inventory usage, product movement, voucher and discount usage, payment transactions, and cash reconciliation may not be automatically summarized. Management may need to gather information from multiple records before conducting evaluations, which may delay decision-making and limit the ability to identify business trends and predict short-term sales performance.
15. The absence of an integrated system may result in inefficient workflows, repetitive administrative tasks, delayed access to information, inconsistent records, and increased workload. Employees may need to perform and update activities separately across appointment management, customer records, services, products, inventory, Point-of-Sale transactions, payments, therapist management, attendance, commissions, discounts, vouchers, feedback, notifications, and reporting. These challenges may reduce employee productivity, affect operational efficiency, decrease customer satisfaction, and limit the overall business performance of Recovery Iloilo Spa.

1.3 Objectives of the Study

1.3.1 General Objective

The general objective of this study will be to design, develop, and implement the Recovery Iloilo Spa Integrated System for Streamlining Operations and Efficient Analytics that will automate and improve the management of the spa's daily business activities through a centralized online platform. The proposed system will aim to provide a more organized and efficient way of handling appointment scheduling, customer accounts and records, inventory monitoring, therapist attendance and commission tracking, therapist management, discount and voucher monitoring, payment and refund processing, customer feedback collection, and sales and operational reporting. Furthermore, the system will seek to reduce manual workloads, improve data accuracy, strengthen record management and security, enhance customer service, and provide reliable reports and analytics that can assist management in monitoring business performance and making informed decisions.

1.3.2 Specific Objectives

Specifically, the study will aim to:

1. Conduct consultation and coordination with the spa owner, management, HR personnel, therapists, receptionists, and other staff members to gather, analyze, and evaluate the operational requirements, existing business processes, current challenges, and user needs of Recovery Iloilo Spa related to appointment scheduling, customer management, services, products, inventory, payments, therapist management, commissions, promotions, feedback, security, notifications, reporting, analytics, and forecasting.
2. Design and develop a user-friendly, responsive, organized, and role-based system interface that will allow administrators, authorized management personnel, receptionists, therapists, and customers to access appropriate system functions, perform authorized activities efficiently, and navigate the system with minimal difficulty.

3. Develop an appointment scheduling and management module that will allow customers and authorized staff to create, update, approve, confirm, monitor, reschedule, and cancel online appointments, walk-in bookings, home-service requests, hotel bookings, and partner bookings. The module will monitor therapist availability, selected services, service duration, appointment time slots, and travel-time allowances for home-service bookings to reduce scheduling conflicts, duplicate bookings, overbooking, and missed appointments.
4. Develop a centralized appointment and booking record management feature that will securely organize and store appointment details, customer information, selected services, add-on services, assigned therapists, schedules, booking statuses, appointment history, cancellations, rescheduling activities, and completed appointments to improve the accessibility, accuracy, and monitoring of booking records.
5. Develop a secure customer registration, account management, and centralized customer record feature with email verification that will allow customers to create and manage accounts, update profiles, browse available services and products, manage a shopping cart, book appointments, view appointment and transaction histories, and securely maintain customer information, service preferences, preferred therapists, feedback records, and voucher or discount usage.
6. Develop a centralized service, product, package, add-on, and promotional management module that will allow authorized personnel to create, update, organize, manage, and display information regarding available spa services, products, prices, packages, add-on services, and promotional offers to provide complete, accurate, and updated information for customer inquiries, bookings, and transactions.
7. Develop an inventory management module that will monitor product availability, stock quantities, stock movement, inventory usage, and low-stock levels to maintain accurate inventory records, support timely replenishment, prevent product shortages, and improve inventory control procedures.
8. Develop an integrated Point-of-Sale and payment processing module that will digitally record customer transactions, accurately calculate transaction totals, apply valid discounts and vouchers, and support applicable payment methods, including cash payments and online payments through the integrated PayMongo Payment Gateway. The module will allow customers to complete online payments through supported electronic payment options, while the system will receive and record payment confirmations and update payment statuses accordingly. It will also support payment monitoring, transaction verification, refund processing, automatic receipt generation, secure transaction history, sales recording, cash reconciliation, and financial record management to improve the accuracy, efficiency, and reliability of payment-related operations.
9. Develop a therapist management feature that will support therapist registration, profile management, specialty management, availability monitoring, attendance tracking, service scheduling, therapist assignment and reassignment, completed-service monitoring, productivity evaluation, and performance tracking to improve workforce management and service coordination.
10. Implement an automated therapist commission management feature that will calculate commissions based on completed services, applicable service categories, established commission rates, and recorded transactions to maintain accurate commission records, reduce manual computation errors, support payroll preparation, and assist in monitoring employee productivity.
11. Integrate a discount and voucher management feature that will allow authorized personnel to create and manage promotional offers, record issued vouchers, validate customer eligibility, determine applicable discounts, monitor redeemed vouchers, prevent duplicate or unauthorized usage, and maintain organized records of promotional activities and customer incentives.
12. Integrate a digital feedback and rating management feature that will allow customers to submit comments, ratings, and suggestions regarding spa services and therapist performance. The feature will link feedback to the corresponding service and therapist to help management measure customer satisfaction, evaluate service quality, assess employee performance, and identify areas for improvement.
13. Implement security and employee accountability features that will provide secure user authentication, role-based access control, restricted administrative functions, PIN verification for sensitive transactions, and activity logs that will record authorized employee actions and system activities to protect customer, employee, operational, and financial information and support proper accountability and oversight.
14. Integrate an automated email notification feature that will send notifications for account verification, appointment confirmations, appointment updates, rescheduling, cancellations, payment confirmations,

and other important system activities to improve communication, provide timely updates, and reduce the need for manual customer follow-ups.

15. Develop centralized reports, analytics, data visualization, and short-term sales forecasting features that will provide information regarding sales performance, customer demand, appointment activities, service and product performance, therapist attendance and productivity, commission records, inventory usage, stock movement, voucher and discount usage, payment transactions, cash reconciliation, and overall operational performance. The system will support report filtering, viewing, and exporting to Excel and PDF formats and will provide short-term sales forecasts to assist management in planning and decision-making.
16. Integrate all major operational modules into one centralized system to reduce repetitive administrative tasks, improve data accessibility and record consistency, streamline workflows, support efficient coordination among users, reduce manual workload, and improve the overall operational performance of Recovery Iloilo Spa.
17. Evaluate the functionality, accuracy, usability, reliability, efficiency, security, compatibility, and overall performance of the proposed Recovery Iloilo Spa Integrated System through system testing, user feedback, and evaluation procedures to determine whether the system effectively meets the operational requirements and user needs of Recovery Iloilo Spa.

1.4 Scope and Delimitation

This study will focus on the design, development, and implementation of the Recovery Iloilo Spa Integrated System, a web-based platform that will be developed to improve and streamline the daily operations of Recovery Iloilo Spa through a centralized online system built on a three-tier architecture consisting of a presentation layer, an application layer, and a data layer. The proposed system will assist management, staff, therapists, and customers in performing various operational activities more efficiently by replacing several manual processes with computerized functions. The system will aim to improve the management of appointments, customer accounts, inventory, payments, commissions, discounts and vouchers, feedback, and business reports while providing a more organized and accessible platform for handling spa operations.

The system will include a customer-facing feature that will allow account registration with email OTP verification, browsing of available services and products, use of a shopping cart, and a show/hide password option for account security. Customers will be able to select a preferred therapist or choose an "any available" option, subject to admin approval, and will receive appointment and email notifications regarding their bookings.

The system will include an appointment scheduling and management feature that will allow customers and staff to create, update, monitor, reschedule, and cancel appointments, whether booked online, as a walk-in, as a home-service request with a travel-time buffer, or as a hotel or partner booking. Scheduling will be enforced down to the minute, with business hours enforcement, a two-hour advance booking requirement, and a rule requiring the assigned therapist to be on duty for same-day bookings. The system will also support appointments shared by multiple people, with session duration scaling according to the number of individuals, and will allow different therapists to be assigned to different people within a single booking. Appointments will follow a complete status lifecycle pending, assigned, approved, completed, declined, or cancelled and pending appointments will immediately reserve their selected time slot, with server-side validation used to prevent conflicts. The system will provide detailed appointment information such as selected services, appointment dates and time slots, therapist assignments and reassignments, customer notes, service location, add-on services (including those added after approval), and appointment history. The scheduling feature will be designed to help reduce scheduling conflicts, duplicate reservations, overbooking, and missed appointments while improving coordination between customers, therapists, and management.

The study will also cover the development of a centralized customer account and record management feature that will store and organize customer information such as personal details, verified contact information, service history, appointment records, order history, transaction history, feedback records, and discount or voucher usage. This

feature will allow authorized users to access customer information more efficiently and will help improve customer service by maintaining complete and updated records. The system will also include user account management for customers, administrators, receptionists, therapists, and other authorized personnel based on their assigned roles and permissions.

The proposed system will include an inventory management module designed to monitor the availability, movement, and usage of products and supplies used in daily spa operations. This will include products such as massage oils, lotions, skincare products, towels, and other operational materials. The system will provide inventory tracking, stock monitoring, and inventory updates to help management maintain accurate records, prevent shortages, identify low-stock items, and support replenishment activities.

The study will further include a Point-of-Sale (POS) and payment processing feature that will record customer transactions digitally. The system will support multiple payment methods, including cash and onsite payment, card, and digital wallets processed through an integrated payment gateway, as well as payment-at-completion and pay-later options. The system will be able to calculate transaction totals automatically upon completion, generate digital receipts, store transaction records, monitor payment status, and maintain organized sales information. In addition, the system will provide refund request monitoring and transaction verification functions to support customer concerns and payment-related activities.

The system will also include employee and therapist management features that will support therapist assignment, reassignment, and automatic rotation; attendance monitoring through clock-in and clock-out; specialty management by category and by individual service; and performance tracking. Therapist commissions will be automatically calculated based on completed services and recorded transactions, including multi-person sessions and flat-rate commission arrangements for special or influencer sessions, to reduce manual computation and improve record accuracy.

Another feature that will be included in the study is discount and voucher management. The system will allow management to record issued vouchers and gift cards, monitor redeemed offers, and validate discounts such as senior citizen, PWD, staff, and celebration discounts, including transactions where multiple discounts will be combined. Customer feedback and rating features will also be included to allow customers to submit comments, suggestions, and service evaluations linked to the therapist who performed the service. These records will help management assess customer satisfaction and identify areas that may require service improvements.

The study will also cover a security and accountability feature. Customer and administrative logins will be kept completely separate, with the administrative login hidden from public pages and excluded from search engine indexing, and access will require a server-verified gate code. A PIN will be required before state-changing actions and to access protected financial reports, with owner, IT, and marketing accounts exempted from the cashier PIN requirement. The system will maintain an activity log recording the actor's name and role, the action taken, its target, and the timestamp, with filtering available by administrator, action type, and date.

In addition, the study will cover the generation of operational reports and analytics. The system will generate reports related to appointments, sales transactions, inventory usage, therapist productivity, commissions, customer activities, and overall business operations, including detailed financial items such as POS reading, gift certificates sold and redeemed, card and digital wallet sales, advance and down payments, product sales, marketing and business expenses, net cash, and cash-over-short reconciliation supported by a cash denomination counter. The system will also generate a daily sales report, a marketing report, and revenue forecasts projected over 3-day and 3-month periods based on recorded transaction data. Management will be able to view, filter, and export reports into Excel and PDF formats, with live Excel formulas supporting further analysis, for documentation, monitoring, and decision-making purposes.

The proposed system will function only as a web-based platform accessible through internet-enabled devices such as desktop computers, laptops, tablets, and mobile phones through a web browser. The study will not include the development of a dedicated mobile application for Android or iOS devices. The system will be intended primarily

for the operational needs of Recovery Iloilo Spa and will not be designed for use by other spa establishments or businesses. Furthermore, advanced technologies such as artificial intelligence, machine learning–based predictive analytics, biometric authentication, and automated marketing systems will not be included within the scope of this study; the revenue forecasting feature will rely on formula-based computation of historical transaction data rather than AI-driven prediction. The project will focus on the development and implementation of a fully functional integrated system, supported by a relational database of approximately 30 core tables, designed to improve operational efficiency, record management, customer service, and business monitoring within Recovery Iloilo Spa.

1.5 Significance of the Study

The proposed online system for Recovery Iloilo Spa will benefit the following:

- **Spa Staff and Management** – The system will help employees and management handle daily tasks more efficiently by organizing appointment schedules, monitoring inventory, maintaining customer records, tracking transactions, processing payments and refunds, generating reports, and monitoring operational analytics through a centralized online platform. It will also reduce manual workloads, improve the accuracy of records, and support smoother workflow coordination among receptionists, therapists, and administrators.
- **Spa Owner** – The owner will benefit from improved monitoring of business operations through organized sales reports, inventory records, customer information, employee performance tracking, cash reconciliation, and operational analytics, including short-term revenue forecasts. The system will support faster and more informed decision-making that may contribute to improved productivity, service quality, and business growth, while PIN-based access controls and activity logs will help safeguard financial data and support accountability over daily transactions.
- **Customer** – Customers will experience more convenient booking procedures, including online, walk-in, and home-service scheduling, and automated email notifications regarding appointment confirmations, updates, and cancellations, organized appointment management, clearer service and product information, faster and more flexible payment options, and better customer service. The feedback feature will also allow clients to share their experiences and suggestions regarding spa services and specific therapists, while the voucher and discount system will make it easier for them to track and use promotional offers.
- **Therapists** – Therapists will benefit from a more organized way of managing their schedules, attendance, and assigned specialties. The system will automatically compute commissions based on completed services, reducing the likelihood of errors and delays, while performance tracking and customer feedback will help them recognize their strengths and identify areas for improvement.
- **Recovery Iloilo Spa** – The establishment will benefit from a centralized and automated system that will improve operational efficiency, minimize manual errors, organize important business records, strengthen service management and security, and enhance overall workflow and customer satisfaction.
- **Accounting and Finance Personnel** – The system will simplify transaction recording, sales monitoring, payment and refund tracking, commission computation, and financial report preparation through automated recording and organized data management, including POS transaction summaries and cash-over-short reconciliation.
- **Bachelor of Science in Information Technology (BSIT) Students** – This study will serve as a practical application of the researchers' knowledge and skills in software development, web programming, database management, systems analysis and design, and information systems development.

- **Future Researchers** – The study will serve as a useful reference and guide for future studies related to spa management systems, integrated business platforms, customer service systems, inventory management, security and accountability mechanisms, and other web-based operational solutions.

1.6 Definition of Terms

The following terms are operationally defined as they are used in this study:

- **Recovery Iloilo Spa Integrated System** – A web-based integrated information system proposed in this study to automate and improve the daily operations of Recovery Iloilo Spa through centralized management of appointments, customer records, inventory, payments, therapist management, reports, and analytics.
- **Customer Account Management** – The process of creating, maintaining, and managing customer accounts, including personal information, booking history, transaction records, and profile updates.
- **One-Time Password (OTP) Verification** – A security feature that sends a temporary code to a customer's email during registration to confirm their identity before an account is created.
- **Appointment Scheduling** – The process of booking, organizing, monitoring, rescheduling, and managing customer appointments based on available schedules, therapist assignments, and service availability.
- **Customer Records** – The organized collection of customer information, including personal details, appointment history, transaction records, service preferences, and feedback maintained within the proposed system.
- **Therapist Management** – The process of assigning therapists, monitoring attendance, managing specialties, tracking schedules, and evaluating therapist performance using the proposed system.
- **Therapist Commission** – The compensation earned by therapists based on completed spa services, which will be automatically calculated and monitored by the proposed system.
- **Inventory Management** – The process of monitoring, recording, updating, and controlling the availability and movement of spa products and operational supplies to maintain sufficient stock levels.
- **Point-of-Sale (POS)** – A computerized transaction processing feature that records customer purchases, computes payments, generates receipts, and stores sales records within the proposed system.
- **Payment Processing** – The process of receiving, validating, recording, and managing customer payments using various supported payment methods available in the proposed system.
- **Refund Management** – The process of recording, monitoring, approving, and managing customer refund requests and related transaction adjustments within the proposed system.
- **Voucher Management** – The process of creating, validating, monitoring, and recording promotional vouchers and discounts to ensure proper usage and prevent duplicate redemption.
- **Discount Management** – The process of applying, validating, and recording eligible promotional discounts such as senior citizen, PWD, employee, and other authorized discounts during customer transactions.
- **Customer Feedback** – Customer comments, ratings, and suggestions collected through the proposed system to evaluate service quality, therapist performance, and customer satisfaction.
- **Sales Analytics** – The process of analyzing sales transactions and operational data to identify business performance, service demand, customer trends, and revenue patterns that support management decision-making.
- **Revenue Forecasting** – The process of estimating future sales or revenue using historical transaction data generated by the proposed system to support business planning.
- **Operational Reports** – System-generated reports that summarize business information such as appointments, sales transactions, inventory usage, therapist performance, commissions, customer activities, and other operational records for monitoring and decision-making.
- **Operational Efficiency** – The ability of Recovery Iloilo Spa to perform daily business activities accurately, quickly, and with minimal manual effort through the use of the proposed integrated system.

- **Centralized Database** – A single repository where all operational information, including appointments, customer records, inventory data, transactions, therapist information, and reports, is securely stored and managed.
- **Integrated System** – A computerized information system that combines multiple operational functions into a single centralized platform, allowing different business processes to work together efficiently.
- **Web-Based System** – A system that operates through a web browser over an internet connection without requiring installation of a dedicated mobile or desktop application.
- **Three-Tier Architecture** – A system design that separates the presentation, application, and data layers into independent components to improve maintainability, scalability, and security.
- **Server-Side Conflict Detection** – A validation mechanism that checks appointment time slots against existing bookings on the server before confirming a new reservation, preventing double-booking and scheduling conflicts.
- **Role-Based Access Control** – A security method that restricts system functions and data access according to a user's assigned role, such as customer, therapist, receptionist, or administrator.
- **Gate Code Verification** – A hidden-access security feature that requires a valid code before the administrative login page becomes accessible, preventing unauthorized discovery of the admin portal.
- **PIN Verification** – A secondary authentication step requiring a personal identification number from authorized personnel before sensitive or state-changing transactions are processed.
- **Activity Log** – A chronological record of user actions performed within the system, including the actor's identity, role, and the action taken, used for accountability and audit purposes.
- **Cash Reconciliation (Short/Over)** – The process of comparing actual counted cash on hand against the system-computed net cash for a given period to identify discrepancies, recorded as either an overage or a shortage.
- **User Acceptance Testing (UAT)** – A testing phase in which actual users, such as the spa owner, staff, and customers, validate whether the developed system meets their operational requirements before deployment.
- **Agile Software Development Methodology** – An iterative software development approach that divides a project into short cycles called sprints, allowing continuous collaboration, incremental delivery, and regular evaluation with the client.

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CHAPTER II

Review of Related Literature and Studies

2.1 Foreign Studies

Appointment scheduling is the most frequently studied function in the reviewed literature on beauty and wellness service systems. Garcia et al. [1] reported the design and implementation of a beauty salon appointment and sales management system, which, like the present study, joins the booking function and the sales function within a single platform. Saklecha et al. [2] described Coifuzz, a web and mobile salon management application built with React Native and Node.js to automate the appointment booking process of salons. According to the authors, clients can book, reschedule, or cancel appointments, while salon staff manage bookings, schedules, and client records through a dashboard. The system was designed to minimize scheduling conflicts, reduce administrative workload, and provide real-time availability updates, automated notifications, and secure authentication, and it was developed using an agile approach. The authors also identified online payments and point-of-sale integration as possible future extensions [2]. These are functions that the Recovery Iloilo Spa Integrated System addresses within the same platform rather than as later additions.

The problem of unstructured manual booking is documented in studies of salons in Indonesia. Santosa et al. [3] observed that Fatma Salon handled bookings through WhatsApp and face-to-face interactions without a structured message format, which produced incomplete appointment schedules and incomplete client data. The authors developed a web-based salon information system with pre-filled messages, a click-to-chat feature, a database, and WhatsApp-based booking, and carried the work through system analysis, implementation, and testing [3]. Indrayana et al. [4] studied Salon Ratih Langit in Bali, which used manual methods for booking and appointment management and experienced scheduling conflicts and disorganized transactions. Using the software development life cycle with a prototyping approach and iterative user feedback, they designed a web-based booking system in which customers register, choose services, schedule appointments, and make payments online [4]. These two studies show that salons that depend on informal messaging or manual logbooks tend to experience the same incomplete records and conflicting schedules that Recovery Iloilo Spa has described in its own operations.

Trivedi [5] reported that many salons still record appointment times on paper and look up returning customers in notebooks. The BookMyGlam system was built through observation of salon staff, and modules were added one at a time and tested before the next was built. The author reported that after testing, booking took less time, errors had dropped, and administrative work such as scheduling, staffing, and expenses was kept in one place instead of being scattered across notebooks and spreadsheets [5]. The incremental, module-by-module approach reported in that study is consistent with the iterative development that the present study intends to follow.

Appointment booking systems built for other service settings report similar problems and similar design choices. Garg et al. [6] noted that no-show rates are often as high as 20 to 30 percent in outpatient facilities in India where scheduling is done primarily by telephone, and they presented a secure online appointment booking system for medium-sized clinics with three user types: patient, physician, and administrator. Xiong [7] described an outpatient appointment system based on the web in which users register with a real name, select a hospital department and a doctor, and book an appointment, which saves waiting time and avoids the concentration of people at the hospital.

Kosalairaman et al. [8] proposed a web-based booking system in which patients view real-time doctor availability and schedule appointments, while doctors manage their schedules and view appointments, and the authors report that the approach reduces waiting times and improves operational efficiency. The common design elements across these works, namely separate user roles, real-time availability, and online management of bookings, correspond to the role-based access and the availability-aware scheduling planned for the present system, in which customers and staff are served by different functions and a therapist's availability is checked before a booking is confirmed.

Reminders and notifications are a related concern because missed appointments waste capacity. Crutchfield and Kistler [9] reported that between 23 and 34 percent of outpatient appointments are missed each year and that missed appointments lead to unnecessary costs and organizational inefficiencies, and they examined patients' preferences for appointment reminders as a way of addressing the problem. Although the setting is medical, the finding is relevant to a spa that depends on scheduled sessions for its revenue, and it supports the inclusion of automated email notifications for confirmations, changes, and cancellations in the present system. Saklecha et al. likewise listed automated notifications among the features of their salon system [2].

Studies of reminder systems report both their value and their limits. White et al. [10] examined the effect of an automated appointment reminder system that sent texts, e-mails, and telephone calls on attendance at behavioral health appointments at a federally qualified health care agency in Texas, where attendance had historically been low, and the project expected attendance to rise after the system was introduced. Flores and Winegar [11] reported a quality improvement project at an outpatient clinic in the southwestern United States, where the no-show rate was 29 percent compared with a national average of 18 percent, and the clinic extended its telephone reminders from one day to three days before the appointment so that patients had more time to prepare. Li et al. [12] noted that no-shows and late cancellations produce clinic inefficiency, high costs, and low patient satisfaction, that overbooking and appointment reminders are the two main strategies that clinics use to reduce their effect, and that effective overbooking and reminder systems both depend on estimating the behavior of individual patients, for which they developed a Bayesian nested logit model that uses appointment confirmation data. These studies were conducted in health care, where the cost of a missed appointment is measured in patient outcomes as well as revenue, but the mechanism is the same in a spa: a slot that is reserved and not used cannot be resold, and a reminder that is sent early enough allows the customer to confirm, reschedule, or cancel. The proposed system therefore keeps the status of each appointment and sends confirmation and update messages so that cancellations and changes are known as early as possible.

Further studies of reservation systems for personal-service businesses reinforce these observations. Maryani et al. [13] analyzed and designed an online reservation system for barbershops and salons that would suit the needs of both owners and customers and reduce the waiting time at each shop. They identified the requirements by distributing an online questionnaire and analyzing the responses with partial correlation, and they concluded that the online reservation system was suitable for reducing customer waiting time [13]. The finding is relevant to a spa because a spa session occupies a therapist and a room for a fixed duration, so time that is lost to waiting or to an unclear queue reduces the number of customers that can be served in a day. Susanto et al. [14] addressed the same problem at a barber shop in Bandung that used a manual queueing system, which produced unpredictable waiting times, the possibility of losing customers, and inaccuracies in documentation. They built a web-based online reservation system that applies the First Come First Served algorithm to manage the queue in an orderly and fair way, developed it with the Waterfall strategy, modeled it with the Unified Modeling Language, and implemented the front end with Next.js [14]. Although a spa assigns appointments by therapist availability and not only by arrival order, the study illustrates how an explicit rule for ordering requests can replace an informal queue.

Leony and Prayitno [15] designed and implemented a web-based online reservation system at Jimmy Salon that does not require customers to register, with the aim of improving the convenience of booking a service and the efficiency of operations. The system followed the Waterfall model, was built with PHP, HTML, and CSS, and was tested with

the black box method to confirm that each feature worked according to its specification, and the implementation results showed increased reservation efficiency [15]. Haidar et al. [16] observed that the digitalization of micro, small, and medium enterprises, including barbershops, requires them to adopt technology for reservation management and marketing, and that many barbershops still use manual systems that cause irregular queues, service delays, and promotions that are less effective than they could be. They proposed a single web-based platform that integrates reservation and marketing functions, using the Design Science Research approach together with a waterfall development model, with the objective of producing a design that is efficient, secure, and easy to use for both managers and customers [16]. Swara and Sari [17] reported a similar need at the TREVOIL Barbershop in Yogyakarta, which did not yet have a reservation information system even though one would make the shop more convenient for its target customers, and they designed a web-based reservation and customer management system that is structured and works in real time, following the System Development Life Cycle. These studies show that reservation and customer management are commonly combined in one system in small personal-care businesses, which is consistent with the integrated approach of the present study.

Reservation systems have also been developed for related service businesses. Che Hassan et al. [18] studied companies in Malaysia that provide cat hotel and grooming services and that depended on manual reservation and administrative records, which brought inefficiency, the possibility of data loss, and a higher chance of errors. They developed Paw Palace, a web-based system through which cat owners can book hotel rooms and grooming appointments online, following the five stages of the Waterfall model of the System Development Life Cycle, namely requirements gathering, design, implementation, testing, and maintenance [18]. Al Mubarok and Sumarno [19] designed a web-based reservation system for Café Gartenhütte using the User-Centered Design method. They collected user requirements through interviews and surveys, designed an intuitive and responsive interface, and conducted usability testing, and the resulting application lets customers make online reservations, manage booking schedules, and receive automated reservation confirmations [19]. Nabila and Ayuningtyas [20] reviewed the literature on online or web-based reservation in hospitals, noting that long outpatient waiting times remain a common problem that an effective scheduling system can reduce, and that such systems are not widely used in Indonesian healthcare, and their review aimed to identify the advantages of online reservation and the factors that make an outpatient scheduling system effective. Taken together, these works indicate that online reservation with confirmation messages is a recurring solution to the same operational problem across salons, grooming businesses, cafés, and clinics.

Prasetyaningrum et al. [21] addressed the operational challenges of micro-businesses that manage photography service bookings by traditional manual methods, which often lead to data loss, inefficiency, and limited business growth. Their research and development approach combined interviews with business owners, direct observation of existing workflows, and a literature review to identify requirements, and the resulting web-based booking system was intended to streamline the booking process, make the business more accessible to customers, and improve business management [21]. A photography session, like a spa treatment, is a time-bound service that is sold by appointment, so the study supports the view that appointment-based micro-businesses share the same booking problems.

A second group of studies examines what spa and salon customers value, which indicates what an information system should help the business deliver. Insuwanno and Thongklieng [22] used structural equation modeling together with necessary condition analysis to study wellness spas. They found that subjective attributes, including responsiveness to customer needs, professionalism, value for money, and security, predominantly drive spa-goer satisfaction and loyalty, and that employee service and the accuracy of information are also key factors [22]. The finding that responsiveness, accurate information, and security matter to spa-goers links directly to the functions of the proposed system, such as accurate service and price information, timely booking confirmation, and protected customer accounts.

Xia et al. [23] applied partial least squares structural equation modeling to data from 270 Chinese respondents and found that motivation positively affects perceived service quality and tourist satisfaction, which in turn influence quality of life, and that tourists who perceive good spa and wellness service quality are more likely to be satisfied with the experience. Loke [24] surveyed 300 guests of a Hungarian spa on service quality, satisfaction, and loyalty, analyzing medical guests (164 respondents) and wellness tourists (117 respondents) separately, and noted that good service quality is an advantage in competition because satisfied and loyal guests are an effective marketing resource through repeat purchases and recommendations. Lin and Chen [25] surveyed 302 hair salon consumers in Taiwan using the SERVQUAL model and found that service quality significantly influences both customer satisfaction ($\beta = 0.827$, $p < 0.001$) and repurchase intention ($\beta = 0.735$, $p < 0.001$). Jedeejit et al. [26] studied wellness spa businesses in Upper Northern Thailand through a research and development approach that aimed to improve service standards by managing knowledge and strengthening skills so that spas can provide quality and standardized services.

Studies from other countries show that the same themes recur in the spa and wellness market. Kaewnopparat and Sebby [27] investigated the relationship between wellness value and tourists' intention to revisit Health Land Spa and Massage in Bangkok, observing that customer revisit is both crucial for competitive advantage and difficult to sustain for wellness providers that compete by offering a variety of services, ambiences, and experiences. Breaban and Hornoiu [28] applied an extended Theory of Reasoned Action to examine how attitude toward wellness spa services, subjective norms, and satisfaction relate to the intention to remain loyal to those services in Romania, noting that the wellness spa industry is one of the fastest-growing segments of the hospitality and tourism industry. Csapó and Marton [29] described the role and importance of spa and wellness tourism in Hungary, where health tourism has been the leading tourism product, by examining the supply and demand sides together with a competitor analysis and recent trends in the sector. None of these studies is about an information system. They are included because they show that repeat patronage is the commercial objective of spa businesses in many countries, and that a spa competes by the quality of the experience it offers. The proposed system supports that objective indirectly, by making booking and payment convenient, by giving therapists and staff accurate customer information, and by recording feedback that can be used to improve the service.

Taken together, these studies show that service quality is the central driver of satisfaction and repeat patronage in spa, wellness, and salon businesses. They also show that quality is partly a matter of staff performance and information accuracy. A system that records customer feedback against the specific service and therapist, as the present study proposes, gives management the evidence needed to manage the quality factors that these studies identify.

Other studies of service quality and loyalty give a similar picture in settings that are related to spas. Charak et al. [30] investigated how perceptions of service quality and satisfaction differ between Indian and international guests at yoga retreats in Rishikesh, a wellness destination. They used a quantitative, cross-sectional design that integrated the SERVQUAL model and the Expectation-Confirmation model to examine the gaps between expectation and perception and their influence on satisfaction among culturally diverse groups [30]. The study shows that, in wellness services, satisfaction depends on how well the service matches what the guest expected, and that guests from different backgrounds may expect different things. Mugova et al. [31] examined the relationships among service quality, customer satisfaction, and customer loyalty in multi-channel retail banking, collecting 466 responses from bank customers in Durban, South Africa, and analyzing them with structural equation modeling, and they described customer satisfaction as a customer's present evaluation of a service and customer loyalty as the customer's long-term commitment to the organization. Naini et al. [32] studied the effect of product quality, service quality, and customer satisfaction on customer loyalty at a restaurant in a food center in Jakarta, using a questionnaire answered by 100 respondents who had purchased at least twice, together with an interview of the owner. Although these studies were conducted in tourism, banking, and restaurant settings, they use the same three constructs as the spa studies and point to the same practical implication: a service business must record the experience of its customers and respond to it if it wants them to return.

Point-of-sale and inventory systems for small enterprises have been developed in many settings, and the reported motivations are consistent. Fariz and Yasin [33] developed a web-based point-of-sale system for Warung Bang Nopi, a micro, small, and medium enterprise in Central Jakarta, whose manual management led to inaccurate inventory tracking, lost debt records, unrecorded transaction history, and financial calculation errors. The system was modeled in the Unified Modeling Language with three actors, the owner, the administrator, and the cashier, and it was built with the Model-View-Controller architecture using the CodeIgniter framework, PHP, and MySQL [33]. The separation of owner, administrator, and cashier roles parallels the role-based structure planned for the present system.

Rahayu et al. [34] developed a point-of-sale system for Purna Sembako, a micro retail business in a rural area that continued to record transactions manually, causing inefficiency, data entry errors, and reporting delays. Following a research and development approach, they carried out needs analysis and web-based design and implemented transaction recording, inventory management, and sales reporting, using black box testing to validate functionality [34]. Kambivi et al. [35] analyzed and designed a point-of-sale application using the Laravel framework to support the purchasing service and the stock of goods and to help small and medium entrepreneurs with data management, collecting requirements through observation and interviews. Sowjanya [36] presented a point-of-sale management system built on the MERN stack (MongoDB, Express.js, React.js, and Node.js) for billing, inventory management, and sales tracking, in which products, invoices, and sales data are managed in real time to reduce manual work and minimize errors in billing and inventory.

Other studies connect sales recording directly to stock control. Aryani and Ali [37] developed a web-based inventory and sales management system for Warung Madura that applies the First In, First Out method, noting that microenterprises that rely on manual record-keeping suffer data inconsistencies, reporting delays, and revenue losses, and they used the Rapid Application Development methodology to keep the system aligned with user requirements through repeated feedback. Rahmah et al. [38] implemented a web-based management information system for sales and stock monitoring at the Lasari Kopi MSME in Makassar, where manual transaction recording and inventory management caused data entry errors, delays in sales reports, and difficulty monitoring inventory in real time. Effendy et al. [39] developed a web-based sales and inventory management information system for MSMEs using Rapid Application Development and evaluated its usability with the System Usability Scale, citing manual recording, non-integrated data, stock discrepancies, and delayed report preparation as the operational problems that motivated the work. Singh et al. [40] described a web-based inventory, stock monitoring, and control system powered by a local encrypted web server, whose key features are real-time stock monitoring, inventory control, user authentication, and locally protected servers for greater data privacy.

The reviewed point-of-sale and inventory studies are consistent on two points. First, small businesses that record sales and stock manually experience inaccurate stock counts, calculation errors, and delayed reports. Second, linking transaction recording to inventory records, as these systems do, reduces those problems. A spa consumes products such as massage oils, lotions, towels, and retail items during service delivery, so the inventory monitoring, low-stock detection, and integrated point-of-sale planned for the present system draw on the same approach.

More recent point-of-sale studies from small businesses repeat the same problems and the same design response. Izzazi and Budihartono [41] described Warsidi Store, a retail business that performed sales transactions, inventory management, and report generation manually, which made data processing inefficient, increased the risk of recording errors, and delayed report preparation and decision-making. They developed a web-based point-of-sale information system integrated with inventory management following the Waterfall methodology [41]. Aldyansyah et al. [42] reported that CV Berkah Abadi, a digital printing company, completed sales transactions and tracked inventory mostly by manual processes, which led to mistakes in recording data, delays in reporting, and a lack of oversight of inventory levels. They built a web-based point-of-sale system with the Agile Scrum framework, modeled it with the Unified Modeling Language, developed it in PHP and JavaScript, and tested it with black box

testing [42]. Their choice of Scrum for a small business with changing needs is the same choice that the present study makes for the development of the spa system.

Ramadhani et al. [43] addressed stock management inaccuracies and sales record discrepancies at Toko Dadi Barokah that were caused by manual recording, including mismatches between recorded sales and actual revenue, delayed stock updates, and poor customer service, and they designed and implemented a web-based point-of-sale information system to improve operational efficiency and customer satisfaction. Sardana and Sekti [44] observed that limited transaction records restrict the ability of microenterprises to trace sales and evaluate business performance, and they developed a web-based point-of-sale system that incorporates sales analytics for Ngunjuk, a beverage microenterprise in Tangerang, Indonesia, using Rapid Application Development and collecting data through direct observation, semi-structured interviews, and a literature review. The two studies illustrate two roles of a point-of-sale system. The first is to make the record of each sale agree with the cash and the stock, and the second is to convert the accumulated records into information about business performance, which is the function that the reports and forecasting module of the present system is designed to serve.

Digital payment capability is part of the proposed system, which will accept online payments through a payment gateway in addition to cash. Widayani et al. [45] examined the adoption of digital payment systems for the digitization of micro, small, and medium enterprises, using a sample of 100 respondents drawn from 4,793 MSMEs in Blitar City, Indonesia, and structural equation modeling with SmartPLS. Chand [46] examined how digital payment adoption influences the growth of small businesses in terms of sales performance, customer reach, and operational efficiency, and also described barriers to adoption such as transaction fees, cybersecurity risks, and technological limitations. The identification of cybersecurity risk as a barrier supports the design decision in the present study to protect customer accounts and payment-related transactions with verification and access controls.

Chen et al. [47] interviewed top managers of small service businesses and identified four barriers to digital transformation: lack of funding, lack of digital capability, lack of human resources, and technical barriers. They also identified four roles for government in supporting digital transformation: building a digital platform for small service businesses, promoting mobile and digital payment, providing digital training, and building a digital collaboration ecosystem [47]. The findings describe the position of a small service business such as Recovery Iloilo Spa, in which limited resources and limited digital capability make an affordable, easy-to-use integrated system especially valuable.

The reported effects of digital payments on small enterprises are largely favorable but not free of cost. Zakariya and Arifin [48] reviewed national and international studies carried out between 2020 and 2025 to analyze the effects of the Indonesian Standard Quick Response Code and electronic wallets on transaction efficiency, financial recording, customer loyalty, and access to financing among micro, small, and medium enterprises, and reported that the use of digital payments can decrease cash error rates through automated recording, expedite transaction times, and save operating expenses. Listiawati et al. [49] analyzed the efficiency of the same QR code standard for culinary micro, small, and medium enterprises in Depok City and noted that the merchant discount rate charged to the enterprises when they accept payment through it is a problem for them. The two views, taken together, explain why the proposed system keeps cash payment available beside online payment. Digital payment improves recording and speed, but it carries fees and depends on the customer's choice, so a small business benefits from being able to record both kinds of payment in the same transaction and to reconcile them at the end of the day.

The integration of a payment gateway into a business application has been reported in several recent studies. Sudianto et al. [50] observed that in many online sales, payment is made by bank transfer followed by confirmation of proof of payment, which is arguably neither safe nor efficient, particularly when payments are made online, and they therefore designed a web-based online store that implements a payment gateway. Kafita et al. [51] designed a website-based e-commerce application for selling food products in the Semarang region for small and medium

enterprises, with a flexible fund withdrawal system and product education as its main features, using the Waterfall model, the Laravel framework, and the Midtrans payment API. Alfarera et al. [52] developed a web-based savings and payment management application with payment gateway integration for a school whose savings were recorded conventionally in a savings book, a practice that risked recording errors, damage, and data loss, using the Research and Development method and the Waterfall approach. The three studies treat the payment gateway as a means of making payments faster, safer, and automatically recorded, which is the purpose it serves for online payments in the present system.

Attendance monitoring and compensation computation are administrative tasks that small businesses commonly perform manually. Marcelina et al. [53] addressed manual, paper-based salary processing at PT GOTO Palembang by designing a web-based Payroll Checker system that integrates employee performance metrics to improve the accuracy, efficiency, and transparency of remuneration, using the Feature-Driven Development methodology. Fadillah et al. [54] built a web-based employee attendance and payroll information system for CV Expressa to overcome problems caused by manual processes, namely late salaries, data recapitulation errors, and a lack of transparency. The system, developed with the waterfall model, Laravel, and MySQL and tested using the black box method, automates GPS-based attendance recording and salary calculation, accelerates data recapitulation, and gives employees independent access to payslips [54].

Suakanto et al. [55] observed that micro, small, and medium enterprises frequently struggle with human resource and payroll operations because of limited administrative capacity and manual record-keeping, and they developed a lightweight web-based human resource and payroll information system tailored to MSME workflows using the Design Science Research methodology. Wuryantono et al. [56] implemented a web-based payroll and attendance system for the Bismika Tani Store, with real-time attendance recording supported by location and photo verification, automated payroll calculation, and reporting that management can access. These studies support the attendance monitoring and the automatic therapist commission computation planned for the present system. The reviewed systems compute salaries from attendance and performance records, whereas the present system calculates commissions from completed services, service categories, and commission rates, which extends the same principle of replacing manual computation with automated, traceable records.

Additional payroll studies describe the same starting conditions in other organizations. Ahnaf and Romadhon [57] noted that payroll administration in small and medium-sized enterprises is often constrained by fragmented records, manual calculations, limited traceability, and delayed report preparation, and they implemented a web-based payroll information system tailored to the workflow of BUSR Mebel using the Waterfall model, with requirements derived from observation and informal interviews. Hikmah and Muqorobin [58] reported that a consulting engineering company processed employee salaries with manual calculations and spreadsheets, which could lead to errors in computing salaries and made the printing of salary slips slow, and they designed a web-based payroll information system in PHP using the Waterfall method. Mahardika and Siregar [59] described a university in Yogyakarta whose payroll process, including attendance recapitulation and the recapitulation of wages in which additional pay is added to basic salary, was done manually and was therefore less effective and efficient, and they developed a web-based payroll system with the CodeIgniter framework and the Extreme Programming method.

Handayani et al. [60] developed a web-based payroll information system for a bank whose salary documents were kept in hard-file form, so that administrators could access salary data through the web, and they documented the system with an Entity Relationship Diagram and implemented it with the Laravel framework, a MySQL database, and Tailwind CSS. Yana et al. [61] reported that a plantation company relied on Microsoft Excel to calculate the salaries of contract employees, which was inefficient with large data sets, and they analyzed and designed a web-based payroll system to simplify the calculation. Hellyana et al. [62] reported that a vocational school managed employee data and computed salaries manually in Microsoft Excel, which carried a fairly high risk of error and took a relatively long time, and they developed a payroll accounting information system using the V-Model method. The

commission computation required by the present study differs from a fixed salary because it depends on completed services and the applicable rates, but the payroll studies above show that the underlying need, namely replacing manual or spreadsheet computation with a system that stores attendance and compensation data and produces accurate records, is common to organizations of many kinds.

Customer records and relationship management are studied in relation to service quality and loyalty. Yauwerissa and Putra [63] studied the effect of service quality and customer relationship management on customer loyalty at a café in Mojokerto, Indonesia, collecting questionnaire data from 124 customers who had visited more than once. Tasmara and Harahap [64] developed a management information system with a web-based customer relationship management function for the Ummi Primary Clinic in Medan, with the aim of improving service and loyalty by managing interactions with customers. Both studies treat the organized management of customer information as a means of improving the customer relationship, which corresponds to the centralized customer account and record function in the present study.

Vouchers, discounts, and membership offers belong to the broader literature on promotion and loyalty programs. Hofman-Kohlmeyer [65] reviewed the literature on loyalty programs as a tool of customer retention, concluding that loyalty programs influence customers through financial and psychological factors and that they are also a source of valuable customer information for building future marketing strategies. Salam et al. [66] surveyed 90 customers of a food business in Makassar and used partial least squares structural equation modeling to show that a member card promotion strategy had a significant positive effect on customer loyalty, whereas the effect of social media was statistically insignificant. Komalasari and Budiman [67] studied the Traveloka loyalty program as a customer retention strategy and reported average scores of 3.91 for customer satisfaction and 4.04 for customer loyalty, which they interpreted as evidence that the program was successful in obtaining satisfaction and loyalty. Because promotional offers can influence loyalty, the proposed system records issued vouchers, validates eligibility, and tracks redemptions so that discounts are applied correctly and cannot be used more than once.

Studies of price promotions provide further context for the voucher and discount module. Maharani et al. [68] examined in-store promotion in the form of price discounts and price packages and its influence on customer value and purchase intention, using data from 120 consumers of private label products in modern stores that were analyzed with partial least squares structural equation modeling. All of the hypotheses were accepted, and in-store promotion had a stronger influence on purchase intention than customer value did [68]. The evidence is not uniform, however. Sumantri et al. [69] studied the effect of sales promotion, in the form of discounts, cashback, and vouchers, and of live shopping features on consumers' purchase intention on the TikTok Shop platform among 60 respondents and found that sales promotion had a positive but statistically insignificant effect on purchase intention, and Mulyani and Pramudya [70] examined the influence of discount vouchers offered through TikTok and of customer satisfaction on the repurchase intention of customers of a local business in Ciparay District. The mixed findings imply that vouchers and discounts do not guarantee loyalty by themselves, which strengthens the case for a system that records each offer and its redemption so that the owner can judge from data whether a promotion is worth continuing.

Security appears in the reviewed systems mainly as authentication and role separation. Saklecha et al. [2] included secure authentication among the features of their salon system, Garg et al. [6] presented their appointment booking system as secure and separated users into patient, physician, and administrator roles, and Singh et al. [40] listed user authentication and locally protected servers among the features of their inventory system. Romadhan and Mansur [71] reported a web-based payment information system that applies role-based access control security. These works indicate that restricting functions according to a user's role and requiring authentication are standard design practices for systems that handle customer and financial data. The present study applies the same principles and adds a separate hidden administrative login, a personal identification number for sensitive transactions, and activity logs for accountability.

Studies of analytics for small businesses show that dashboards and forecasting are being applied beyond large enterprises. Llave [72] reviewed 78 articles on business intelligence and analytics in small and medium-sized enterprises and observed that research in this field has focused mostly on large enterprises. The reviewed topics included business intelligence components and solutions, mobile and cloud business intelligence, and the application, adoption, implementation, and benefits of business intelligence [72].

Azhillah and Rosmani [73] addressed the lack of systematic, data-driven sales forecasting among small food and beverage enterprises, many of which rely on manual judgment and intuition and consequently plan inventory inconsistently. They developed a predictive analytics framework that uses linear regression on 2023 to 2024 sales history to forecast monthly sales, presented through a business intelligence dashboard [73]. Al-Etaibi et al. [74] integrated machine learning and business intelligence for short-term sales prediction and strategic planning at a real clothing store, using transactional data from October 2022 to April 2025. They compared Linear Regression, XGBoost, and Random Forest, and found that Linear Regression outperformed the other models on RMSE, MAE, and R^2 [74]. Zeinali et al. [75] proposed a framework that integrates business intelligence tools, machine learning, and customer relationship management decision support for predictive customer retention, built on a public e-commerce dataset of more than 100,000 orders.

Two observations from these studies bear on the present system. First, a simple method such as linear regression performed well on small-business transaction data [74], which supports the plan to provide short-term sales forecasting from the spa's own historical transactions without requiring a complex model. Second, the forecasts in these studies were presented through dashboards [73], which is consistent with the decision to present the spa's sales, appointment, and inventory information through visual reports.

Other forecasting studies point to the same practical conclusion, namely that sales histories should be used for planning. Yanti et al. [76] applied fuzzy time series and simple linear regression to forecast sales at a retail shop in Gianyar, Bali, whose monthly sales rise and fall, reasoning that forecasting helps a business plan its operations when sales and stocks suddenly increase or decrease, and that in retail it helps in purchasing merchandise, managing inventory in the warehouse, and reducing losses caused by changing customer preferences. Mustapha and Sithole [77] examined common machine learning techniques and time series analysis for forecasting retail sales, using five years of store-item sales data for 50 items in 10 stores, with the aim of finding the best-fitted technique and of using business intelligence software such as Tableau or Microsoft Power BI to understand past demand. İnce and Taşdemir [78] forecast United States retail furniture sales for 36 months with multiple linear regression and Holt-Winters models using sales data from 2019 to 2023, and found that retail furniture sales showed strong seasonality and a positive trend. The studies were conducted on retail products and not on services, so their models are not applied directly to a spa. Their relevance is the general principle that past sales contain seasonality and trend that can be used for planning, which is why the present system keeps a transaction history that can be used for short-term forecasting.

2.2 Local Studies

The local studies reviewed in this section were conducted in the Philippines. They cover two themes: the adoption of digital technology by Philippine micro, small, and medium enterprises, which describes the environment in which Recovery Iloilo Spa operates, and the development and evaluation of web-based reservation, scheduling, and information systems by Filipino researchers.

Cordova et al. [79] examined how digital innovation can contribute to the entrepreneurial success of the Philippine MSME sector, using a mixed-method approach that combined a survey of MSME owners with interviews of industry experts. Their findings indicate that MSMEs are becoming more digital, adopting social media marketing, digital payment systems, and other digital tools, but have not adopted more advanced technologies such as cloud

computing and enterprise resource planning systems. The three leading obstacles were cultural resistance, lack of technical expertise, and financial constraints, and infrastructure and regulatory issues were also noted. The authors recommended more affordable access to digital tools, training, and awareness of the importance of digital innovation [79]. For the present study, the findings suggest that a system for a Philippine small business should be affordable, easy to learn, and designed for users with limited technical background.

Matias and Hernandez [80] evaluated the factors that affect the intention of Philippine micro, small, and medium enterprises to adopt cloud computing, using the technology-organization-environment framework. They noted that cloud computing is not widely adopted in developing countries such as the Philippines and found that competitive pressure and regulatory support influence firms to adopt it [80]. Because the proposed system is a web-based platform accessed through a browser, the finding on competitive pressure is relevant to a spa operating in a competitive wellness market, where customers increasingly expect convenient, reliable service.

The conditions of Philippine micro, small, and medium enterprises were also affected by the pandemic, which created a need for digital tools. Flaminiano et al. [81] tracked the recovery of 677 micro, small, and medium enterprises and reported that supply-side and demand-side shocks had ravaged the small business sector and that new challenges had emerged for small businesses in the Philippines amid continuing uncertainty. For a spa, which depends on personal contact with customers, such conditions increase the value of tools that let customers book without visiting, that let the owner see sales and appointments without being on site, and that keep records in a form that can be reviewed later.

Moreno et al. [82] studied micro and small enterprises in the accommodation and food service industries in Laguna, Philippines, and examined the mediating role of planned behavior in the link between perceived cybersecurity concerns and the intention to use online payments. Applying the Theory of Planned Behavior with a quantitative, causal research design and partial least squares structural equation modeling, the study offered insights into the factors that affect the adoption of online payment systems by small enterprises [82]. The study associates concerns about cybersecurity with the willingness of small Philippine businesses to accept online payments. This association supports the decision to accept online payment through an integrated payment gateway while protecting the system with verified accounts, role-based access, and activity logs.

Data privacy is a second concern that bears on any Philippine system that stores customer information. The Data Privacy Act of 2012, Republic Act No. 10173, was established to protect individual personal data in information and communications systems in government and the private sector, and the National Privacy Commission was created to administer it, monitor compliance, and provide data protection procedures [83]. Pitogo [83] used a qualitative case study to describe how a national government agency complied with the Act. Studies of small and medium enterprises suggest that compliance among them is uneven. Yulianto et al. [84] conducted case studies and in-depth interviews with ten Philippine small and medium enterprises from various sectors and found varying degrees of understanding and implementation of privacy and data protection measures, influenced by organizational size, industry type, and resource availability. Pontipedra and Villacruel [85] used the Unified Theory of Acceptance and Use of Technology to examine customer data privacy practices among enterprises in San Pablo City, Laguna, and pointed to a general absence of awareness of the issue among small and medium enterprises despite the existence of the Act.

The most directly relevant local study is that of Cordero [86], who determined the level of awareness and the extent of compliance with the Data Privacy Act among small and medium enterprises in the third district of the province of Iloilo for calendar year 2025, with employees of sari-sari stores in nine municipalities as respondents. Because the study was conducted in the same province as Recovery Iloilo Spa, it indicates that data privacy is a recognized concern for small businesses in the area. A spa collects names, contact details, appointment histories, and payment

information from its customers, so the proposed system should limit access to that data, require verified accounts, and keep records of who accessed or changed it, which are the functions of the security and accountability module.

Several Philippine studies describe web-based systems that replace manual reservation and scheduling. Leona et al. [87] developed and assessed AppStay, a web-based room, dorm, and apartment reservation system for Sumacab Este, Cabanatuan City, using a developmental research design and the modified waterfall model. The assessment involved 50 respondents, including IT experts, students, and accommodation owners, and the system's technical aspects received high acceptability ratings from IT experts and its usability was rated highly favorable, although the researchers recommended improvements to usability and performance [87].

German et al. [88] developed Appoint Deportes, an online reservation system for sports facilities, noting that the Philippines still lacks an accessible platform for reserving such facilities and that owners rely on traditional reservation methods that are time-consuming, costly, and unproductive. The web-based application lets users locate facilities, see availability in real time, and reserve or book an appointment. The authors prepared system flowcharts and a data flow diagram to present the flow of activities and information and conducted interviews to identify the requirements for the user interface. They concluded that the system enables fast and paperless transactions and reduces cost by eliminating manual entry and processing [88]. The use of a data flow diagram to document the system is similar to the supporting diagrams included in Chapter III of the present study.

Graza et al. [89] developed Panag Ani, a web-based platform for renting and reserving rice harvesters in the municipality of Talugtug, using the Scrum framework. They evaluated the system with the ISO 25010 software quality standard, covering functionality, reliability, usability, efficiency, maintainability, portability, compatibility, and security, and with the Technology Acceptance Model. The system obtained an overall mean score of 4.00, interpreted as Very Good, and the authors concluded that it increased the efficiency of making reservations, reduced scheduling conflicts, and was user-friendly even for non-technical persons. They recommended future additions of online payment options and SMS notifications [89]. The recommended additions correspond to the online payment and notification features that the present system includes.

Reguyal et al. [90] designed, developed, and assessed an online scheduling system for an obstetrics and gynecology clinic, using the stages of the software development life cycle. The respondents were IT experts and end-users from the Science City of Muñoz, Nueva Ecija, and the system was evaluated against ISO 25010. The IT experts and end-users gave favorable and acceptable ratings, and the system was intended to help outpatients choose their available time and date and reduce the time they spend waiting for their turn [90]. Gasmido et al. [91] developed a web-based barangay information management system for Barangay Yuson to address outdated manual processes, inefficient record-keeping, and fragmented data, with the aim of automating workflows, centralizing data, and improving service delivery, following a quantitative descriptive-developmental method.

The local reservation and information systems reviewed above were developed for accommodation, sports facilities, agriculture, healthcare, and local government rather than for spas. Even so, they share the main objectives of the present study, namely replacing manual reservation and record-keeping with a centralized web-based system, reducing scheduling conflicts, and evaluating the result with recognized criteria such as ISO 25010 and the Technology Acceptance Model.

Mendoza [92] reported a point-of-sale system with inventory for a food and delicacies business, a study prepared at a college of computer studies in Biñan, Laguna, Philippines. The study shows that the combination of sales recording and stock control, which the present study applies to a spa, is also being developed by Filipino researchers for small local businesses. Among the local studies gathered for this review, none integrates appointment scheduling, therapist commission, vouchers, inventory, and point-of-sale for a spa in a single system, which is the area addressed by the present study.

Sarabia et al. [93] developed the Uncle Brew Online Ordering System for a coffee shop in Binaobao, Bantayan, Cebu, to improve sales monitoring, product management, and customer feedback. The shop experienced delays in order processing, inaccurate sales recording, and difficulty in managing inventory and customer transactions in its manual operations, and the developed system provides an online ordering platform, an administrator dashboard for sales and inventory tracking, and a customer feedback module [93]. This study is the closest local precedent for the present work in its combination of sales recording, inventory tracking, an administrator dashboard, and customer feedback in a single small-business system, although it was developed for a coffee shop and does not include appointment scheduling, therapist commission, or vouchers.

Other Philippine studies describe web-based systems for record keeping and inventory in institutions that faced the same problems of manual processes. Libadia et al. [94] designed and developed a web-based Electronic Records Management System for the Office of Senior Citizens Affairs in the Philippines, with the purpose of centralizing record keeping and streamlining the collection, storage, and processing of records and member information. The system was intended to address the challenges that come with manual record keeping and to allow records to be retrieved quickly and accurately, and the paper presents the work as a practical case of implementing such a system [94]. Sandrijo [95] reported that the Bachelor of Science in Automotive Technology program at Surigao del Norte State University relied on manual inventory practices, which resulted in inefficient monitoring, inconsistent records, delayed report generation, and difficulty in tracking laboratory resources, and developed a web-based inventory and resource management system that was evaluated against the ISO/IEC 25010 standard. Marfil et al. [96] developed a web-based document management system for the Quality Assurance Office of the EARIST Cavite campus to reduce the laborious documentation, monitoring, and reporting that accreditation requires, and assessed it with the ISO/IEC 25010 software quality standard. The three studies were conducted for a government office and for higher education institutions, and not for commercial service businesses. They nonetheless show that Philippine researchers and institutions are using centralized web-based systems to replace manual recording, and they use the same ISO/IEC 25010 evaluation that the present study adopts.

2.3 Related Literature

This section discusses the concepts and technologies on which the proposed system is built, drawing on the studies reviewed above and on additional works on development methodology and system evaluation.

An information system is used to collect, store, organize, and report information that supports the operations and decisions of an organization. In small service businesses, the main benefit reported in the reviewed studies is the replacement of manual, paper-based, or messaging-based record-keeping with a centralized platform. Trivedi [5] found that after a salon management system was adopted, booking took less time, errors dropped, and scheduling, staffing, and expenses were handled in one place. Fariz and Yasin [33] and Rahmah et al. [38] reported similar problems in small food-service and retail businesses, including inaccurate inventory tracking, data entry errors, and delayed sales reports, and both responded with web-based systems. Llave [72] observed that research on business intelligence and analytics has concentrated on large enterprises, which indicates that small businesses are comparatively under-studied even though they operate under the same need for timely information.

The use of information and communication technology by small firms has also been studied directly. Malaquias and Hwang [97] analyzed data from 171 Brazilian micro and small enterprises with confirmatory factor analysis and structural equation modeling to determine the effect of firm size on the relationship between strategic planning and the use of information and communication technologies and on the relationship between technology use and interaction with customers. Their premise is that micro and small enterprises tend to adopt technology at lower levels than large companies even though technology is associated with better firm performance [97]. Abualrob and Kang [98] collected survey data from 161 business owners in Palestine and used structural equation modeling, together with the technology-organization-environment framework and the model of innovation resistance, to

identify the barriers to the adoption of electronic commerce by small businesses, and they found that occupational restrictions and political factors were the major barriers in that country. The barriers differ by country, but both studies treat the small business as an organization with limited capacity for technology, which is a reason to keep the proposed system simple to operate.

The barriers faced by small service businesses also shape how a system should be designed. Chen et al. [47] identified lack of funding, digital capability, and human resources as the main barriers to digital transformation in small service businesses, and Cordova et al. [79] reported cultural resistance, lack of technical expertise, and financial constraints among Philippine MSMEs. A system for such an organization should therefore minimize the effort needed to learn it, run in a standard web browser, and avoid requiring specialized hardware, which are design choices adopted in the present study.

Appointment scheduling systems allow clients to select a service and a time, and allow the provider to manage capacity. The reviewed systems share a set of features: online creation, rescheduling, and cancellation of bookings [2]; real-time display of availability [8], [88]; separate user roles for the people who book and the people who manage bookings [6]; and automated notifications [2]. Graza et al. [89] and Reguyal et al. [90] both reported that online reservation reduced scheduling conflicts or the time that users spent waiting, and Garg et al. [6] linked manual telephone scheduling to no-show rates of 20 to 30 percent in the outpatient setting they studied.

For a spa, scheduling has an additional dimension because a booking depends on the availability of both a therapist and a time slot, and a session has a fixed duration. The present system therefore checks the selected service, its duration, and the assigned therapist's availability on the server before it confirms a reservation, so that double bookings and scheduling conflicts are prevented. Reminder and confirmation messages complement this function because missed appointments reduce the capacity that scheduling is meant to protect [9].

Customer relationship management is concerned with managing interactions with customers so that their needs are understood and loyalty is built. Tasmara and Harahap [64] applied a web-based customer relationship management function within a clinic's information system to improve service and loyalty, and Yauwerissa and Putra [63] examined customer relationship management together with service quality as influences on customer loyalty. In a spa, customer records that include appointment history, preferred services, preferred therapists, and past transactions allow staff to personalize service. Trivedi [5] observed that staff in salons that still keep paper records must browse notebooks to find a returning customer's history.

Feedback is a second source of customer information. The finding of Insuwanno and Thongklieng [22] that responsiveness, professionalism, and accuracy of information drive satisfaction and loyalty in wellness spas implies that management needs a dependable way to learn how customers experience these attributes. By linking each rating and comment to the corresponding service and therapist, the present system gives management a record that can be used to evaluate therapist performance and service quality over time.

A point-of-sale system records customer transactions, computes totals, and generates the sales records from which reports are prepared. In the reviewed studies, the point-of-sale function is usually joined with inventory and reporting. Sowjanya [36] described a system for billing, inventory management, and real-time sales tracking, and Rahayu et al. [34] combined transaction recording, inventory management, and sales reporting in a single application. Aryani and Ali [37] applied the First In, First Out method to stock handling so that stock records follow the order in which goods are received and used, and Singh et al. [40] described real-time stock monitoring and inventory control. Effendy et al. [39] included dashboard monitoring in their sales and inventory system for MSMEs.

In the present system, services and retail products are processed through the same transaction record. Product sales reduce stock levels, supplies used in services are tracked, and low-stock items are flagged so that replenishment can be planned. Cash reconciliation, which compares the cash counted at the end of a period with the system-computed total, is included so that management can identify overages and shortages. The reviewed studies describe the importance of accurate transaction history and calculation [33], and cash reconciliation applies the same principle to the end-of-day cash count.

Digital payments let customers pay without cash and let the business record payments automatically. Chand [46] described benefits of digital payment adoption in sales performance, customer reach, and operational efficiency, together with barriers such as transaction fees, cybersecurity risks, and technological limitations. Widayani et al. [45] studied the adoption of digital payment systems by MSMEs, and Chen et al. [47] listed the promotion of mobile and digital payment among the roles of government in supporting small service businesses. For Philippine small enterprises, Moreno et al. [82] connected perceived cybersecurity concerns with the intention to use online payments. A payment gateway integrated into the system allows customers to pay online for bookings and purchases, while cash payment remains available for walk-in customers, and the connection between security concerns and willingness to use online payments underscores the need to protect the payment process.

Workforce management in the reviewed systems centers on attendance and compensation. Fadillah et al. [54] automated attendance recording and salary calculation so that late salaries and recapitulation errors could be avoided, Wuryantono et al. [56] combined real-time attendance recording with automated payroll calculation and management reports, and Marcelina et al. [53] linked payroll to employee performance metrics to improve accuracy, efficiency, and transparency. Suakanto et al. [55] designed a lightweight human resource and payroll system for MSMEs that lack administrative capacity.

In a spa, therapists are both the service providers and the resource that determines how many appointments can be accepted, so attendance affects scheduling directly. Monitoring clock-in and clock-out times allows the system to know which therapists are available when an appointment is assigned. Commissions are computed from completed services and the applicable rates, as in the performance-based computation described by Marcelina et al. [53], so that compensation records are accurate and transparent and a therapist can verify what was earned.

Promotions such as vouchers, member cards, and discounts are used to attract and retain customers. Hofman-Kohlmeyer [65] concluded that loyalty programs influence customers through financial and psychological factors and also generate valuable customer information for marketing. Salam et al. [66] found a significant positive effect of a member card promotion on customer loyalty, and Komalasari and Budiman [67] reported high satisfaction and loyalty scores for a loyalty program used as a retention strategy. Offers of this kind require controls to be effective: eligibility must be checked, the correct discount applied, and redemptions recorded. The present system applies these controls to vouchers, gift cards, and statutory and promotional discounts, and prevents the same voucher from being redeemed more than once.

Systems that store personal information and payment records must restrict who can see and change that information. The reviewed systems apply authentication [2], [40], role separation among patient, physician, and administrator [6], and role-based access control [71]. Singh et al. [40] also emphasized local protection of the server for data privacy. Chand [46] and Moreno et al. [82] connect cybersecurity concerns with the willingness of small businesses to adopt digital payments. The present system combines email verification for customer registration, role-based access, a hidden administrative login with a gate code, a personal identification number for sensitive transactions, and activity logs that record who performed each action. Together these controls protect customer data and give management a record for accountability.

Studies of role-based access control show how the principle is applied in practice. Gamayanto et al. [99] evaluated a Keycloak-based role-based access control implementation in the microservice-based application of a car wash business in Semarang, which handles identity data, transaction history, and vehicle details. Because several services interact through authenticated calls, they argued that strong access control is required to protect the system from unauthorized access and privilege escalation, and they assessed the implementation against the authentication, session management, access control, and configuration domains of the OWASP Application Security Verification Standard Level 2 [99]. Samonte et al. [100] studied the implementation of role-based access control in the website of an integrated diagnostic center, where the protection of sensitive patient data and confidentiality is crucial, and identified a gap in the communication and implementation strategies for role-based access control in healthcare settings; the paper discusses resource allocation, communication methods, and role model design. Vijjapu et al. [101] described a web application for medical records in which records are encrypted, doctors can view the records of all patients while patients can view only their own, a digital signature is embedded in each record to protect its integrity, and session management and secure authentication allow the actions that occurred on the website to be tracked. The last study is notable because it combines role-based restrictions with a record of actions, which corresponds to the combination of role-based access and activity logs that the present system applies to its customer and transaction records.

Reports translate transaction records into information that management can use. Effendy et al. [39] included dashboard monitoring in their system, and Azhillah and Rosmani [73] delivered sales forecasts through a business intelligence dashboard for a small food and beverage enterprise whose owners had relied on judgment and intuition. Al-Etaibi et al. [74] showed that, on a real store's transaction data, Linear Regression gave better RMSE, MAE, and R^2 values than XGBoost and Random Forest for short-term daily sales prediction. Zeinali et al. [75] connected business intelligence with customer relationship management data to predict customer retention. On this basis, the present system generates operational reports and presents a short-term sales forecast based on the spa's own transaction history, so that the owner can plan staffing, inventory, and promotions with evidence instead of guesswork.

The development methods reported in the reviewed studies include agile, Scrum, prototyping, Rapid Application Development, Feature-Driven Development, Design Science Research, and the waterfall model [2], [4], [37], [53], [55], [54]. Hema et al. [102] described Scrum as an incremental, time-boxed, iterative agile approach and explained that traditional models suit small products with stable requirements but are less suitable when requirements change quickly. They noted that Scrum allows the product to be changed in response to customer requests, since feedback is taken regularly through review meetings [102].

Prabowo and Wiguna [103] applied Scrum to a web-based information system for an MSME workshop that had no synchronization between existing stock and sales stock, and the Scrum artifacts of product backlog, sprint backlog, sprint, and working increment helped resolve time, scope, and cost issues. Rachmawati et al. [104] reported that a well-documented Scrum implementation made it easier to track progress and served as a guide and a history of the development of a mobile application. Graza et al. [89] used Scrum for a Philippine reservation system. The present study adopts the Agile approach because the requirements of a spa, such as service rules, discount rules, and report formats, are likely to be refined as the staff and owner review working versions of the system.

Further studies confirm that Scrum is a common choice for small systems that serve businesses and institutions. Basri et al. [105] applied the Scrum framework to the development of a system for D'Laundry, a micro, small, and medium enterprise that provides laundry services and had operated with conventional transaction methods, using an object-oriented analysis modeled with the Unified Modeling Language and a PHP and MySQL implementation. Simangunsong et al. [106] developed GoClass, a web-based classroom reservation system that supports efficient scheduling, approval, and analysis of classroom usage, following Agile Scrum with sprint planning, daily stand-ups, sprint reviews, and sprint retrospectives, and modeled it with use case, activity, and class diagrams. Manisha et al.

[107] discussed the effect of Agile Scrum on a team's productivity and on client satisfaction, noting that changes are organized into user stories for each sprint, developed in iterations, and reviewed by the customer at regular intervals under the supervision of a Scrum master. The regular customer review that these studies describe is the reason that Scrum fits the present project, in which the owner and staff of the spa review working versions of the system and the feedback is applied in the next iteration.

The reviewed studies evaluate systems with functional testing, usability measures, and software quality standards. Black box testing was used to validate system functions [34], [54]. Effendy et al. [39] used the System Usability Scale, and Wijayanti and Asri [108] combined the System Usability Scale with User Acceptance Testing among 13 respondents, obtaining a score of 82.11 (very good) and acceptance of all system functions. Ichsani [109] reviewed the concurrent think-aloud method as a way of performing usability evaluation as part of user acceptance testing, in which users' spoken opinions are recorded while they use the system.

The ISO/IEC 25010 standard was used in several studies. Graza et al. [89] applied it across eight quality characteristics, and Reguyal et al. [90] used it to evaluate a clinic scheduling system. Parhusip et al. [110] evaluated a web portal with four ISO/IEC 25010 characteristics, namely usability, functionality, performance efficiency, and reliability, using a Likert survey of actual users together with black box testing of login, browsing, search, filtering, and download scenarios. Petiwi and Jatmiko [111] evaluated a web-based claims portal using black box testing with 152 scenarios, a task-based evaluation, a System Usability Scale survey of 250 respondents, and reliability analysis, and found that functional suitability was the highest improvement priority. The evaluation of the present system follows the same approach: functional testing, user acceptance testing with the spa owner, staff, and customers, and the assessment of functionality, usability, reliability, efficiency, and security described in Chapter III.

Additional studies illustrate how functional, usability, and acceptance testing are combined. Riansyah and Syahputra [112] developed a web-based tracer study application with the Waterfall method and the Model-View-Controller approach using PHP, MySQL, HTML5, and CSS3. Functional testing showed that all features succeeded, the System Usability Scale produced an average score of 82 in the Excellent category, performance testing showed a login response time of 1.5 seconds with stability at 25 simultaneous users, and the user acceptance test showed that all users found the system easy to use [112]. Azkia et al. [113] applied black box testing with the state transition technique to a web-based clinic appointment system, mapping the flow of user states in a transition diagram and writing a test scenario for each change of state, and found that all the functions of the system worked according to the specification. State transitions are relevant to the present system because an appointment passes through states such as pending, confirmed, rescheduled, completed, and cancelled, and each change of state must be tested.

Balintag et al. [114] developed SORMS, a web-based online reservation management system for a garden resort that had a manual reservation system, integrating real-time booking and digital payments. They used Agile Scrum with 50 participants, of whom 18 were information technology professionals and 32 were end users, and reported mean scores of 4.68 on the ISO 25010 evaluation and 4.83 on the Technology Acceptance Model, together with 94 percent fewer booking errors and 80 percent faster processing than the manual method. A compatibility rating of 4.18 was identified as an area for improvement in later iterations [114]. The pairing of reservation with digital payment, the use of Scrum, and the combined use of ISO 25010 and the Technology Acceptance Model are close to the approach of the present study. As in the other reviewed development studies, the participants were the researchers' own respondents and the improvements reported are those of one resort, so the figures are evidence of feasibility and not a forecast of the result for a spa.

2.4 Synthesis of the Reviewed Literature

The foreign and local studies reviewed in this chapter come from different industries and countries, but they point to the same set of conclusions about what a system for a small service business such as Recovery Iloilo Spa must do and how it should be built.

First, manual and informal record-keeping produces the same problems wherever it is used. Salons that took bookings by WhatsApp or from notebooks ended up with incomplete schedules and client data [3], [5], small retail and food businesses reported inaccurate stock counts, lost records, calculation errors, and delayed reports [33], [34], [38], and businesses that computed salaries by hand experienced late pay and errors [54]. These are the problems that Chapter I identifies in the operations of Recovery Iloilo Spa.

Second, the reviewed systems that replaced manual work with a centralized web-based system reported improvements in scheduling, record accuracy, and efficiency. Online booking reduced scheduling conflicts and waiting time [89], [90], [88], integrated sales and inventory records improved stock accuracy and reporting [37], [39], and automated attendance and payroll improved timeliness and transparency [54], [56]. Several of these studies are small-scale development projects evaluated by their own developers and a limited number of respondents, so their reported results should be read as indications of feasibility and not as proof of a particular size of improvement.

Third, what customers value in spa and wellness services is closely tied to the quality of service and the accuracy of information. Responsiveness, professionalism, value for money, security, employee service, and accurate information were found to drive satisfaction and loyalty [22], and service quality significantly influenced satisfaction and repurchase intention in the hairdressing industry [25]. A system that supports quick confirmation, accurate service and price information, protected accounts, and structured feedback addresses these attributes directly.

Fourth, security and the willingness to use digital payments are connected. Cybersecurity risk was identified as a barrier to digital payments [46], and perceived cybersecurity concerns were linked to the intention of Philippine micro and small enterprises to use online payments [82]. Authentication, role separation, and access control were standard in the reviewed systems [6], [71], [40], which supports the security and accountability features planned for the present system.

Fifth, the development and evaluation approaches reported in the literature are consistent with those chosen for this study. Agile and Scrum approaches were used where requirements were expected to change [102], [103], [89], and quality was assessed with black box testing, the System Usability Scale, user acceptance testing, and ISO/IEC 25010 characteristics [39], [108], [110], [111]. Finally, the local studies show that Philippine small enterprises are adopting digital tools but are constrained by cost, limited technical expertise, and cultural resistance [79], which means that an affordable, easy-to-use system with clear role-based functions is appropriate.

Sixth, the additional studies gathered for this review give the same answer from other types of business. Reservation systems for barbershops, a salon, a café, a cat hotel and grooming business, a resort, and outpatient clinics were each developed because manual booking caused queues, errors, or delays [14], [15], [19], [18], [114], [20]. Point-of-sale systems for a retail store, a printing company, a small shop, and a beverage microenterprise were developed because manual sales and stock records produced mistakes, delayed reports, and mismatches between recorded sales and revenue [41], [42], [43], [44]. Payroll systems for a furniture business, an engineering consultancy, a university, a bank, a plantation, and a school were developed because manual or spreadsheet computation was error-prone and slow [57], [58], [59], [60], [61], [62]. The consistency of these accounts across different organizations supports the conclusion that the problems of Recovery Iloilo Spa are typical of small businesses that rely on manual records, and that the proposed remedy is well established.

Eighth, the literature on the spa and wellness market itself supports the reasons for building the system. Wellness spa businesses in different countries compete on the quality of the experience and on securing repeat visits [27], [28], [29], and retaining a customer depends on the same service attributes that the system is designed to support, namely timely confirmation, accurate information, and consistent service [22]. The systems literature adds that sales histories are used for planning in retail businesses [76], [77], [78], and that Scrum is widely used to build small business systems in short iterations with customer review [105], [107], [106]. Together, the market and systems literatures describe a business that depends on repeat patronage and an information system that keeps its records accurate and its decisions informed.

Ninth, the local studies bring the problem close to the setting of the present study. A study of data privacy awareness among small and medium enterprises in the province of Iloilo [86], studies of privacy practices among Philippine enterprises [84], [85], and a Philippine case study of compliance with the Data Privacy Act [83] show that the protection of customer data is a recognized obligation that small businesses do not always meet. Philippine systems for reservation, records, inventory, and sales were developed for institutions and small businesses that faced manual processes [94], [95], [96], [93], and Philippine reservation systems were evaluated with the ISO 25010 standard and the Technology Acceptance Model [89], [90], [87]. The proposed system is therefore consistent with the direction of local practice, and it adds the combination of functions that a spa requires.

Tenth, the reviewed studies on reminders and payments explain two design details of the present system. Reminders are used to reduce missed appointments in clinics [10], [11], [12], and the same reasoning applies to a spa because an unused appointment slot cannot be recovered. Digital payments speed up transactions and reduce cash errors but involve fees [48], [49], which supports the combination of cash recording, online payment, and end-of-day reconciliation in a single transaction record. These two details show how the conclusions drawn from the reviewed literature were converted into specific functions of the system.

Seventh, the studies that evaluated promotions and service quality warn that benefits should be measured and not assumed. Discounts and in-store promotions influenced purchase intention in one study [68] but had a positive and statistically insignificant effect in another [69], and service quality was related to satisfaction and loyalty in wellness, banking, and restaurant settings [30], [31], [32]. For this reason, the proposed system records the redemption of each voucher and the ratings that customers give, so that the owner can examine the results with the spa's own data. Finally, the security studies show that access control is applied by assigning each user a role and by keeping a record of actions [99], [101], [100], which is how the proposed system protects customer and financial records.

Although the reviewed works cover appointment booking, point-of-sale, inventory, payroll, customer relationship management, loyalty programs, security, and analytics, they typically address these areas one at a time or in pairs. The salon and spa studies concentrate on appointment booking and customer experience [2], [3], [4], [5], the point-of-sale and inventory studies concentrate on retail and food businesses [33], [34], [37], and the payroll studies concentrate on attendance and salary computation for general employees [54], [56]. Among the studies gathered for this review, none combines in a single platform appointment scheduling with therapist-aware availability, therapist rotation and commission computation, voucher and multiple-discount validation, an integrated point-of-sale with payment gateway and cash reconciliation, inventory monitoring, customer feedback linked to services and therapists, a separated and protected administrative portal with PIN verification and activity logs, and reports with short-term sales forecasting. The local literature is further limited by the fact that the Philippine reservation and information systems found were developed for accommodation, sports facilities, agriculture, healthcare, and local government rather than for spa operations [87], [88], [89], [90], [91].

The present study addresses this gap by developing and evaluating one integrated, web-based system for the specific operations of Recovery Iloilo Spa in Iloilo City. By building on the design practices that the reviewed studies report

as successful, namely role-based access, server-side conflict checking, integrated sales and stock records, automated commission computation, and dashboard reporting, the study applies them to a Philippine spa and combines them into a single platform.

The reviewed literature was used to formulate the requirements of the proposed system. Each requirement below follows from the findings summarized above and is tied to the studies that support it.

Requirement 1: Online appointment booking with conflict checking. The reviewed booking systems provide online creation, rescheduling, and cancellation of appointments and show availability in real time [2], [8], [88]. Studies of online reservation in the Philippines reported fewer scheduling conflicts and shorter waiting times [89], [90]. The proposed system therefore lets customers book online and verifies the selected service, its duration, and the therapist's availability on the server before confirming a reservation.

Requirement 2: Centralized appointment records and notifications. Missed appointments waste capacity [9], and the reviewed systems used automated notifications to keep clients informed [2]. The proposed system therefore keeps every appointment in one record with a status and sends confirmation, update, and cancellation messages by email.

Requirement 3: Customer accounts and a complete customer history. Studies of customer relationship management treat the organized management of customer information as a means of improving service and loyalty [64], [63], and staff in salons that keep paper records must search notebooks to find a returning customer's history [5]. The proposed system therefore keeps a customer profile with appointment history, preferences, and past transactions.

Requirement 4: An integrated point-of-sale with cash and online payments. The reviewed point-of-sale systems join transaction recording with inventory and reporting [36], [34], [33], and digital payments are associated with benefits in sales performance and operational efficiency as well as with security concerns [46], [82]. The proposed system therefore records services and retail products in one transaction, accepts cash and online payment through a payment gateway, and reconciles the cash count at the end of a period.

Requirement 5: Inventory linked to sales and usage. Integrated sales and stock records improved stock accuracy and reporting in the reviewed systems [37], [38], [39], and real-time stock monitoring was a feature of an inventory system built for a small enterprise [40]. The proposed system therefore reduces stock when products are sold, tracks supplies used in services, and flags items that are running low.

Requirement 6: Attendance and commission records for therapists. Automated attendance and payroll computation improved timeliness and transparency in the reviewed studies [54], [56], and payroll linked to performance measures improved accuracy [53]. The proposed system therefore records clock-in and clock-out times and computes commissions from completed services so that therapists can verify what they earned.

Requirement 7: Controlled vouchers and discounts. Loyalty and promotion programs influence customer loyalty [65], [66], [67], and their effect depends on the offers being applied correctly. The proposed system therefore validates the eligibility of a voucher or discount, applies the correct amount, and prevents the same voucher from being redeemed more than once.

Requirement 8: Structured customer feedback. Service quality and the accuracy of information drive satisfaction and loyalty in spa, wellness, and salon businesses [22], [23], [25]. The proposed system therefore collects ratings and comments that are linked to the service and the therapist so that management can monitor quality over time.

Requirement 9: Security and accountability. The reviewed systems applied authentication and role separation [2], [6], [40], and role-based access control was reported for a payment information system [71]. The proposed system

therefore applies role-based access, a hidden administrative login, a personal identification number for sensitive transactions, and activity logs that record who performed each action.

Requirement 10: Reports and short-term forecasting. Dashboards and simple forecasting methods were applied to small-business transaction data in the reviewed studies [73], [74], [39]. The proposed system therefore presents operational reports and a short-term sales forecast based on the spa's own transaction history.

Requirement 11: An agile development process and recognized evaluation criteria. Agile and Scrum approaches were used in the reviewed studies where requirements were expected to change [102], [103], [89], and evaluation relied on black box testing, user acceptance testing, usability scales, and ISO/IEC 25010 characteristics [108], [110], [111]. The proposed system is therefore developed iteratively and evaluated against these criteria, as described in Chapter III.

Several limitations of the reviewed works should be noted. First, many of the system-development studies were conducted by their own developers and evaluated with a limited number of respondents, as in the case of a usability study with 13 respondents [108] and an assessment with 50 respondents [87]. Their results show that a system was feasible and acceptable in a particular setting, but they do not establish how large the improvement would be in other settings.

Second, the businesses studied differ from a spa in important ways. The point-of-sale and inventory studies were conducted for retail and food businesses [33], [34], [37], and the booking studies for clinics, sports facilities, and farm equipment [6], [88], [89]. A spa must also coordinate the availability of therapists with the duration of each service, compute commissions, and control vouchers, which most of these studies did not address.

Third, several of the studies were published in Indonesian or conducted in Indonesian settings [3], [4], [33], [34], and the number of studies conducted in the Philippines is small in comparison. The findings from other countries are used in this paper to identify good design practices, and the local studies are used to describe the Philippine setting. The present study therefore evaluates the proposed system directly with the spa owner, staff, and customers rather than assuming that the results reported elsewhere will be repeated.

A comparison of the foreign and local studies adds a final observation. The foreign studies are more numerous and more specific to the functions of the proposed system, because they include salon and barbershop reservation systems, point-of-sale and payroll systems, and studies of the spa and wellness market. The local studies, by contrast, are fewer and tend to concern the environment in which a Philippine small business operates, namely the digital adoption of micro, small, and medium enterprises, concern for cybersecurity and data privacy, and the use of Scrum and ISO 25010 in locally developed systems. The two groups complement each other. The foreign studies show what a system for a salon, spa, or small retailer should contain, and the local studies show the conditions under which such a system must be adopted, secured, and evaluated in the Philippines, including in the province of Iloilo where the spa is located [86].

The reviewed literature also helped to define the scope of the present study. Features that were reported repeatedly, such as online booking, role-based access, integrated sales and stock records, attendance and payroll computation, and dashboards, were treated as core functions. Features that were reported in only a few studies or that depend on conditions that the spa may not have, such as location verification for attendance [56] or complex machine learning models for forecasting [74], [75], were treated as optional or were replaced by simpler alternatives, such as manual clock-in records and linear forecasting from the spa's own history. This approach follows the observation in the reviewed studies that small businesses are limited by funds, human resources, and technical expertise [47], [79], and that a system for them should therefore be practical to operate and maintain.

The literature also guides how the proposed system will be judged. The development studies that were reviewed typically assess a system with functional testing and with the opinions of the people who use it. Black box testing was used to confirm that each function works according to its specification [15], [113], usability was measured with the System Usability Scale [112], [108], and software quality was rated against ISO/IEC 25010 together with the Technology Acceptance Model in studies conducted in the Philippines [114], [89]. Several of the studies also recorded concrete operational measures, such as the share of booking errors and the speed of processing compared with a manual method [114], and login response time under simultaneous users [112]. The evaluation of the present system therefore combines functional testing, user acceptance testing with the spa owner, staff, and customers, and ratings of the software quality characteristics, and it reports the results as evidence from one business and not as a general claim. This position is consistent with the limitations noted above, since the reported improvements in the reviewed studies come from single organizations and relatively small groups of respondents.

Taken as a whole, the literature gives the present study three kinds of support. The first is evidence of need, because small service and retail businesses in many settings report the same consequences of manual records, namely scheduling conflicts, inaccurate stock and sales figures, delayed reports, and payroll errors [3], [33], [41], [58]. The second is evidence of method, because web-based systems built with Waterfall, Scrum, and related approaches, and tested with black box testing, usability scales, and ISO/IEC 25010, have been reported for many small organizations [42], [105], [110]. The third is evidence of context, because the local studies describe the digital adoption, privacy obligations, and constraints of Philippine small enterprises [79], [86], [82]. None of these bodies of evidence shows that a particular system will succeed in a particular spa, and the present study does not claim that they do. They provide the background against which the design choices of the proposed system are justified and its results are interpreted in the chapters that follow.

In relation to the studies reviewed, the contribution of the present study is therefore specific and modest. It does not propose a new technique for booking, point-of-sale, payroll, or forecasting, since each of these has been reported in simpler forms by the studies above. Its contribution is to combine these functions around the actual workflow of one Philippine spa, in which a booking leads to a service, the service leads to a sale and a therapist commission, the sale may involve a voucher or a discount and a change in stock, and every step leaves a record that the owner can review in a report. The salon systems [2], [5], the retail systems [34], [43], and the payroll systems [54], [57] each handle one part of this workflow, and the present study connects the parts so that the information entered at one step is available at the next without being typed again. The practical value of that connection is what the evaluation described in Chapter III is designed to examine.

This chapter reviewed foreign and local studies and related literature on appointment scheduling, service quality, point-of-sale and inventory, digital payments, attendance and commission, customer records and loyalty, security, analytics, and system development and evaluation. The reviewed works show that manual and informal record-keeping leads to scheduling conflicts, inaccurate stock and sales records, and payroll errors in small businesses, and that centralized web-based systems with role-based access, integrated sales and stock records, automated payroll computation, and dashboard reporting were developed to address these problems. They also show that research on salon, spa, and small-business systems is usually limited to one or two functions and rarely addresses the combined operations of a spa. The present study responds to this gap by developing one integrated system for Recovery Iloilo Spa, using an agile approach and recognized evaluation criteria, as described in the chapters that follow.

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CHAPTER III

METHODOLOGY

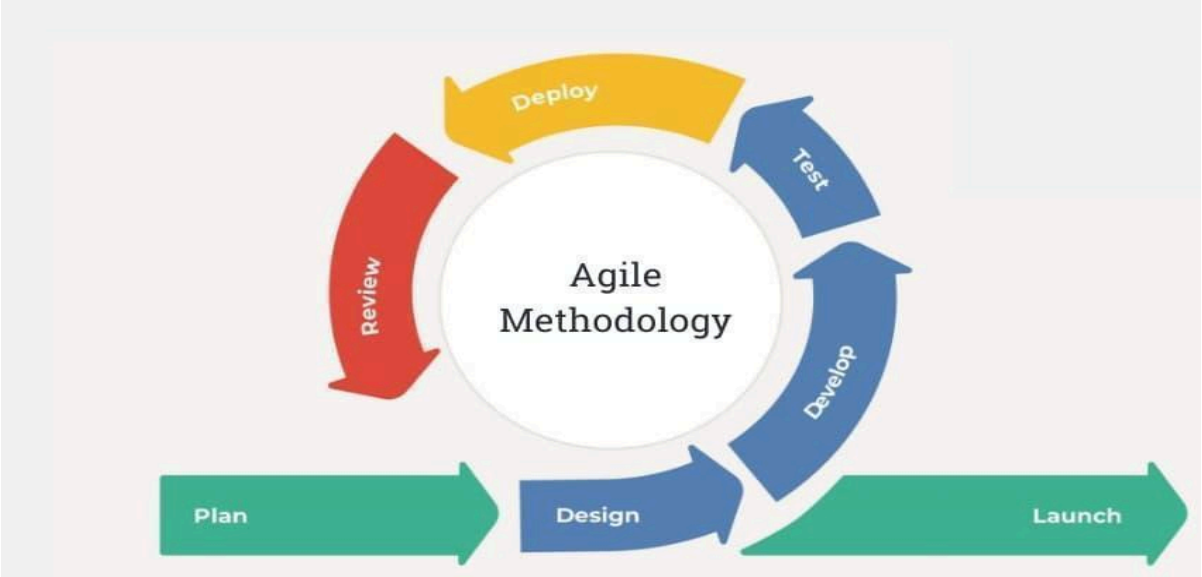
3.1 Research Design

This study will employ a developmental research design in the design, development, implementation, and evaluation of the Recovery Iloilo Spa Integrated System for Streamlining Operations and Efficient Analytics. This research design is appropriate because the primary objective of the study is to develop a web-based information system that will improve the spa's daily business operations through the integration of computerized processes. The proposed system will provide a centralized online platform for managing appointment scheduling, customer accounts and records, therapist management, inventory monitoring, Point-of-Sale (POS) transactions, payment and refund processing, commission computation, discount and voucher management, customer feedback, operational reports, and business analytics. The system will be developed using a three-tier architecture consisting of the presentation layer, application layer, and data layer to ensure organized system structure, maintainability, and efficient data management.

The developmental research design will guide the researchers through the systematic stages of requirements gathering, analysis of existing business processes, system planning, interface and database design, system development, testing, implementation, and evaluation. Information will be gathered through consultations and interviews with the spa owner, management, receptionists, therapists, and other staff to identify operational requirements, existing challenges, and user needs. Based on the collected information, the researchers will develop the various modules of the proposed system, including appointment management, customer account management, therapist management, inventory management, payment processing, Point-of-Sale (POS), commission management, voucher management, feedback management, security features, and operational reporting and analytics. These modules will be designed to improve operational efficiency, reduce manual workloads, strengthen record management, and support better customer service.

Upon completion of the development phase, the proposed system will undergo comprehensive testing and evaluation to determine whether its features and functions operate according to the identified requirements. The researchers will evaluate the system's quality based on the applicable software quality characteristics (such as those in ISO/IEC 25010, if required by your institution), including functionality, usability, reliability, performance efficiency, and security. Feedback and recommendations from the spa owner, management, receptionists, therapists, and other users will also be gathered to identify areas for improvement and determine whether the proposed system effectively supports appointment management, therapist coordination, inventory monitoring, payment processing, customer record management, operational reporting, business analytics, and overall service delivery at Recovery Iloilo Spa.

3.2 System Development Methodology



The study will use the **Agile Development Model in developing the Recovery Iloilo Spa Integrated System for Streamlining Operations and Efficient Analytics.** Agile is a software development methodology that emphasizes continuous improvement, iterative development, regular feedback, and close collaboration between developers and stakeholders throughout the system development process. This approach will allow the researchers to work closely with the spa owner, management, receptionists, therapists, and other authorized personnel to ensure that the proposed system will effectively address the operational requirements of Recovery Iloilo Spa. Through continuous testing, evaluation, and user feedback, the researchers will be able to refine system features, resolve identified issues, and implement necessary improvements throughout each development iteration. The proposed web-based system will be

developed using a three-tier architecture consisting of the presentation layer, application layer, and data layer to promote organized system structure, maintainability, scalability, and efficient data management.

The Agile Development Model will be used, consisting of the following phases:

1. Planning and Requirements Analysis – This phase will involve identifying, gathering, and analyzing the operational requirements, existing business processes, and challenges encountered by Recovery Iloilo Spa through consultation and coordination with the spa owner, management, HR personnel, receptionists, therapists, and other staff members. The researchers will examine the current workflow and identify issues related to customer registration and account management, email OTP verification, appointment scheduling, therapist assignment and attendance, inventory monitoring, Point-of-Sale (POS) transactions, payment and refund processing, customer record management, therapist commission computation, voucher and discount management, customer feedback collection, security controls, operational reporting, and revenue forecasting. The information gathered during this phase will serve as the basis for identifying the functional and non-functional requirements, determining project objectives, and planning the modules to be included in the proposed system.

2. System Design Phase – During this phase, the researchers will prepare the overall architecture and design of the proposed web-based system. The system will be designed using a three-tier architecture consisting of the presentation layer, application layer, and data layer to ensure proper separation of system components and efficient processing of information. The researchers will develop the database structure, system architecture, process flow diagrams, navigation structure, user interface layouts, and module designs to ensure that the platform will be organized, responsive, user-friendly, and secure. The design will include modules for customer registration and authentication, customer account management, appointment scheduling and management, therapist management, inventory management, Point-of-Sale (POS), payment processing, refund management, commission computation, discount and voucher management, customer feedback, activity logging, security controls, notifications, operational reports, business analytics, and revenue forecasting.

3. Development Phase – During this phase, the researchers will develop the proposed system by coding and integrating each module based on the approved system requirements and design specifications. Development will follow an iterative approach, allowing each completed feature to be reviewed, tested, and improved before proceeding to the next iteration. The researchers will develop customer registration with email OTP verification, secure user authentication, profile management, shopping cart functionality, appointment scheduling for online, walk-in, home-service, hotel, and partner bookings, therapist assignment and reassignment, therapist attendance monitoring, specialty management, inventory monitoring, Point-of-Sale (POS) transactions, payment gateway integration, refund processing, commission computation, voucher and discount validation, customer feedback collection, role-based access control, gate code verification, PIN verification, activity logging, automated notifications, operational reporting, business analytics, and formula-based revenue forecasting. Continuous collaboration with the spa owner, management, receptionists, therapists, and other intended users will ensure that each module will satisfy operational requirements and business needs.

4. Testing Phase – This phase will focus on verifying and evaluating the quality and performance of the proposed system before deployment. The researchers will conduct various testing procedures to determine whether all modules and system functions operate according to the specified requirements. The testing process will evaluate customer registration and authentication, appointment scheduling, therapist assignment, attendance monitoring, inventory management, Point-of-Sale (POS) transactions, payment processing, refund handling, commission computation, voucher validation, customer feedback, notifications, security mechanisms, operational reports, analytics, and revenue forecasting. The system will also be evaluated in terms of functionality, usability, reliability, performance efficiency, security, and compatibility. Feedback and recommendations from the spa owner, management, receptionists, therapists,

and customers will be gathered to identify necessary improvements and ensure that the proposed system effectively supports daily business operations.

5. Deployment Phase – This phase will involve implementing the completed web-based system within Recovery Iloilo Spa for actual operational use. The spa owner, administrators, receptionists, therapists, accounting personnel, and customers will be allowed to access and utilize the appropriate system features based on their assigned roles and permissions. The deployed system will support customer registration, appointment scheduling, therapist management, inventory monitoring, Point-of-Sale (POS) transactions, payment processing, refund management, commission computation, voucher and discount validation, customer feedback collection, operational reporting, business analytics, and revenue forecasting. During implementation, users will provide additional comments and recommendations regarding the system's functionality, usability, efficiency, and overall effectiveness. The collected feedback will support further refinements and ensure that the proposed system will improve operational efficiency, strengthen record management, enhance customer service, and support informed decision-making within Recovery Iloilo Spa.

3.3 System Architecture

The **Recovery Iloilo Spa Integrated System for Streamlining Operations and Efficient Analytics** will adopt a **three-layer architecture** consisting of the **Presentation Layer**, **Application Layer**, and **Data Layer**. This architectural approach will separate the system into three interconnected components, each having a specific responsibility in processing, managing, and presenting information. By separating the user interface, business logic, and database, the system will become more organized, maintainable, scalable, and secure. This architecture will also simplify future maintenance, updates, and feature enhancements since modifications made to one layer will have minimal impact on the other layers.

The proposed system will also follow a **hub-and-spoke communication model**, wherein the **Application Layer** will serve as the central processing hub between the Presentation Layer and the Data Layer. All requests initiated by users—including customer registrations, appointment bookings, inventory transactions, payment processing, therapist management, report generation, and other operational activities will first be processed and validated by the Application Layer before being stored in or retrieved from the database. Likewise, all information requested by users will pass through the Application Layer before being displayed on the user interface. This approach will ensure proper validation, secure processing, and consistent handling of all system transactions.

The system will be structured into three main layers:.

1.Presentation Layer (User Interface)

The Presentation Layer will serve as the part of the system that users directly interact with through a web browser. It will be responsible for displaying information, receiving user input, and providing access to the different features of the proposed system. The interface will be designed to be responsive, organized, user-friendly, and accessible through desktop computers, laptops, tablets, and mobile devices with internet access.

For customers, the Presentation Layer will provide pages for customer registration with Email OTP verification, secure login, password recovery, profile management, browsing available spa services and products, shopping cart management, therapist selection or "Any Available Therapist" option, online appointment booking, walk-in appointment viewing, home-service booking requests, hotel and partner bookings, appointment rescheduling and cancellation, viewing appointment history, payment processing, refund request submission, viewing digital receipts, receiving appointment notifications, and submitting customer feedback and ratings after completed services.

For the spa owner, administrators, receptionists, accountants, HR personnel, and therapists, the Presentation Layer will provide dashboards and interfaces for managing appointments, therapist assignments and reassignments, therapist attendance, therapist specialties, customer records, inventory monitoring, Point-of-Sale (POS) transactions, payment and refund processing, commission monitoring, voucher and discount management, customer feedback review, activity logs, operational reports, business analytics, and revenue forecasting. Authorized users will also be able to monitor appointment status, approve bookings, manage products and services, update inventory, verify transactions, and generate reports through the system interface.

The Presentation Layer will be developed using HTML, CSS, and Vanilla JavaScript to provide a responsive and interactive user experience across supported web browsers.

2. Application Layer

The Application Layer will serve as the core processing component of the proposed system. It will receive requests from the Presentation Layer, validate user input, execute business rules, process transactions, communicate with the database, and return processed information to the Presentation Layer. This layer will ensure that all business operations are executed accurately, consistently, and securely.

Specifically, the Application Layer will:

- Process customer registration, Email OTP verification, secure authentication, password management, and profile updates.
- Manage appointment scheduling for online, walk-in, home-service, hotel, and partner bookings while validating therapist availability, operating hours, travel-time allowances, booking conflicts, duplicate reservations, and overbooking.
- Process appointment approvals, cancellations, rescheduling, therapist assignments and reassignments, appointment status updates, and automated notifications.
- Manage therapist attendance, therapist specialties, workload balancing, automatic therapist rotation, and performance monitoring.
- Process shopping cart transactions, Point-of-Sale (POS) transactions, payment processing through supported payment methods, refund requests, receipt generation, and transaction verification.
- Automatically calculate therapist commissions based on completed services, service categories, transaction records, multi-person sessions, and applicable commission rules.
- Monitor inventory by updating stock quantities whenever products are used, sold, restocked, or adjusted while identifying low-stock items requiring replenishment.
- Manage customer records, appointment history, service preferences, transaction history, voucher usage, and feedback records.
- Validate discounts and vouchers, including Senior Citizen, PWD, Employee, Celebration Discounts, Gift Certificates, and promotional vouchers while preventing duplicate redemption and unauthorized usage.
- Process customer ratings, comments, and feedback linked to completed appointments and assigned therapists.
- Implement security mechanisms including role-based access control, hidden administrative login with Gate Code verification, PIN verification for sensitive transactions, activity logging, and secure session management.
- Generate operational reports, dashboards, business analytics, sales summaries, therapist productivity reports, inventory reports, POS readings, cash reconciliation reports, and formula-based revenue forecasts.
- Control all communication between the Presentation Layer and the Data Layer to ensure that every transaction is validated, processed securely, and recorded accurately.

The Application Layer will be developed using PHP, which will implement the business logic and facilitate communication between the user interface and the MariaDB relational database.

3. Data Layer (Database)

The Data Layer will serve as the centralized repository of all information managed by the proposed system. It will be responsible for storing, organizing, updating, retrieving, and protecting all operational data required by Recovery Iloilo Spa. The database will ensure data integrity, consistency, security, and efficient retrieval while supporting business operations, reporting, and analytics.

The Data Layer will store:

- Customer account information, login credentials, Email OTP verification records, user roles, and profile information.
- Appointment records including booking details, selected services, assigned therapists, schedules, booking status, service locations, appointment history, and add-on services.
- Therapist information including attendance records, specialties, assignments, schedules, commission records, and performance-related transaction history.
- Customer records including contact information, service preferences, transaction history, feedback records, and voucher usage.
- Inventory records including products, supplies, stock quantities, stock movements, restocking history, and low-stock monitoring.
- Point-of-Sale (POS) transactions, payment records, refund transactions, payment methods, digital receipts, discounts applied, and complete transaction history.
- Voucher records, gift certificates, promotional campaigns, redeemed discounts, eligibility validation, and redemption history.
- Customer feedback, ratings, comments, and service evaluations linked to completed appointments and therapists.
- Notifications, appointment confirmations, payment confirmations, and email notification history.
- Security records including activity logs, PIN verification records, role permissions, and audit trails.
- Operational reports, business analytics, sales summaries, inventory reports, therapist productivity reports, cash reconciliation records, and formula-based revenue forecasting data.

The proposed system will utilize MariaDB as its relational database management system to ensure efficient data storage, secure information management, reliable transaction processing, and fast retrieval of records. The centralized database will support approximately 30 core database tables, allowing the system to maintain organized business information while supporting the operational, reporting, and analytical requirements of Recovery Iloilo Spa.

3.4 Tools and Technologies Used

The development of the Recovery Iloilo Spa Integrated System for Streamlining Operations and Efficient Analytics will utilize various software tools and technologies to support the design, development, implementation, and maintenance of the proposed web-based system. These technologies will work together to provide a responsive user interface, efficient business logic, secure data management, reliable transaction processing, and organized operational reporting.

- **Frontend: HTML, CSS, and Vanilla JavaScript** - HTML (HyperText Markup Language), CSS (Cascading Style Sheets), and Vanilla JavaScript will be used to develop the Presentation Layer of the proposed system. HTML will provide the structure and content of the web pages, CSS will define the visual appearance, layout, and responsiveness of the user interface, while Vanilla JavaScript will add interactivity and dynamic functionality to the system. These technologies will enable the development of a responsive, organized, and user-friendly interface that can be accessed through desktop computers, laptops, tablets, and mobile devices using a web browser. The frontend will support customer registration, appointment booking,

shopping cart management, therapist selection, payment processing, dashboard visualization, notifications, and other interactive system features.

● **Backend: PHP** - PHP will be used as the primary server-side programming language for developing the Application Layer of the proposed system. It will process business logic, validate user input, manage user authentication, handle appointment scheduling, therapist assignment, inventory management, Point-of-Sale (POS) transactions, payment processing, refund management, commission computation, voucher validation, customer feedback, operational reports, business analytics, and revenue forecasting. PHP will also facilitate secure communication between the user interface and the MariaDB database while implementing security features such as role-based access control, Gate Code verification, PIN authentication, session management, and activity logging.

● **Database Management System: MariaDB** - MariaDB will serve as the relational database management system of the proposed Recovery Iloilo Spa Integrated System. It will store, organize, retrieve, and manage all operational data required by the system, including customer accounts, Email OTP verification records, appointments, therapist assignments, attendance records, inventory information, Point-of-Sale (POS) transactions, payment records, refund transactions, customer feedback, voucher and discount records, commission records, activity logs, notifications, operational reports, business analytics, and revenue forecasting data. MariaDB will ensure data integrity, security, consistency, and efficient retrieval of information while supporting approximately thirty core database tables within the proposed system.

● **Integrated Payment Gateway: PayMongo Payment Gateway** - The proposed system will integrate an online payment gateway to support secure electronic transactions. The payment gateway will enable customers to pay for appointments using supported digital payment methods such as GCash, QR Ph, and debit or credit cards. It will also facilitate payment verification, transaction recording, digital receipt generation, and secure communication between the system and electronic payment services.

● **Email Service** - An email service utilizing PHPMailer will be integrated into the proposed system to support automated email notifications. This service will be responsible for sending Email OTP verification codes during customer registration, appointment confirmations, booking updates, payment notifications, password recovery messages, and other system-generated communications to registered users.

● **Reporting and Export Libraries: PhpSpreadsheet and mPDF** - PhpSpreadsheet and mPDF will be used as reporting and export libraries to support the daily sales and cash reconciliation report of the proposed system. PhpSpreadsheet will generate downloadable Excel workbooks that preserve live formulas for figures such as net cash and short/over computation, while mPDF will generate downloadable PDF copies of the same report for printing and record-keeping. Together, these libraries will allow management and the receptionist to export accurate, formatted daily and sales reports directly from the system.

● **Development Environment: Visual Studio Code -** Visual Studio Code (VS Code) will be used as the primary Integrated Development Environment (IDE) during system development. It provides features such as syntax highlighting, intelligent code completion, debugging tools, extension support, integrated terminal access, and Git version control integration. These capabilities will assist the researchers in writing, testing, debugging, and maintaining the source code more efficiently throughout the software development process.

● **Web Browser** - Modern web browsers such as Google Chrome, Microsoft Edge, Mozilla Firefox, and Safari will be used for testing and accessing the proposed web-based system. These browsers will allow users to utilize the system without installing additional software, ensuring accessibility across different devices and operating systems.

● **Three-Tier Architecture** - The proposed system will be developed using a three-tier architecture consisting of the Presentation Layer, Application Layer, and Data Layer. This architectural design will separate the user interface, business logic, and database management into independent layers to improve maintainability, scalability, security, and overall system performance.

3.5 Data Gathering Techniques

The researchers will use the following data gathering techniques to identify the operational requirements of Recovery Iloilo Spa, analyze the existing business processes, and gather the information necessary for the design, development, implementation, and evaluation of the proposed **Recovery Iloilo Spa Integrated System for Streamlining Operations and Efficient Analytics**.

● **Survey and Questionnaire -** The researchers will prepare and distribute structured questionnaires to the spa owner, management, receptionists, therapists, and other authorized staff members to collect information regarding the current operational practices, existing problems, user requirements, and suggested system improvements. The survey will gather measurable data related to appointment scheduling, customer account management, therapist assignment, inventory monitoring, Point-of-Sale (POS) transactions, payment processing, commission computation, voucher and discount management, customer feedback, operational reporting, security requirements, and business analytics. The collected responses will serve as one of the primary bases for identifying the functional and non-functional requirements of the proposed system.

● **Observation of Current Operations** – The researchers will conduct direct observation of the daily operations of Recovery Iloilo Spa to obtain firsthand information regarding the current workflow and operational procedures. The observation will focus on customer registration, appointment scheduling, therapist assignment, attendance monitoring, inventory management, payment processing, Point-of-Sale (POS) transactions, commission computation, voucher validation, customer service, and report preparation. This technique will enable the researchers to identify delays, repetitive tasks, manual errors, bottlenecks, and other operational inefficiencies that the proposed system will address.

● **Interview –** The researchers will conduct interviews with the spa owner, management, HR personnel, receptionists, therapists, and other authorized staff members to obtain detailed information regarding the operational processes, business policies, user expectations, and specific system requirements. The interviews will provide a deeper understanding of existing challenges involving appointment management, therapist scheduling, payment procedures, inventory control, commission computation, promotional discounts, customer feedback, reporting practices, and overall business operations. Information gathered during the interviews will help validate the data collected through surveys and observations.

● **Review of Existing Systems** – The researchers will review existing spa management systems, appointment booking systems, and other related web-based information systems to examine their features, workflows, user interfaces, and best practices. This technique will enable the researchers to identify useful functionalities, effective design approaches, and operational strategies that may be adapted and incorporated into the proposed Recovery Iloilo Spa Integrated System while ensuring that the developed system addresses the specific operational requirements of the client.

● **Document and Record Review** –The researchers will review existing business documents and operational records maintained by Recovery Iloilo Spa to understand the current methods of data management and transaction processing. Documents such as appointment records, customer information, inventory lists, therapist schedules, attendance records, commission records, sales reports, payment records, voucher records, promotional activities, and other operational documents will be examined to identify existing data structures, workflow processes, and areas requiring improvement. The information obtained through document review will assist the researchers in designing an organized database structure, accurate system modules, and reliable reporting features for the proposed system.

3.6 Testing Procedures

To ensure that the **Recovery Iloilo Spa Integrated System for Streamlining Operations and Efficient Analytics** functions properly and satisfies the operational requirements of Recovery Iloilo Spa, the researchers will conduct several testing procedures throughout the development process. These testing procedures will help identify and correct system errors, verify that all modules operate according to the specified requirements, improve system performance, and ensure that the proposed system is reliable, secure, and suitable for actual implementation.

- **Unit Testing** – Each module of the proposed system will be tested individually to verify that its functions operate correctly before being integrated with other modules. The researchers will evaluate customer registration and Email OTP verification, user authentication, appointment scheduling, therapist assignment and attendance, shopping cart, inventory management, Point-of-Sale (POS) transactions, payment processing, refund management, commission computation, voucher validation, customer feedback, notifications, operational reports, business analytics, and revenue forecasting. Unit testing will ensure that every module performs its intended functions accurately and independently.

- **System Testing** – After all system modules have been integrated, the researchers will conduct system testing to evaluate the proposed system as a complete web-based platform. This testing procedure will verify the interaction among the Presentation Layer, Application Layer, and Data Layer, ensuring that information flows correctly throughout the system. It will also confirm that all integrated modules—including customer account management, appointment management, therapist management, inventory monitoring, POS transactions, payment processing, refund handling, commission computation, voucher management, customer feedback, security mechanisms, notifications, operational reports, analytics, and revenue forecasting—function together properly without errors or inconsistencies.

- **User Acceptance Testing (UAT)** – The proposed system will undergo User Acceptance Testing (UAT) with the participation of the spa owner, administrators, receptionists, therapists, accounting personnel, and selected customers. The purpose of this testing is to determine whether the system satisfies user requirements, supports actual business operations, and provides a user-friendly and efficient experience. Participants will evaluate the system based on its functionality, ease of use, accuracy, convenience, and overall effectiveness in improving appointment scheduling, customer management, therapist coordination, inventory control, payment processing, reporting, and other operational activities.

- **Performance Testing** – Performance testing will be conducted to evaluate the system's responsiveness, stability, and efficiency while handling multiple users, concurrent appointments, continuous Point-of-Sale (POS) transactions, inventory updates, report generation, and other operational activities. The researchers will determine whether the system can maintain acceptable response times and reliable performance under normal operating conditions while processing large volumes of business data.

- **Security Testing (Optional)** – Security testing will be conducted to verify that the proposed system adequately protects user accounts, confidential business information, and customer records from unauthorized access. The researchers will evaluate authentication mechanisms, Email OTP verification, role-based access control, Gate Code verification, PIN verification, session management, activity logging, payment processing security, and database access controls. This testing will ensure that sensitive information is handled securely and that only authorized users can access restricted system functions.

- **Final Evaluation** - After all testing activities have been completed, the researchers will evaluate the overall performance of the proposed system to determine whether it meets the operational requirements of Recovery Iloilo Spa. The evaluation will focus on the system's **functionality, usability, reliability, performance efficiency, and security**, as well as its ability to improve appointment management, therapist coordination, inventory monitoring, customer record management, payment processing, operational

reporting, business analytics, and overall service delivery. Feedback and recommendations gathered from the spa owner, management, receptionists, therapists, accounting personnel, and customers will be used to identify necessary improvements before the final implementation of the proposed system.

3.7 Evaluation Criteria

The developed **Recovery Iloilo Spa Integrated System for Streamlining Operations and Efficient Analytics** will be evaluated to determine its overall quality, performance, reliability, and usefulness in improving the daily operations of Recovery Iloilo Spa. The evaluation will assess whether the proposed system effectively satisfies the operational requirements of the establishment and provides an organized, efficient, and secure platform for managing spa services.

- **Functionality** – This criterion will evaluate whether the proposed system performs all intended functions correctly and completely. It will determine whether the different modules including customer registration, appointment scheduling, therapist management, inventory monitoring, Point-of-Sale (POS) transactions, payment processing, refund management, commission computation, voucher and discount management, customer feedback, notifications, operational reports, analytics, and revenue forecasting operate according to the specified requirements. The system should produce accurate outputs and successfully complete user transactions without major errors.

- **Accuracy** – This criterion measures how correctly the system processes, stores, and displays information. It evaluates the correctness of appointment schedules, inventory records, transaction details, commission computations, customer information, and generated reports. Accurate information helps reduce errors and supports better daily operations.

- **Usability** – This criterion will determine how easy the proposed system is to learn, understand, and operate for customers, receptionists, therapists, administrators, management, and other authorized users. It will evaluate the clarity of menus, forms, navigation, instructions, dashboards, and overall user interface design. A user-friendly system will allow users to complete their tasks efficiently with minimal training and confusion.

- **Reliability** – This criterion will evaluate the ability of the proposed system to operate consistently and continuously without unexpected failures, crashes, or interruptions. It will determine whether appointments, transactions, reports, notifications, and other system functions can be processed accurately during regular business operations while maintaining stable performance and dependable service.

- **Performance Efficiency** – This criterion will assess how efficiently the proposed system performs its operations while utilizing available system resources. It will evaluate the speed of appointment processing, payment transactions, inventory updates, report generation, analytics processing, database retrieval, and other system functions. An efficient system should respond quickly while maintaining smooth performance during normal daily operations.

- **Security** – This criterion will evaluate the effectiveness of the security mechanisms implemented within the proposed system to protect confidential business information and customer data. It will assess user authentication, Email OTP verification, role-based access control, Gate Code verification, PIN authentication, activity logging, payment security, and database protection. The system should ensure that only authorized users can access restricted functions while maintaining the confidentiality, integrity, and security of stored information.

- **Maintainability** – This criterion will evaluate how easily the proposed system can be modified, updated, corrected, and maintained after implementation. It will assess whether the three-tier architecture, organized

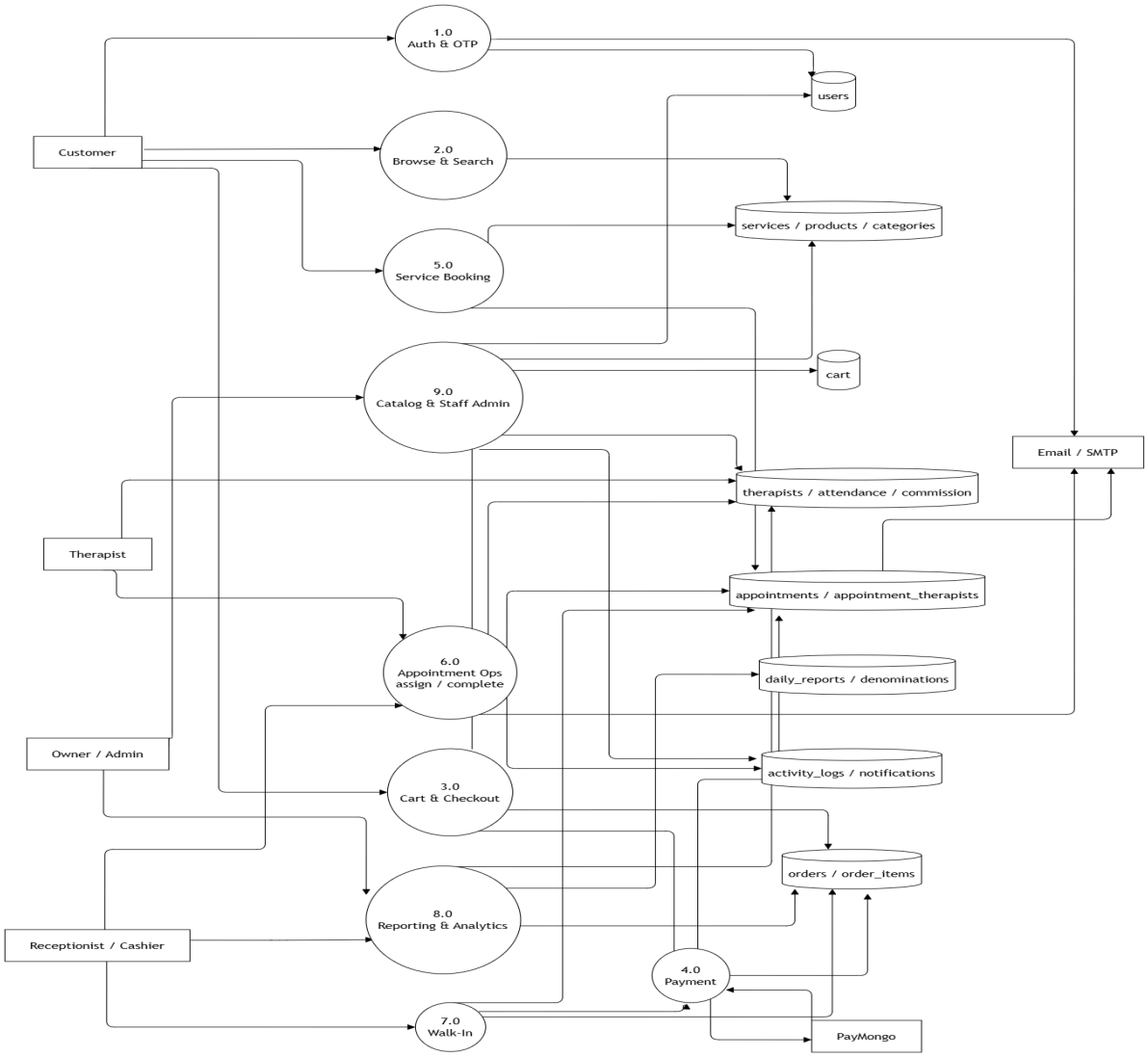
source code, database structure, and modular system design support future enhancements, bug fixes, and feature improvements without significantly affecting other components of the system.

● **Feedback Integration** – This criterion evaluates how effectively the system collects, stores, and manages customer feedback and ratings. It checks whether customer comments and evaluations are properly recorded and made available for review by management. This information can help identify areas for improvement and support better service quality.

● **Overall User Satisfaction** – This criterion will evaluate the overall satisfaction of the spa owner, management, receptionists, therapists, accounting personnel, and customers regarding the proposed system. It will determine whether users believe that the system improves appointment management, therapist coordination, inventory monitoring, payment processing, customer service, operational reporting, business analytics, and the overall efficiency of Recovery Iloilo Spa.

SUPPORTING DIAGRAMS (INCLUDED WITHIN CHAPTER 3)

Data Flow Diagram (DFD – Level 1)



Integrated System DFD Level 1 Overview

User (Customer/Staff) → Integrated System → Database

1.0 Authentication and OTP

- Customers, therapists, receptionists, and administrators will submit their registration or login credentials to gain authorized access to the Recovery Iloilo Spa Integrated System according to their assigned user roles.
- The system will validate the submitted credentials against **D1: Users**, verify the account status and assigned permissions, and establish a secure session that will control access to authorized system functions.
- A one-time password (OTP) will be generated and transmitted through the **Email / SMTP** service to verify the identity of the user before access to the system will be granted.

2.0 Browse and Search

- Customers will browse the spa catalog and search for available wellness services, products, and service categories offered by Recovery Iloilo Spa.
- The system will retrieve the requested information from **D2: Services, Products, and Categories** and will display updated details regarding service descriptions, pricing, session duration, available products, and service classifications.
- The process will allow customers to review accurate service information before proceeding with appointment booking or purchasing spa products.

3.0 Cart and Checkout

- Customers will select spa services and retail products and will place their chosen items into their shopping cart before confirming their purchase or reservation.
- The system will record the selected items in **D3: Cart** and, after checkout has been confirmed, will create the corresponding transaction in **D4: Orders and Order Items** for further processing.
- The completed transaction will then be forwarded to the Payment process so that the selected payment method can be processed and the customer's order can be finalized.

4.0 Payment

- The system will receive payment requests originating from online checkout transactions and walk-in transactions processed by the receptionist or cashier.
- Payment requests for online transactions will be transmitted securely to **PayMongo**, and the payment confirmation returned by the gateway will be used to update the payment status and transaction details stored in **D4: Orders and Order Items**.
- After successful payment processing, the system will record the completed transaction and will send payment confirmations, receipts, or related notifications through the **Email / SMTP** service whenever applicable.

5.0 Service Booking

- Customers will submit appointment requests by selecting their preferred spa service, appointment schedule, therapist preference, number of persons, and other booking details required by the establishment.
- The system will retrieve service information, pricing, therapist availability, and service categories from **D2: Services, Products, and Categories** to validate the booking request and calculate the applicable service charges.
- Confirmed booking information will be stored in **D5: Appointments and Appointment Therapists**, where the reservation will remain pending until it is reviewed and processed by the receptionist or authorized staff.

6.0 Appointment Operations

- Receptionists will review pending appointments, approve or decline booking requests, assign or reassign therapists, update appointment status, and mark appointments as completed once the requested services have been rendered.
- Therapists will submit their attendance and duty information, while the system will update **D6: Therapists, Attendance, and Commission** to record attendance, therapist assignments, and the commissions earned from completed services.
- Throughout the appointment process, the system will continuously update **D5: Appointments and Appointment Therapists**, generate notifications through the **Email / SMTP** service whenever necessary, and record administrative activities in **D8: Activity Logs and Notifications** to maintain an accurate audit trail.

7.0 Walk-In

- Receptionists or cashiers will encode appointment information for customers who will avail themselves of spa services without making prior online reservations.
- The system will create the corresponding appointment record in **D5: Appointments and Appointment Therapists** and simultaneously generate the related transaction in **D4: Orders and Order Items** for proper sales recording.
- The completed walk-in transaction will then be forwarded to the Payment process so that payment can be processed using the same workflow applied to online transactions, ensuring consistency in transaction recording.

8.0 Reporting and Analytics

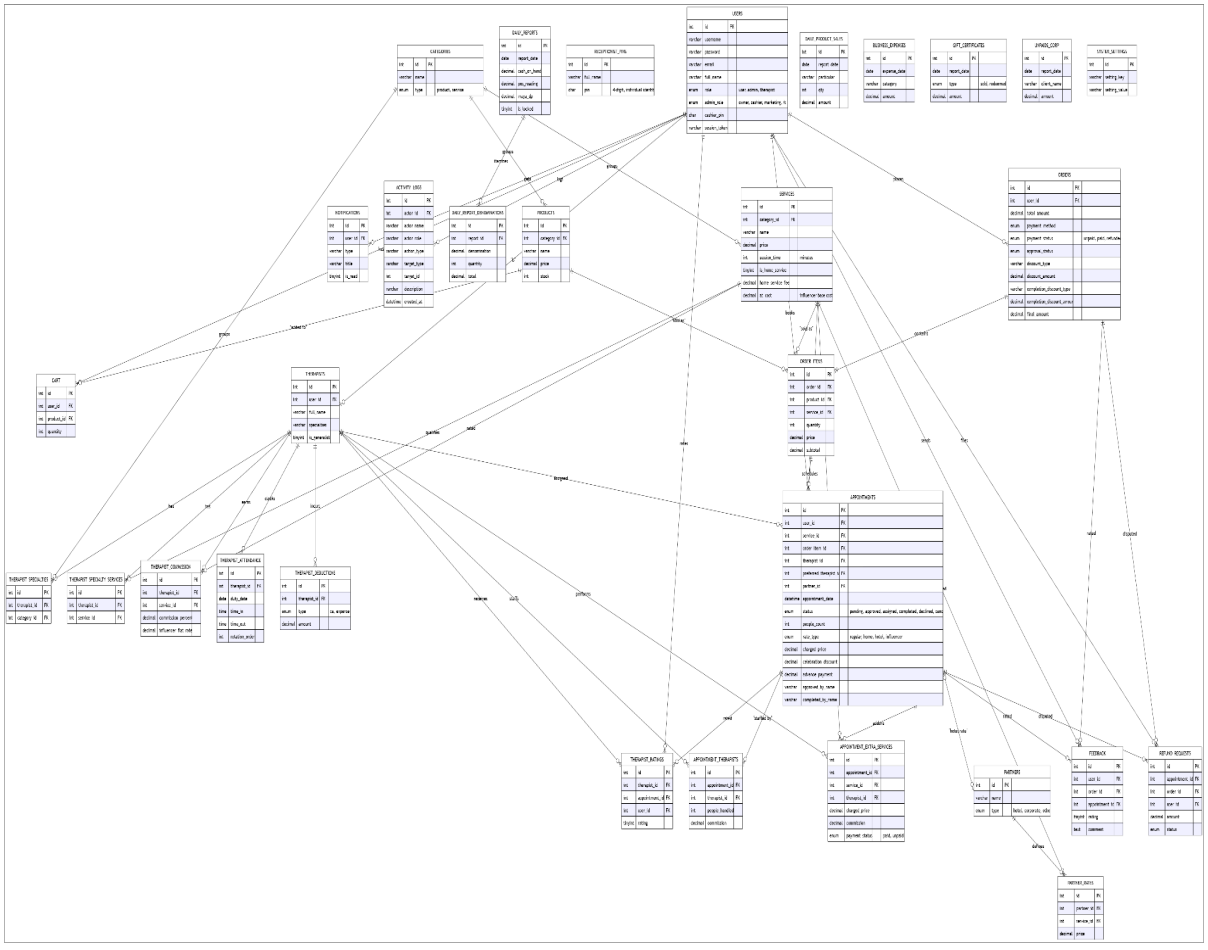
- Receptionists will prepare daily cash counts, cash denomination records, and end-of-day summaries, while the Owner or Administrator will review the consolidated operational performance of the spa through the reporting dashboard.
- The system will store daily cash reconciliation records in **D7: Daily Reports and Denominations** and will retrieve operational data from **D4: Orders and Order Items**, **D5: Appointments and Appointment Therapists**, and **D6: Therapists, Attendance and Commission** to generate comprehensive reports.
- The reporting and analytics module will generate sales reports, appointment reports, inventory movement summaries, therapist productivity reports, commission reports, customer activity reports, payment summaries, voucher utilization reports, and operational analytics. It will also produce short-term revenue forecasts based on recorded historical transactions and allow management to filter, view, and export reports into Excel and PDF formats. These analytics will provide management with timely insights that will support operational monitoring, performance evaluation, and informed business decision-making.

9.0 Catalog and Staff Administration

- The Owner or Administrator will manage the spa catalog by adding, updating, or maintaining available services, products, categories, pricing, inventory information, therapist profiles, specialties, commission rates, attendance records, and user accounts.
- The system will store and update catalog information in **D2: Services, Products, and Categories**, maintain therapist information within **D6: Therapists, Attendance, and Commission**, and manage user accounts and access permissions through **D1: Users**.

All administrative modifications performed within the system will be recorded in **D8: Activity Logs and Notifications** to maintain accountability, preserve an accurate audit trail, and support secure management of system activities.

Entity Relationship Diagram (ERD)



Entity Attributes

- users (id, username, password, email, full_name, role, admin_role, cashier_pin, session_token)
- therapists (id, user_id, full_name, specialties, is_generalist)
- receptionist_pins (id, full_name, pin)
- categories (id, name, type)
- services (id, category_id, name, price, session_time, is_home_service, home_service_fee, at_cost)
- products (id, category_id, name, price, stock)
- cart (id, user_id, product_id, quantity)
- orders (id, user_id, total_amount, payment_method, payment_status, approval_status, discount_type, discount_amount, completion_discount_type, completion_discount_amount, final_amount)
- order_items (id, order_id, product_id, service_id, quantity, price, subtotal)
- appointments (id, user_id, service_id, order_item_id, therapist_id, preferred_therapist_id, partner_id, appointment_date, status, people_count, rate_type, charged_price, celebration_discount, advance_payment, approved_by_name, completed_by_name)
- appointment_therapists (id, appointment_id, therapist_id, people_handled, commission)
- appointment_extra_services (id, appointment_id, service_id, therapist_id, charged_price, commission, payment_status)
- therapist_specialties (id, therapist_id, category_id)
- therapist_specialty_services (id, therapist_id, service_id)
- therapist_commission (id, therapist_id, service_id, commission_percent, influencer_flat_rate)
- therapist_attendance (id, therapist_id, duty_date, time_in, time_out, rotation_order)
- therapist_deductions (id, therapist_id, type, amount)
- therapist_ratings (id, therapist_id, appointment_id, user_id, rating)
- partners (id, name, type)
- partner_rates (id, partner_id, service_id, price)
- feedback (id, user_id, order_id, appointment_id, rating, comment)
- refund_requests (id, appointment_id, order_id, user_id, amount, status)
- notifications (id, user_id, type, title, is_read)
- activity_logs (id, actor_id, actor_name, actor_role, action_type, target_type, target_id, description, created_at)
- daily_reports (id, report_date, cash_on_hand, pos_reading, maya_dp, is_locked)

- daily_report_denominations (id, report_id, denomination, quantity, total)
- daily_product_sales (id, report_date, particular, qty, amount)
- business_expenses (id, expense_date, category, amount)
- gift_certificates (id, report_date, type, amount)
- unpaids_corp (id, report_date, client_name, amount)
- system_settings (id, setting_key, setting_value)

Relationships

- A **User** will create an account in the system in order to access the features corresponding to the assigned role, which may be a customer, therapist, receptionist, or administrator. The administrative role will further identify whether the account belongs to the owner, cashier, marketing personnel, or information technology staff, while the session token will maintain secure access throughout system usage.
- A **User** account may optionally be linked to one **Therapist** record. This relationship will allow therapists to authenticate using their user account while maintaining a separate professional profile containing therapist information, specialties, and assignment details.
- A **User** will be able to schedule multiple **Appointments** for spa services. Each appointment will store the selected service, appointment date and time, preferred therapist, assigned therapist, partner information, number of persons, pricing details, discounts, advance payment, booking status, and approval information.
- A **User** will be able to place multiple **Orders** involving spa services and wellness products. Each order will record the payment method, payment status, approval status, applied discounts, total amount, and final payable amount for proper transaction management.
- A **User** will be able to add and manage selected **Products** inside the **Cart** before confirming checkout. The cart will temporarily store the selected products together with their quantities, allowing customers to review or modify their selections before completing the purchase.
- A **User** will receive **Notifications** regarding appointment confirmations, payment updates, refund requests, promotional announcements, and other important system activities. Each notification will store its type, title, and read status to help users monitor system-generated messages.
- A **User** will be able to submit **Feedback** after completing a service transaction by providing a rating and written comment regarding the quality of service and overall customer experience. The feedback record will be linked to the corresponding appointment and order for proper reference.
- A **User** may submit **Refund Requests** for cancelled appointments or payment-related concerns. Each refund request will record the corresponding appointment, order, requested amount, and processing status to support organized refund management.
- A **User** will generate **Activity Logs** whenever important administrative actions are performed within the system. Each activity log will record the actor's identity, role, action performed, affected record, description, and timestamp to provide accountability and auditing.
- **Categories** will organize both **Services** and **Products** to simplify classification, searching, filtering, and management within the system. Each category will identify whether it belongs to a service classification or a product classification, allowing one category to contain multiple services or products.
- **Categories** will also be connected to **Therapist Specialties** to identify the service categories in which therapists are qualified to perform treatments.
- **Services** will be connected to **Appointments** because every appointment will require at least one selected spa service. Service information will also be linked to **Order Items** so that booked treatments can be recorded as part of customer transactions.
- **Services** will be connected to **Therapist Commission** records to determine therapist earnings based on either commission percentages or predefined flat rates assigned to particular services.

● **Services** will also be connected to **Therapist Specialty Services** to identify the individual services that each therapist will be authorized and qualified to perform.
● **Services** will be associated with **Partner Rates** to support customized pricing agreements established for hotel partners, corporate clients, and other affiliated organizations.
● **Products** will be connected to the **Cart** and **Order Items** so that product selections, purchased quantities, inventory movement, and sales transactions can be properly recorded and monitored.
● **Orders** will contain multiple **Order Items**, representing the individual services and products included within a customer transaction. Each order item will store the corresponding product or service, quantity, unit price, and subtotal for accurate billing.
● **Order Items** may generate corresponding **Appointments** whenever spa services are included within a customer order. This relationship will establish a connection between the payment transaction and the scheduled spa session.
● **Appointments** may include additional treatments through **Appointment Extra Services**. These records will store the additional service selected, assigned therapist, charged amount, commission earned, and payment status of every add-on service rendered during the appointment.
● **Appointments** will be assigned to one or more therapists through **Appointment Therapists**, allowing multiple therapists to participate in servicing group appointments. Each assignment will record the therapist involved, the number of persons handled, and the commission earned.
● **Appointments** will also store both the assigned therapist and the customer's preferred therapist whenever applicable, allowing management to accommodate customer preferences whenever therapist availability permits.
● **Appointments** will be connected to **Therapist Ratings** and **Feedback** to measure customer satisfaction, evaluate therapist performance, and support continuous service quality improvement after completed appointments.
● **Appointments** may also generate **Refund Requests** whenever cancellations, payment adjustments, or customer disputes occur. Each refund record will reference the related appointment, associated order, requested refund amount, and approval status.
● **Appointments** will be connected to **Partners** whenever bookings are made under corporate accounts, hotel partnerships, or special business arrangements so that negotiated pricing can automatically be applied.
● **Therapists** will perform the spa services associated with scheduled appointments. Therapist records will store professional information, listed specialties, and a generalist indicator identifying whether the therapist may perform all available services.
● **Therapists** will receive earnings through **Therapist Commission** records, which will define either commission percentages or fixed influencer rates corresponding to specific services.
● **Therapists** may also receive deductions recorded in **Therapist Deductions** for salary adjustments, cash advances, penalties, or other authorized deductions affecting commission computation.
● **Therapists** will maintain attendance records through **Therapist Attendance**, which will monitor duty dates, clock-in time, clock-out time, and therapist rotation order used during appointment assignment.
● **Therapists** will establish their service qualifications through **Therapist Specialties** and **Therapist Specialty Services**, allowing management to identify therapist expertise by both service category and specific spa treatment.
● **Therapists** will receive customer evaluations through **Therapist Ratings**, which will record the rating submitted for each completed appointment to assist management in monitoring therapist performance.
● **Partners** will be connected to both **Appointments** and **Partner Rates** to support hotel partnerships, corporate clients, and negotiated service pricing. Each partner record will identify the organization type and the corresponding pricing arrangements available within the system.
● **Daily Reports** will store the overall operational summary prepared at the end of each business day, including report date, cash on hand, POS reading, Maya digital payment collections, and report lock status to preserve finalized financial records.

● **Daily Reports** will contain multiple **Daily Report Denominations**, which will record the breakdown of cash denominations counted during cashier balancing and end-of-day reconciliation.
● **Daily Product Sales** will record the products sold during each business day, including the product description, quantity sold, and total sales amount, to support inventory monitoring and sales reporting.
● **Business Expenses** will store operational expenditures such as supplies, maintenance costs, utilities, marketing expenses, and other authorized business expenses to support financial monitoring and profitability analysis.
● **Gift Certificates** will store records of gift certificates issued, sold, and redeemed by the spa, including transaction dates, certificate types, and corresponding monetary amounts for promotional monitoring and financial reporting.
● **Unpaids Corp** will store outstanding balances of corporate clients whose transactions remain unpaid. Each record will contain the client name, report date, and outstanding amount to support receivable monitoring and collection activities.
● **Receptionist Pins** will store the unique PIN assigned to each receptionist for transaction verification and cashier authorization. These records will operate independently and will primarily be referenced during payment processing, cashier verification, and end-of-day financial reporting.
● **System Settings** will store the configurable parameters that control various operational behaviors of the Recovery Iloilo Spa Integrated System through individual setting keys and corresponding values. These records will operate independently and will be referenced whenever system-wide configuration values are required during system execution.

Use Case Diagram

Actors:

● **Customer (User)** – The **Customer** will be the primary user of the system and will represent any client who avails of the services offered by Recovery Iloilo Spa. Customers will be able to register an account with email OTP verification, log in securely, browse available spa services and wellness products, manage their shopping cart, place orders, and schedule appointments through the online booking facility. They will be able to select a preferred therapist or choose any available therapist, request home-service appointments, hotel or partner bookings, and add extra services whenever applicable. Customers will also be able to monitor the status of their appointments and payments, receive email and in-system notifications, view transaction history and digital receipts, submit feedback, rate the therapist who rendered the service, redeem vouchers and discounts, and submit refund requests whenever necessary.
● **Receptionist / Cashier** – The **Receptionist / Cashier** will be responsible for managing daily front-desk operations and customer transactions within the spa. This role will review, approve, decline, reschedule, and cancel appointment requests, assign or reassign therapists, process walk-in customers, record additional services rendered during appointments, and verify therapist availability before confirming bookings. Receptionists will process onsite payments, verify cashier PINs before sensitive financial transactions, handle refund requests, validate vouchers and discounts, and generate receipts. At the end of each business day, they will prepare the daily report, record cash denominations, POS readings, Maya collections, product sales, business expenses, gift certificate transactions, and unpaid corporate accounts for financial reconciliation.
● **Owner / Administrator** –The **Owner / Administrator** will oversee the overall operation, configuration, and maintenance of the Recovery Iloilo Spa Integrated System. This role will manage user accounts and role assignments, therapist records, therapist specialties, commission rates, attendance records, service categories, products, inventory, partner establishments, partner pricing, vouchers, gift certificates, and system settings. The administrator will also monitor appointments, transactions, refund requests, customer feedback, therapist ratings, operational reports, sales analytics, revenue forecasts, and activity logs. In

addition, the administrator will configure business rules, review financial reports, monitor employee productivity, and supervise the overall performance and security of the system.

- **Therapist** – The **Therapist** will be responsible for providing spa treatments based on appointments assigned through the system. Therapists will be able to record their attendance by clocking in and out, monitor their assigned schedules, review customer booking details, and identify additional services requested before or during treatment sessions. They will also monitor completed appointments, automatically computed commissions, deductions, assigned specialties, customer ratings, and feedback related to the services they have rendered. Attendance records and therapist rotation information will also be maintained to support fair workload distribution and efficient scheduling.
- **PayMongo Payment Gateway** – The **PayMongo Payment Gateway** will serve as an external payment processing service responsible for handling online payment transactions initiated by customers. Whenever a customer chooses an online payment method such as GCash, QR Ph, or bank card, the system will securely forward the payment request to PayMongo for processing. After payment verification, PayMongo will return the transaction result to the system, allowing payment records and order status to be updated automatically. This external actor will not directly access the system and will interact only through secure payment requests and responses.
- **Email / SMTP Service** – The **Email / SMTP Service** will function as an external communication service responsible for sending electronic messages generated by the Recovery Iloilo Spa Integrated System. It will deliver one-time passwords (OTP) for customer account verification, appointment confirmations, booking updates, payment acknowledgements, refund notifications, password reset instructions, and other system-generated announcements. This external actor will not access the system directly and will function solely as the communication channel for outgoing email notifications.

Use Cases:

Customer Use Cases

- **Register and Log In Account** – The system will allow customers to create an account and securely access the Recovery Iloilo Spa Integrated System using their registered credentials. During registration, customers will provide the required personal information, and a one-time password (OTP) will be sent to their registered email address for verification before the account can be activated. After successful login, customers will be able to access the services, products, bookings, payments, notifications, and other features available to their account.
- **Manage User Profile Information** –The system will allow customers to view and update their personal profile information whenever necessary. Customers will be able to modify details such as their full name, contact number, email address, and password while maintaining accurate customer records. Keeping profile information updated will help improve communication, booking management, and transaction processing.
- **Browse Spa Services and Products** – The system will allow customers to browse all spa services, wellness products, packages, and promotional offerings available at Recovery Iloilo Spa. Customers will be able to view service descriptions, session duration, pricing, product availability, and other relevant details before making a booking or purchase. This feature will help customers select the services and products that best suit their preferences.
- **View Categories and Pricing** – The system will allow customers to browse services and products according to their assigned categories. Customers will be able to compare available services, products, prices, promotional packages, and home-service availability before confirming a booking or purchase. This feature will help customers make informed decisions based on their preferred treatments.
- **Add Products to Cart** – The system will allow customers to add wellness products into their shopping cart before completing a purchase. Customers will be able to specify product quantities, review selected items, and update their selections before proceeding to checkout. The shopping cart will temporarily store all selected products until the transaction is confirmed.

● **Manage Cart and Checkout** – The system will allow customers to review the contents of their shopping cart before placing an order. Customers will be able to modify quantities, remove unwanted products, and verify the total amount due before proceeding with checkout. Once confirmed, the system will generate the corresponding order and transaction records. .

● **Place Orders for Products and Services** – The system will allow customers to purchase wellness products and confirm service transactions through the integrated platform. Each order will contain complete information regarding purchased services, products, quantities, unit prices, discounts applied, payment details, and computed totals to ensure accurate billing and transaction monitoring.

● **Book Spa Appointments** – The system will allow customers to schedule spa appointments by selecting their preferred services, appointment date, available time slot, number of persons, and preferred service location. Customers may create online bookings, while walk-in appointments will be encoded by authorized reception staff. Submitted appointments will remain pending until reviewed and processed by authorized personnel.

● **Select Home Service or In-Store Service** – The system will allow customers to choose whether the appointment will be performed as an in-spa service, home-service booking, hotel booking, or partner booking whenever applicable. The system will automatically compute any additional travel fees or partner pricing based on the selected appointment type.

● **Select Preferred Therapist** – The system will allow customers to select their preferred therapist or choose the "Any Available Therapist" option during appointment booking. The system will record the customer's preference and will assign the requested therapist whenever available. If the selected therapist is unavailable, management will be able to assign another qualified therapist.

● **Add Extra Appointment Services** – The system will allow customers to request additional spa services before or after an appointment has been approved. Each additional service will be recorded separately together with the assigned therapist, corresponding charges, commission, and payment status. This feature will allow customers to customize their spa experience without creating another appointment.

● **Pay Through the Online Payment Facility** – The system will allow customers to settle payments using available payment options, including cash, digital wallets, payment gateway transactions, payment upon completion, or approved pay-later arrangements. The system will automatically calculate transaction totals, generate digital receipts, record payment history, and update payment status after successful processing.

● **Track Order and Appointment Status** – The system will allow customers to monitor the progress of their appointments throughout the booking lifecycle. Customers will be able to view whether an appointment is pending, assigned, approved, completed, declined, or cancelled. This feature will provide customers with real-time updates regarding their reservations.

● **View Payment and Refund Status** – The system will allow customers to monitor payment confirmations and refund requests associated with their transactions. Customers will be able to review the processing status of submitted refund requests and determine whether the request remains pending, approved, or declined.

● **Receive System Notifications** – The system will allow customers to receive notifications regarding account verification, appointment confirmations, therapist assignments, payment confirmations, refund updates, reminders, promotional announcements, and other important system activities. Notifications will also be sent through the integrated email service whenever necessary.

● **View Transaction History** – The system will allow customers to review their complete transaction history, including previous appointments, purchased products, completed services, payments, refunds, and booking activities. This feature will provide customers with convenient access to their historical records whenever needed.

● **Submit Service Feedback and Ratings** – The system will allow customers to submit comments, suggestions, and overall evaluations regarding the spa services they received after completing an appointment. These feedback records will assist management in monitoring customer satisfaction and identifying opportunities for service improvement.

- **Rate Assigned Therapist** – The system will allow customers to evaluate the therapist who performed their completed spa session by submitting ratings and comments. Therapist evaluations will help management monitor employee performance, recognize outstanding service, and improve service quality.
- **Request Refund or Appointment Cancellation** – Allows customers to file a request for the cancellation of a booking or the reimbursement of a transaction. The system records the amount involved and holds the request in a pending state. Management may then review the request and take the appropriate action.

For Receptionist / Cashier:

- **Log In and Verify Security PIN** – The Receptionist or Cashier will log in using the assigned account credentials before accessing the operational functions of the system. A cashier PIN will be required before performing protected financial transactions, approving sensitive operations, or finalizing the daily report. This process will help strengthen accountability and ensure that only authorized personnel can perform secured activities.
- **Manage Walk-In Appointments** – The Receptionist or Cashier will record appointments for customers who visit the spa without making an online reservation. The system will create the corresponding appointment record, assign the selected services, and prepare the transaction for payment processing. This process will ensure that walk-in customers will follow the same workflow as online bookings.
- **Review and Process Appointment Requests** – The Receptionist or Cashier will review appointment requests submitted by customers before confirming the reservation. The system will allow staff to approve, decline, reschedule, or cancel appointments while recording the personnel responsible for each action. This process will help maintain organized appointment scheduling and proper operational control.
- **Assign and Reassign Therapists** – The Receptionist or Cashier will assign available therapists to approved appointments based on therapist availability, specialty, rotation order, and customer preference whenever possible. The system will also allow therapist reassignment when scheduling conflicts or operational changes occur. This process will ensure proper workload distribution and efficient service coordination.
- **Manage Appointment Details and Add-on Services** – The Receptionist or Cashier will update appointment information whenever necessary, including modifying schedules, adding extra services, updating customer notes, and recording service completion details. Additional services rendered during or after the appointment will be recorded together with the assigned therapist, corresponding charges, commission, and payment status. This process will ensure that all services provided are properly documented.
- **Process Point-of-Sale (POS) Transactions** – The Receptionist or Cashier will process customer payments through the Point-of-Sale module using supported payment methods such as cash, card, digital wallet, payment at completion, or pay-later arrangements. The system will automatically calculate transaction totals, apply eligible discounts, generate receipts, and update payment records. This process will ensure accurate financial recording and faster transaction processing.
- **Process Refund Requests** – The Receptionist or Cashier will review customer refund requests and perform the necessary verification before forwarding them for approval or processing according to company policies. The system will update the refund status and maintain complete refund transaction records for monitoring purposes.
- **Manage Customer Transactions** – The Receptionist or Cashier will monitor customer orders, appointment payments, payment status, and transaction history. The system will maintain complete transaction records to support customer inquiries, financial reconciliation, and operational monitoring.
- **Monitor Daily Sales Transactions** – The Receptionist or Cashier will review all completed transactions recorded throughout the business day. The system will automatically consolidate payment information,

service sales, product sales, discounts, refunds, and transaction summaries to support daily financial monitoring.

- **Prepare Daily Reports** – The Receptionist or Cashier will prepare the end-of-day operational report by reviewing recorded sales, POS readings, payments, gift certificates, business expenses, unpaid corporate accounts, and other financial records. The system will generate the required report for management review and documentation.
- **Record Cash Denominations and Cash Reconciliation** – The Receptionist or Cashier will record the quantity of each cash denomination at the close of business. The system will automatically calculate the total cash on hand, compare it with recorded POS collections, and generate cash-over or cash-short reconciliation results to improve financial accuracy.
- **Record Daily Product Sales** – The Receptionist or Cashier will review and confirm product sales recorded during the business day. The system will maintain daily product sales records to support inventory monitoring, sales analysis, and operational reporting.
- **Record Business Expenses** – The Receptionist or Cashier will encode business expenses incurred during daily operations, including the expense category, amount, and transaction date. The system will include these records in the daily financial reports and analytics.
- **Manage Gift Certificate Transactions** – The Receptionist or Cashier will record gift certificates that are sold or redeemed during the business day. The system will maintain accurate gift certificate balances and transaction history for monitoring promotional activities.
- **Manage Corporate Unpaid Transactions** – The Receptionist or Cashier will record unpaid transactions for corporate clients and partner establishments. The system will maintain outstanding balances and support monitoring of accounts receivable for future collection.
- **Submit and Lock Daily Reports** – The Receptionist or Cashier will finalize the daily operational report after verifying all recorded transactions, payments, expenses, and cash balances. Once submitted, the system will lock the report to prevent unauthorized modifications while preserving the integrity of financial records for management review.

For Owner / Administrator:

- **Log In Account** – The Owner or Administrator will securely access the system using assigned account credentials. The login process will restrict administrative functions to authorized personnel based on assigned roles and permissions. After successful authentication, authorized users will be able to access system management, monitoring, reporting, and configuration functions.
- **Manage User Accounts and Roles** – The Owner or Administrator will create, update, activate, deactivate, and manage customer and employee accounts within the system. Administrative roles such as owner, administrator, receptionist, therapist, marketing personnel, and information technology personnel will also be assigned according to operational responsibilities. This process will ensure that users will only have access to the system features appropriate to their assigned roles.
- **Manage Therapist Records and Specialties** – The Owner or Administrator will maintain therapist profiles, specialties, service qualifications, and generalist status. Therapist specialty categories and individual service qualifications will also be managed to support proper therapist assignment during appointment scheduling. This process will help ensure that therapists will only be assigned to services they are qualified to perform.
- **Manage Services, Products, Categories, and Pricing** – The Owner or Administrator will maintain the complete service and product catalog by creating, updating, or deactivating categories, services, products, prices, session durations, home-service fees, and inventory information. This process will ensure that customers and staff will always access accurate and updated information during bookings and transactions.
- **Manage Inventory and Product Availability** – The Owner or Administrator will monitor inventory levels, product availability, stock movements, and replenishment activities. The system will assist in

identifying low-stock items and maintaining accurate inventory records to support uninterrupted spa operations.

- **Configure Therapist Commission and Deduction Rules** – The Owner or Administrator will define therapist commission percentages, influencer flat-rate commissions, and other compensation rules based on completed services. Administrative users will also record therapist deductions such as cash advances, penalties, or other adjustments to ensure accurate payroll computation.
- **Manage Partners and Partner Service Rates** – The Owner or Administrator will maintain records of hotels, corporate clients, and other partner establishments. Negotiated service rates for each partner will also be configured to ensure that partner bookings will automatically apply the appropriate pricing during appointment processing.
- **Manage Receptionist Security PINs** – The Owner or Administrator will create, update, and maintain the security PIN assigned to each receptionist or cashier. These PINs will be required before protected financial transactions, cashier verification, and daily report submission to strengthen accountability and transaction security.
- **Review and Manage Customer Transactions** – The Owner or Administrator will review customer orders, appointment transactions, payment records, discounts, vouchers, and transaction approvals whenever administrative intervention is required. This process will help maintain accurate financial records and ensure that transactions comply with company policies.
- **Review and Process Refund Requests** – The Owner or Administrator will examine refund requests submitted by customers by reviewing the corresponding appointment, order, payment information, and refund amount. The request status will then be updated based on management approval to support organized refund processing and customer service.
- **Monitor Appointments and Therapist Operations** – The Owner or Administrator will monitor appointment schedules, therapist assignments, attendance records, completed services, commissions, and operational workload. This process will help ensure efficient workforce management and proper service coordination throughout daily operations.
- **Generate Operational Reports and Analytics** – The Owner or Administrator will generate reports covering appointments, sales transactions, inventory usage, therapist productivity, commissions, customer activities, payment summaries, refunds, and overall business performance. The system will also generate short-term revenue forecasts and analytical summaries to support business monitoring and informed decision-making.
- **Generate Daily Financial Reports** – The Owner or Administrator will review daily operational reports that include Point-of-Sale readings, cash-on-hand balances, cash denomination summaries, business expenses, gift certificate transactions, unpaid corporate accounts, and cash reconciliation results. These reports will support financial monitoring and end-of-day verification.
- **View Customer Feedback and Therapist Ratings** – The Owner or Administrator will review customer comments, ratings, and service evaluations submitted after completed appointments. The system will organize these records by therapist and transaction to assist management in evaluating customer satisfaction, therapist performance, and opportunities for service improvement.
- **Review Activity Logs and Employee Accountability** – The Owner or Administrator will examine activity logs that record significant actions performed throughout the system. Each record will identify the responsible user, assigned role, action performed, affected record, description, and timestamp to support auditing, accountability, and operational monitoring.
- **Manage System Settings** – The Owner or Administrator will configure and maintain system-wide settings that control the behavior of the application, including operational parameters, configurable options, and other business rules. These settings will allow management to adjust system behavior without modifying the underlying program code.

For Therapist:

- **Log In Account** – The Therapist will securely access the system using the assigned account credentials. The login process will restrict therapist functions to authorized users and protect therapist information from unauthorized access. After successful authentication, therapists will be able to access their schedules, attendance records, assigned services, commissions, ratings, and notifications.
- **Record Attendance and Duty Schedule** – The Therapist will record daily attendance by logging the time in and time out for each assigned duty. The system will maintain attendance records together with the therapist's rotation order to support fair workload distribution and proper therapist assignment. This process will assist management in monitoring attendance and workforce availability.
- **View Assigned Appointments** – The Therapist will view appointments assigned by the Receptionist or Administrator through the appointment management module. The system will display appointment schedules, assigned services, customer information, appointment location, number of persons, and current booking status. This process will help therapists prepare for scheduled appointments before rendering the service.
- **View Appointment Details** – The Therapist will review the complete details of each assigned appointment before the scheduled session begins. The system will display the selected services, additional services, customer notes, preferred service location, appointment schedule, applicable pricing, and other booking information. This process will help therapists understand customer requirements and prepare the necessary treatments.
- **View Assigned Specialties and Qualified Services** – The Therapist will review the service categories and individual treatments assigned based on qualifications and specialties maintained by management. The system will identify whether the therapist is assigned as a specialist or generalist for particular services. This process will ensure that therapists will only perform services appropriate to their qualifications.
- **View Commission and Earnings** – The Therapist will monitor commissions earned from completed appointments, additional services, influencer sessions, and other eligible transactions. The system will automatically calculate commission records based on completed services and approved transactions to provide accurate earnings information.
- **View Deduction Records** – The Therapist will review deductions applied to commission earnings, including cash advances, operational expenses, penalties, or other authorized adjustments. The system will maintain complete deduction records to provide transparency and accurate payroll computation.
- **View Customer Ratings and Feedback** – The Therapist will review customer ratings, comments, and service evaluations submitted after completed appointments. The system will organize feedback records according to completed services to help therapists evaluate their performance and identify opportunities for professional improvement.
- **Receive System Notifications** – The Therapist will receive notifications generated by the system regarding newly assigned appointments, schedule updates, therapist reassignments, appointment cancellations, commission updates, attendance reminders, and other work-related announcements. This process will help therapists remain informed about daily responsibilities and operational activities.

For PayMongo Payment Gateway:

- **Process Online Payment Request** – The PayMongo Payment Gateway will receive online payment requests forwarded by the Recovery Iloilo Spa Integrated System whenever a customer selects an electronic payment method during checkout. The gateway will securely process the payment transaction without exposing sensitive payment information to the system. This process will allow customers to complete online payments through a secure and reliable payment service.

- **Return Payment Confirmation** – The PayMongo Payment Gateway will return the payment result to the Recovery Iloilo Spa Integrated System after the payment transaction has been completed or declined. The system will use the returned confirmation to update the payment status, record the transaction, generate the corresponding receipt, and continue the appropriate booking or order workflow. This process will ensure that only verified payment transactions will be recorded within the system.

For Email / SMTP Service:

- **Deliver One-Time Password (OTP)** – The Email / SMTP Service will receive the one-time password generated by the Recovery Iloilo Spa Integrated System during customer registration or account verification. The generated code will be transmitted to the registered email address of the customer to verify account ownership before access is granted. This process will strengthen account security and help prevent unauthorized registration and access.
- **Deliver Booking and Transaction Notifications** – The Email / SMTP Service will receive notification messages generated by the Recovery Iloilo Spa Integrated System and deliver them to the intended recipients. Messages may include appointment confirmations, appointment approvals, booking updates, payment acknowledgements, refund status updates, password verification messages, and other important transaction notifications. This process will help ensure that customers remain informed about significant activities related to their accounts and transactions.

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CHAPTER IV

SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULTS, AND DISCUSSION

The **Recovery Iloilo Spa Integrated System for Streamlining Operations and Efficient Analytics** is a web-based platform developed by the proponents to automate, centralize, and organize the daily operational activities of Recovery Iloilo Spa. The system integrates major business processes that were previously handled through manual records and separate procedures into a single centralized environment. These processes include appointment scheduling and management, customer registration and account management, service and product information management, therapist management, attendance monitoring, inventory monitoring, Point-of-Sale (POS) transactions, payment and refund processing, therapist commission computation, discount and voucher management, customer feedback and ratings, automated notifications, security and accountability, operational reporting, and business analytics. By bringing these functions together, the system provides users with a more organized method of recording, retrieving, updating, and monitoring operational information. It also reduces repetitive administrative tasks and provides management with consolidated information that can be used for monitoring business performance and supporting operational decision-making.

The developed system goes beyond a conventional standalone booking or Point-of-Sale application because it integrates multiple operational functions within one centralized platform. The system was implemented using a three-tier architecture consisting of the presentation layer, application layer, and data layer, allowing the user interface, business processes, and database operations to be managed as interconnected but organized components. Operational information is captured as customers and staff perform activities within the system and is processed according to the corresponding business rules. Appointment information, customer records, transactions, inventory movements, therapist activities, commissions, discounts, vouchers, and feedback are stored in the centralized database and can subsequently be used for monitoring and reporting. The system also transforms accumulated operational data into organized reports, dashboards, analytical information, and formula-based revenue forecasts, allowing management to review business activities using recorded data rather than relying solely on manually maintained records.

The system was designed and developed based on the operational requirements of Recovery Iloilo Spa, located at G&R Building, M.H. Del Pilar Street, Molo, Iloilo City. The establishment provides various wellness and personal-care services, including body massage treatments, facial services, body scrubs, nail care services, waxing treatments, and other therapeutic procedures. During the development process, the proponents gathered information regarding the existing operational procedures and identified the requirements that needed to be addressed by the system. The spa owner, management, HR personnel, receptionists, therapists, and customers provided information regarding their respective activities, requirements, and experiences with the existing processes. These requirements served as the basis for designing the system modules, workflows, user interfaces, database structure, security controls, and reporting functions. The completed system was subsequently tested and evaluated by the proponents and respondents to determine whether the implemented functions addressed the identified operational requirements.

The developed system consists of several interconnected modules that support different users and operational activities. Customers can register and manage their accounts, verify their email addresses through OTP verification, browse available services and products, manage their shopping cart, book appointments, view appointment and

transaction histories, monitor payment and refund information, and provide feedback and ratings. Staff members can manage appointments, customer records, therapist assignments, inventory, transactions, discounts and vouchers, and other operational activities according to their assigned access levels. Management and authorized personnel can access reports, analytics, commission information, cash reconciliation records, and other administrative functions. Through these interconnected modules, information generated from one operational activity can be utilized by other related functions, reducing the need to repeatedly record or search for the same information across separate records.

The major functions and capabilities achieved by the developed system are as follows:

● **Automated Minute-Level Appointment Scheduling** — The appointment management module supports minute-level scheduling and server-side conflict detection to help prevent duplicate reservations, overbooking, and scheduling conflicts. It accommodates different appointment sources, including online bookings, walk-in appointments, home-service requests, and hotel or partner bookings. The module records important appointment information such as customer details, selected services, therapist assignments, appointment date and time, booking status, and additional services. Staff members can create, update, reschedule, cancel, and monitor appointments while considering therapist availability and applicable scheduling requirements.

● **Centralized Customer Accounts and History** — The customer management module maintains centralized customer profiles containing relevant account information, verified contact details, appointment history, transaction records, service preferences, feedback, and voucher or discount usage. Customers can manage their profiles and access relevant records through their accounts, while authorized staff can retrieve customer information when needed for appointment processing and service coordination. The centralized structure reduces the need to search through separate manual records and supports more organized customer information management.

● **Centralized Service and Product Information** — The system organizes information about the spa's available services and products in a centralized environment. Service records may contain information such as service names, descriptions, categories, prices, durations, and availability, while product records contain relevant product information and pricing. This allows customers to view available services and products when making inquiries or bookings and allows authorized staff to maintain updated information. Centralizing these records also helps reduce inconsistencies when service or product information is updated.

● **Dynamic Point-of-Sale and Multi-Channel Payment Processing** — The POS and payment module supports the recording and processing of customer transactions using available payment methods. The system calculates transaction totals, applies applicable discounts or vouchers, records payments, generates receipts, maintains transaction histories, and supports refund requests. Online payments are processed through the integrated PayMongo payment gateway, supporting the configured digital payment options such as GCash, QR Ph, and cards. These functions provide a more organized transaction workflow and reduce the need for repetitive manual computation and recording.

● **Automated Therapist Workload and Commission Management** — The therapist management module supports therapist assignment and reassignment, scheduling, attendance monitoring, specialty information, and performance-related records. The system also supports therapist rotation and workload coordination based on the implemented scheduling rules. The commission module calculates applicable therapist commissions based on completed services and recorded transactions, including applicable service categories and commission arrangements. This reduces the need for repeated manual commission calculations and provides management with organized commission records for monitoring and payroll-related preparation.

● **Inventory Replenishment and Stock Control** — The inventory module monitors spa supplies, consumable materials, and retail products used or sold by the establishment. It records stock information and inventory movements and provides low-stock monitoring to help staff identify products that may require replenishment. By connecting inventory information with relevant product and transaction activities, the system provides management with more organized information regarding product availability and stock movement. This supports more systematic inventory monitoring and helps reduce the possibility of relying on outdated manual stock records.

● **Discount, Voucher, and Promotional Management** — The system provides functions for managing applicable discounts, vouchers, gift certificates, and promotional offers. Authorized personnel can record and validate applicable discounts, monitor voucher redemption, and maintain records of promotional transactions. The system supports the identification of applicable customer incentives and helps prevent duplicate voucher usage according to the implemented business rules. This provides a more organized method of monitoring promotional activities and their corresponding transactions.

● **Customer Feedback and Rating Management** — The feedback module allows customers to submit comments, ratings, and suggestions regarding their experiences with the spa's services. Feedback can be associated with the therapist who performed the service, allowing management to review customer evaluations and identify areas that may require improvement. The collected feedback provides an additional source of information for monitoring customer satisfaction and evaluating service performance.

● **Automated Notifications and Customer Communication** — The system provides automated email notifications for important activities such as account verification, appointment confirmations, appointment updates, cancellations, payment confirmations, refund-related activities, password reset instructions, and other system-generated events. These notifications provide customers with timely information regarding their accounts, appointments, and transactions and reduce the need for staff to manually communicate every system-related update.

● **Role-Based Security and Accountability** — The system separates public customer functions from administrative functions and applies access controls based on user roles. Sensitive transactions require additional verification through the implemented gate code and PIN mechanisms, while system activities are recorded through activity logs containing relevant user actions and timestamps. These controls provide greater accountability by allowing authorized personnel to monitor important actions and help protect sensitive customer, transaction, and administrative information.

● **Financial Reconciliation and Business Analytics** — The system generates reports and analytical information related to sales performance, appointment activities, inventory usage, therapist productivity, commissions, payment records, cash reconciliation, and other operational activities. The system also supports cash-over and cash-short calculations and provides formula-based short-term revenue forecasting using recorded transaction data. Generated reports can be filtered, viewed, and exported into Excel and PDF formats for documentation, monitoring, and further analysis. These functions provide management with organized information that can support business evaluation and operational decision-making.

The development of the system incorporated software engineering principles, responsive interface design, centralized database management, user authentication, access control, transaction processing, and analytical processing. The interconnected modules allow information generated from customer activities, appointments, services, products, transactions, therapist activities, inventory movements, and feedback to be stored and processed within the centralized database. The resulting system provides Recovery Iloilo Spa with an organized platform for managing its operational activities and accessing relevant business information.

Overall, the developed system established an integrated workflow that connected customer-facing activities with internal spa operations. Instead of maintaining separate records for appointments, customers, transactions, inventory, therapists, commissions, promotions, and feedback, the system consolidated these functions within one platform. The completed implementation therefore provided the establishment with a more systematic approach to managing its daily operations while also supporting reporting, analytics, accountability, and data-informed decision-making.

Figure 4.1 illustrates the overall workflow of the developed system.

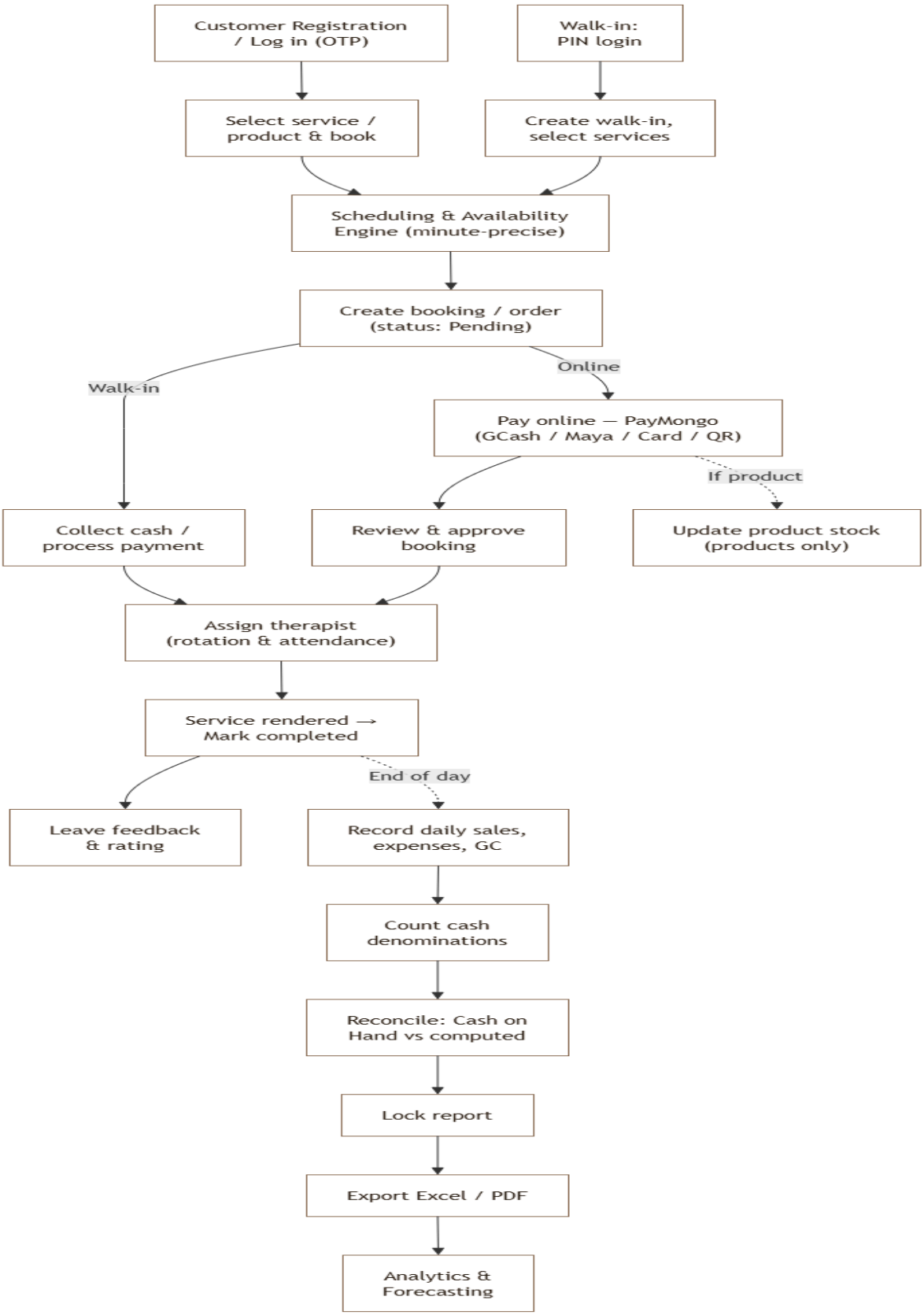


Figure 4.1. System Workflow Diagram of the Recovery Iloilo Spa Integrated System

Figure 4.1 presents the comprehensive end-to-end operational workflow of the Recovery Iloilo Spa Integrated System, encompassing both the customer-facing service lifecycle and the administrative back-office processes that were developed and refined throughout the study. The workflow commences at two distinct points of ingestion: an online customer who has completed verified registration and authentication through a one-time password (OTP), and an on-site walk-in client whose transaction is initiated by a receptionist via PIN-authenticated access. Both entry paths converge at the conflict-aware scheduling and availability engine, which evaluates therapist qualifications, duty rotation, and existing bookings at minute-level precision to determine feasible time slots before a booking or order record is generated with a pending status.

Following record creation, the workflow bifurcates according to the payment channel. Online transactions are routed to the multi-channel PayMongo payment gateway, which processed electronic settlements through GCash, Maya, card, and QR Ph facilities, after which the booking underwent administrative review and approval; product purchases additionally triggered a conditional stock-deduction routine. Walk-in transactions, in contrast, were settled directly through in-person cash or electronic collection at the counter. Both branches subsequently reconverged at the therapist-assignment stage, wherein staff allocated a service provider according to the attendance and rotation records, culminating in the rendering of the service and its marking as completed. Upon completion, the customer was afforded the opportunity to submit feedback and a corresponding therapist rating.

The latter portion of the workflow reflects the daily-report and reconciliation subsystem that constituted the principal development effort of this study. Each completed transaction was recorded within the daily-report spreadsheet, which supported both the automatic importation of completed appointments—safeguarded against duplication through source-transaction referencing—and the manual entry of off-system sales for troubleshooting and correction purposes. The subsystem automatically computed therapist commissions and net sales and disaggregated revenue by payment method. The receptionist thereafter counted the physical cash denominations and reconciled the computed cash-on-hand against the recorded figures. Upon completing the report, the receptionist submitted it through a validation procedure that verified the presence of essential fields, whereupon the system locked the report against further modification, dispatched an in-app and electronic-mail notification to the proprietor, and rendered the finalized report available for review and for exportation to Microsoft Excel and PDF formats in the client's prescribed layout. The consolidated data ultimately fed the analytics and forecasting module, which aggregated operational figures to support managerial decision-making.

4.3 System Development Methodology

The development of the Recovery Iloilo Spa Integrated System adopted the Agile Software Development Methodology, an iterative and incremental approach that emphasizes continuous collaboration, adaptive planning, and regular evaluation throughout the software development life cycle. Unlike traditional sequential models, Agile permits requirements and solutions to evolve through the cooperative effort of the proponents and the client. The proponents selected this methodology because it accommodated the evolving requirements of Recovery Iloilo Spa and promoted frequent feedback from the respondents, thereby ensuring that the resulting application remained aligned with the actual operational needs of the establishment rather than with assumptions made at the beginning of the project.

The Agile methodology enabled the proponents to divide the project into manageable iterations, commonly referred to as sprints. Each sprint was dedicated to a defined portion of the system, and at its conclusion a functional component was produced, tested, and reviewed before the proponents proceeded to the next iteration. This incremental delivery minimized development risks, facilitated the early detection and correction of software defects, and allowed the client to observe tangible progress at regular intervals. Because each iteration built upon the outcomes of the previous one, the proponents were able to refine both the design and the functionality of the system continuously as their understanding of the spa's business processes deepened. The development process consisted of the following phases, which were revisited iteratively across successive sprints rather than performed in a single linear pass:

The development process consisted of the following phases:

Requirements Gathering

The first phase involved the systematic collection of information from the client through interviews, surveys, direct observation, and document review. The proponents consulted the spa owner, human-resource personnel, receptionists, and therapists in order to determine the recurring challenges encountered in managing appointments, customer records, inventory, payments, commissions, discounts, and daily reports. Particular attention was given to the establishment's existing practices, such as the manual preparation of the daily sales report in a spreadsheet, the manual computation of therapist commissions, and the reconciliation of cash at the end of each business day. The respondents supplied detailed information regarding these processes and articulated their operational needs, which the proponents subsequently analyzed to define the functional requirements such as online booking, payment processing, and automated reporting and the non-functional requirements such as security, reliability, and ease of use. The insights obtained in this phase formed the foundation upon which all succeeding phases were built and were periodically revisited whenever new or revised requirements emerged.

System Planning

After the requirements had been identified, the proponents established the project scope, objectives, system specifications, development schedule, and allocation of resources. The planning stage incorporated a feasibility analysis that assessed the technical, operational, and economic viability of the proposed system, confirming that the project could be accomplished with the available tools, skills, and time frame. During this stage, the proponents decomposed the system into the principal modules to be developed—namely, appointment scheduling, therapist management, inventory monitoring, point-of-sale and payment processing, discounts and voucher validation, feedback collection, security and access control, and reporting and analytics. These modules were then organized into a prioritized backlog, from which the tasks of each sprint were drawn. This planning ensured that the most critical functionalities were addressed early in the development timeline while allowing lower-priority features to be scheduled for later iterations.

System Design

During this phase, the proponents designed the overall architecture of the application together with its constituent components. Database structures, process flows, user-interface layouts, and module interactions were prepared to serve as the blueprint for software construction. The proponents produced the Data Flow Diagram, the Entity Relationship Diagram, and the interface wireframes to model the movement of data and the relationships among the system's entities. The application was designed around a three-tier architecture consisting of the presentation, application, and data layers, an arrangement that separates the user interface from the business logic and the underlying database in order to promote maintainability, scalability, and security. The design phase also addressed the encapsulation of complex logic—most notably the conflict-aware scheduling engine, which determines therapist availability at minute-level precision and the incorporation of security measures such as role-based access control, prepared database statements, and cross-site request forgery protection. Because the design evolved iteratively, these artifacts were refined in successive sprints as the proponents' comprehension of the client's reconciliation and commission procedures matured.

Software Development

The development phase involved translating the approved designs into functional program code using the selected technologies. The proponents implemented the presentation layer using HTML, CSS, and Vanilla JavaScript; the application layer using the PHP programming language; and the data layer using the MariaDB relational database management system. Individual functionalities were constructed incrementally, including customer registration with electronic-mail one-time-password (OTP) verification, secure authentication, appointment scheduling with automated conflict detection, therapist assignment and attendance recording, inventory monitoring, point-of-sale and payment processing through the multi-channel PayMongo payment gateway, refund handling, commission computation, discount and voucher validation, feedback collection, activity logging, and formula-based revenue forecasting. A substantial portion of the development effort was devoted to the daily-report and reconciliation subsystem, which enabled the automatic importation of completed transactions, the manual entry and correction of off-system sales, the automatic computation of commissions and net sales, the disaggregation of revenue by payment method, cash reconciliation, and the exportation of the finalized report to Microsoft Excel and PDF formats in the client's prescribed layout. Throughout this phase, the proponents employed version control through Git and GitHub to manage the source code, track incremental changes, and safeguard the project against data loss, thereby allowing completed features to be integrated into the larger system without disrupting previously functioning components.

Testing

Each completed module underwent functional testing to verify that it operated in accordance with its specified requirements before it was integrated into the wider system. The proponents subsequently performed integration testing to confirm that the interactions among modules functioned correctly, and system testing to evaluate the

overall behavior and performance of the application. Testing was conducted iteratively: defects identified during a sprint were documented, corrected, and re-verified before the associated feature was considered complete, in keeping with the Agile principle of early and continuous defect resolution. Upon the stabilization of the system, User Acceptance Testing (UAT) was carried out with the client, wherein the spa owner, receptionists, therapists, accounting personnel, and selected customers participated as respondents. These respondents exercised the live functionalities such as booking an appointment, processing a payment, recording and submitting the daily report, and exporting it in the required format and evaluated whether the application satisfied their operational expectations. The feedback derived from this testing informed the final refinements to the system.

Deployment

Following the successful completion of testing, the proponents deployed the application within the target environment of Recovery Iloilo Spa. The deployment activities encompassed database configuration, server setup, and application installation, which together rendered the system accessible to authorized users in accordance with their assigned roles. During this phase, the respondents accessed the live system under realistic operating conditions and provided additional observations, which the proponents used to fine-tune the final configuration and to resolve any environment-specific issues that had not surfaced during earlier testing.

Maintenance

The maintenance phase concerns the continued monitoring of system performance, the correction of newly identified issues, the implementation of software updates, and the enhancement of features in response to feedback obtained from the client after deployment. Through routine maintenance, the proponents ensure the long-term reliability, security, and usability of the developed application, and provide a mechanism through which the system may be adapted to future operational requirements of the establishment.

Through the repeated execution of these phases across successive sprints, the proponents were able to deliver the Recovery Iloilo Spa Integrated System incrementally, to validate each component against the expectations of the client, and to progressively enhance the application until it satisfied the intended operational and functional requirements. The iterative and collaborative character of the Agile methodology thus proved instrumental in producing a system that is both technically sound and closely aligned with the real-world needs of Recovery Iloilo Spa.

Figure 4.2 presents the Agile Software Development Life Cycle adopted by the proponents in the development of the system

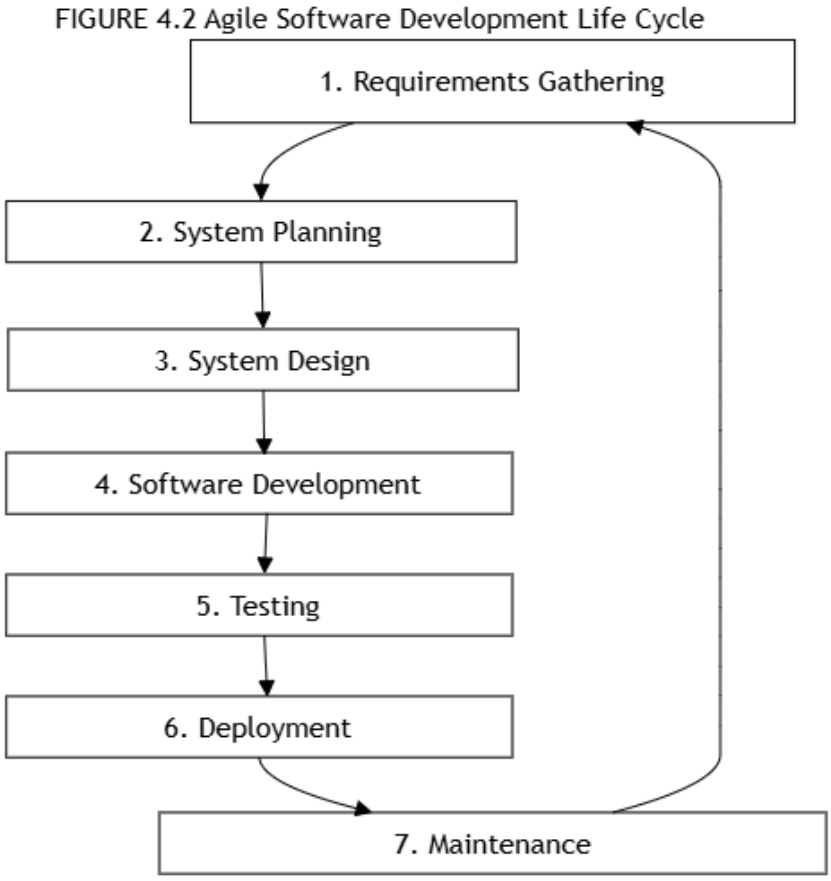


Figure 4.2. Agile Software Development Life Cycle Adopted in the Study

4.4 System Architecture

The Recovery Iloilo Spa Integrated System utilizes a Three-Tier Client-Server Architecture that separates the presentation layer, the application layer, and the data layer. This architectural approach improves system maintainability, scalability, and security by organizing the software components according to their respective responsibilities, so that a change introduced within one tier exerts minimal impact upon the others. The system further adopts a hub-and-spoke communication model in which the application layer serves as the central processing hub between the presentation layer and the data layer. Every request initiated by a user is first authenticated, validated, and processed by the application layer before information is stored in or retrieved from the database, thereby ensuring that no client is permitted to access the underlying data directly.

Figure 4.3 illustrates the architecture of the developed system.

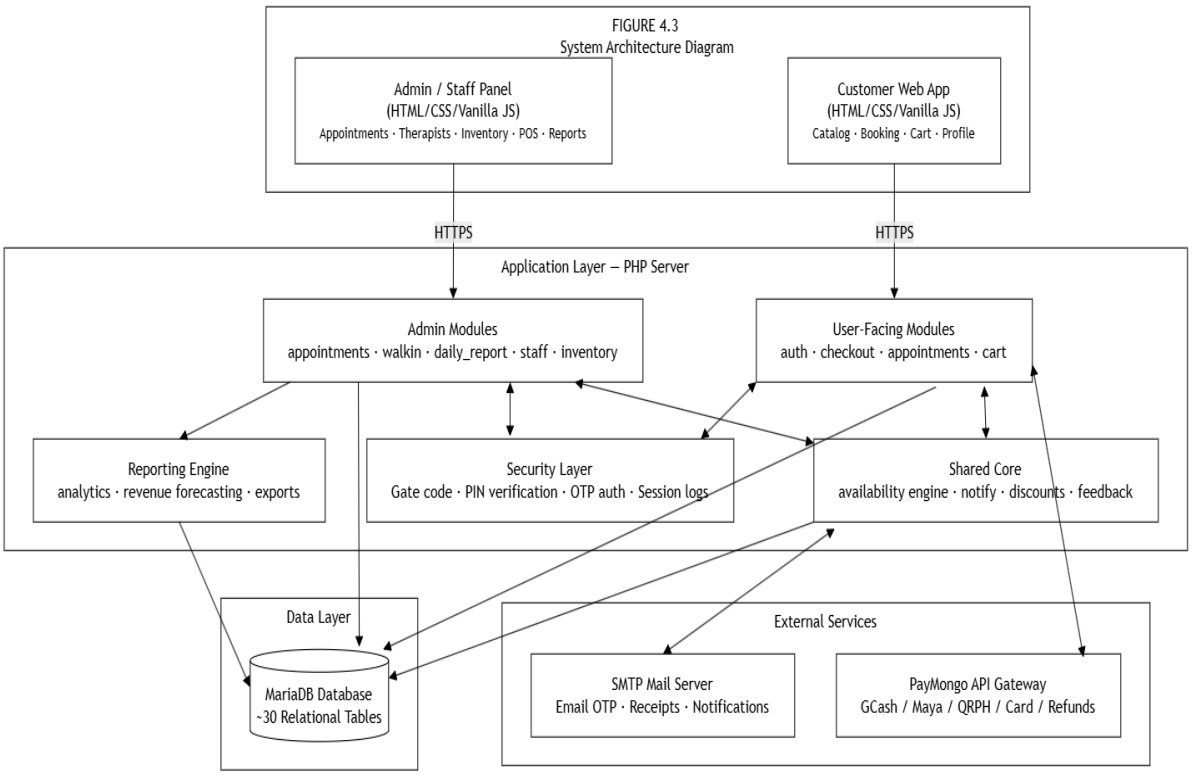


Figure 4.3. Three-Tier System Architecture of the Recovery Iloilo Spa Integrated System

The architecture consists of the following components:

Presentation Layer

The presentation layer serves as the interface between the users and the application. It is responsible for displaying information, receiving user input, and presenting reports and dashboards to the intended users. Users access the system through desktop computers, laptops, tablets, or smartphones by means of a modern web browser, thereby eliminating the need for any additional software installation on the client device. The proponents developed this layer using HTML, CSS, and Vanilla JavaScript in order to provide a responsive and interactive user experience, and all communication between this layer and the application layer is transmitted over the Hypertext Transfer Protocol Secure (HTTPS) to protect the data exchanged.

The major interface components include the following:

- Customer registration and login pages
- Service and product catalog
- Appointment booking and shopping cart
- Customer dashboard and profile management
- Administrative dashboards for appointments, therapists, inventory, and point-of-sale operations
- Reports, analytics, and revenue-forecasting interfaces

Application Layer

The application layer contains the business logic responsible for processing user requests, validating input, executing business rules, and generating reports. The proponents developed this layer using the PHP programming language, which implements the business logic and facilitates secure communication between the user interface and the MariaDB database. The application server acts as the intermediary between the presentation layer and the data layer, and it constitutes the sole pathway through which all data operations are performed.

Its primary responsibilities include the following:

- User authentication, electronic-mail one-time-password (OTP) verification, and secure session management
- Appointment scheduling with conflict detection and status management
- Therapist assignment, reassignment, rotation, attendance recording, and commission computation
- Inventory monitoring and stock updates
- Point-of-sale and payment processing through the PayMongo payment gateway, together with refund handling
- Discount, voucher, and gift-certificate validation
- Feedback processing linked to completed appointments and assigned therapists
- Security enforcement through gate-code and personal-identification-number (PIN) verification and activity logging
- Report generation, business analytics, and formula-based revenue forecasting

Data Layer

The data layer stores all business information within a centralized MariaDB relational database. It is responsible for maintaining data integrity, enforcing the relationships among entities, and supporting the efficient retrieval of the information required by the reporting and analytics modules. The database comprises approximately thirty core tables, and access to it is granted exclusively to the application layer through secure, parameterized statements that safeguard the system against structured query language injection.

The information stored within this layer includes the following:

- User accounts, roles, and credentials
- Services, products, and categories
- Appointments and appointment assignments
- Therapist records, attendance, and commissions
- Orders, order items, payments, and refunds
- Discounts, vouchers, and gift certificates
- Customer feedback and ratings
- Activity logs, notifications, and daily reports

Through this layered and hub-centered organization, the architecture ensures secure communication between the application server and the database while restricting unauthorized access to sensitive business and financial information. It consequently promotes maintainability by localizing changes within individual tiers, supports scalability by permitting each tier to be extended independently, and enhances security by channeling every operation through a single, validated point of control.

4.5 Hardware and Software Requirements

The successful implementation of the Recovery Iloilo Spa Integrated System required appropriate hardware and software resources. These specifications ensure that the application performs efficiently during appointment processing, transaction recording, report generation, and analytics computation. The requirements are presented according to the role that each machine fulfills within the three-tier architecture namely, the server that hosts the application and database, and the client devices through which the users access the system.

4.5.1 Hardware Requirements

The hardware requirements specify the physical computing resources necessary for the proper operation of the Recovery Iloilo Spa Integrated System. These requirements were determined on the basis of the system's processing, memory, and storage demands, particularly during the execution of operational transactions and the rendering of graphical dashboards and reports. The minimum and recommended hardware specifications are presented in Table 4.1.

Table 4.1 presents the recommended hardware specifications.

Component	Minimum Specification	Recommended Specification
Processor	Dual-core, 2.0 GHz (Core i3 / AMD equiv.)	Quad-core, 2.4 GHz+ (Core i5 / Ryzen 5)
Memory (RAM)	4 GB	8 GB or higher
Storage	20 GB free (SSD preferred)	50 GB+ SSD
Monitor	1366 × 768 Resolution	Full HD Display
Internet Connection	10 Mbps	50 Mbps
Mobile Device	Android 10	Android 13 or higher

The recommended hardware configuration provides sufficient processing capability for executing operational transactions and for rendering the graphical dashboards and reports of the system without significant delay. The minimum configuration remains adequate for standard operation, whereas the recommended configuration ensures optimal performance during periods of concurrent usage and report generation.

4.5.2 Software Requirements

The software requirements specify the programs, development tools, and services employed in the construction and operation of the Recovery Iloilo Spa Integrated System. Each component was selected for its role within the development environment and the running application. The software requirements are presented in Table 4.2.

Table 4.2 presents the software requirements.

Software	Purpose
Windows 11	Operating System
Visual Studio Code	Source Code Editor
MariaDB	Database Management System
PHP	Backend Development
HTML, CSS, Vanilla JavaScript	Frontend Development
PayMongo Payment Gateway	Online Payment Processing
PHPMailer	Email and OTP Notification Service
Google Chrome / Microsoft Edge / Mozilla Firefox / Safari	Web Browser
GitHub Repository	Version Control
XAMPP	Local Development Server

The selection of these software tools was based upon their stability, community support, compatibility, and widespread adoption in modern web-application development. Collectively, these tools provided a dependable and well-supported environment for the design, implementation, testing, and deployment of the system.

4.6 Database Design

The database serves as the foundation of the Recovery Iloilo Spa Integrated System by organizing and storing appointment records, customer information, product inventories, transactions, and analytical data. A well-designed database ensures efficient data retrieval, minimizes redundancy, and maintains data integrity throughout the operation of the system. The proponents employed a relational database model using MariaDB, and database normalization techniques were applied to reduce duplication and maintain consistency among related tables. The database supports approximately thirty core tables.

Figure 4.4 presents the Entity Relationship Diagram (ERD) of the system.

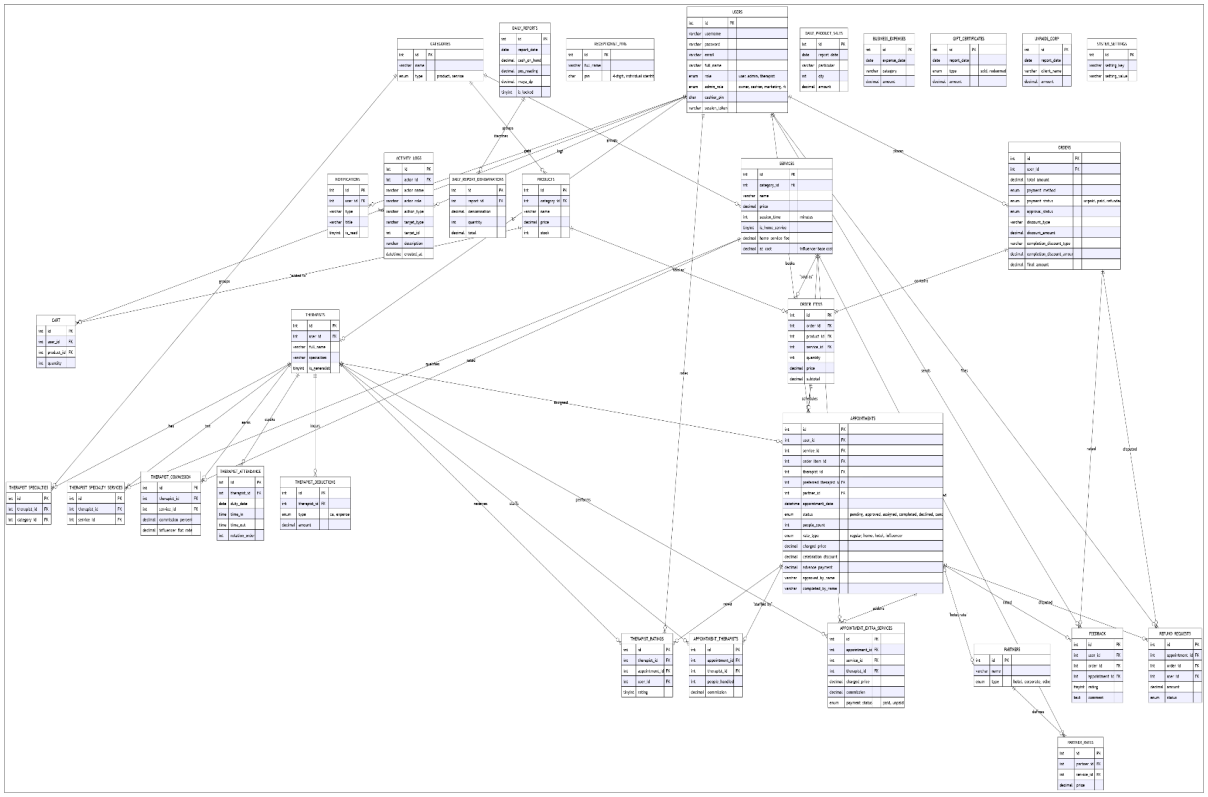


Figure 4.4. Entity Relationship Diagram of the Recovery Iloilo Spa Integrated

The primary entities of the database include the following:

- Users
- Therapists
- Categories

- Services
- Products
- Cart
- Orders
- Order Items
- Appointments
- Appointment Therapists
- Therapist Commission
- Therapist Attendance
- Feedback
- Refund Requests
- Notifications
- Activity Logs
- Daily Reports

Users Table

This table stores the account credentials and role information of all system users, including customers, therapists, receptionists, and administrators.

Field	Description
id	Primary Key
username	Login Username
password	Encrypted Password
email	Registered Email Address
full_name	Complete Name of the User
role	User Role (customer, therapist, receptionist, admin)
admin_role	Administrative Role (owner, cashier, marketing, IT)
cashier_pin	Cashier Verification PIN
session_token	Secure Session Identifier

Therapists Table

This table stores the professional profile of each therapist and links the record to a corresponding user account.

Field	Description
id	Primary Key
user_id	Foreign Key referencing Users
full_name	Therapist Name
specialties	Assigned Service Specialties
is_generalist	Indicates whether the therapist can perform all services

Categories Table

This table classifies both services and products to simplify searching, filtering, and management.

Field	Description
id	Primary Key
name	Category Name
type	Classification (service or product)

Services Table

This table stores the spa services offered by the establishment, including pricing and home-service details.

Field	Description
id	Primary Key
category_id	Foreign Key referencing Categories
name	Service Name
price	Service Price
session_time	Duration of the Service
is_home_service	Home-Service Availability Indicator
home_service_fee	Additional Travel Fee
at_cost	At-Cost Indicator

Products Table

This table stores retail and consumable product information used in daily spa operations.

Field	Description
id	Primary Key
category_id	Foreign Key referencing Categories
name	Product Name
price	Selling Price
stock	Available Inventory

Cart Table

This table temporarily stores the products selected by a customer before checkout is confirmed.

Field	Description
id	Primary Key
user_id	Foreign Key referencing Users
product_id	Foreign Key referencing Products

quantity	Quantity Selected

Orders Table

This table records customer transactions involving spa services and wellness products.

Field	Description
id	Primary Key
user_id	Foreign Key referencing Users
total_amount	Total Transaction Amount
payment_method	Selected Payment Method
payment_status	Payment Status
approval_status	Transaction Approval Status
discount_type	Applied Discount Type
discount_amount	Discount Value
completion_discount_type	Discount Applied at Completion
completion_discount_amount	Completion Discount Value
final_amount	Final Payable Amount

Order Items Table

This table records the individual services and products included in each customer order.

Field	Description
id	Primary Key
order_id	Foreign Key referencing Orders
product_id	Foreign Key referencing Products
service_id	Foreign Key referencing Services
quantity	Quantity Purchased
price	Unit Price
subtotal	Item Total

Appointments Table

This table records scheduled spa appointments together with their complete booking details and status lifecycle.

Field	Description
id	Primary Key
user_id	Foreign Key referencing Users
service_id	Foreign Key referencing Services
order_item_id	Foreign Key referencing Order Items

therapist_id	Assigned Therapist
preferred_therapist_id	Customer's Preferred Therapist
partner_id	Partner or Hotel Reference
appointment_date	Scheduled Date and Time
status	Status (pending, assigned, approved, completed, declined, cancelled)
people_count	Number of Persons in the Booking
rate_type	Applicable Rate Type
charged_price	Charged Amount
celebration_discount	Celebration Discount Applied
advance_payment	Advance Payment Recorded
approved_by_name	Personnel Who Approved the Appointment
completed_by_name	Personnel Who Completed the Appointment

Appointment Therapists Table

This table records the therapists assigned to each appointment, supporting multi-person and group bookings.

Field	Description
id	Primary Key
appointment_id	Foreign Key referencing Appointments
therapist_id	Foreign Key referencing Therapists
people_handled	Number of Persons Served
commission	Commission Earned

Therapist Commission Table

This table defines the commission rules applied to each therapist for particular services.

Field	Description
id	Primary Key
therapist_id	Foreign Key referencing Therapists
service_id	Foreign Key referencing Services
commission_percent	Commission Percentage
influencer_flat_rate	Flat Rate for Influencer or Promotional Sessions

Therapist Attendance Table

This table records therapist duty and attendance information used during appointment assignments.

Field	Description
id	Primary Key

therapist_id	Foreign Key referencing Therapists
duty_date	Duty Date
time_in	Clock-In Time
time_out	Clock-Out Time
rotation_order	Assignment Rotation Order

Feedback Table

This table stores customer feedback and ratings linked to the corresponding appointment and order.

Field	Description
id	Primary Key
user_id	Foreign Key referencing Users
order_id	Foreign Key referencing Orders
appointment_id	Foreign Key referencing Appointments
rating	Numerical Service Rating
comment	Customer Comment

Refund Requests Table

This table records customer refund requests and their processing status.

Field	Description
id	Primary Key
appointment_id	Foreign Key referencing Appointments
order_id	Foreign Key referencing Orders
user_id	Foreign Key referencing Users
amount	Requested Refund Amount
status	Request Status (pending, approved, declined)

Notifications Table

This table stores the system-generated notifications delivered to users.

Field	Description
id	Primary Key
user_id	Foreign Key referencing Users
type	Notification Type
title	Notification Title
is_read	Read Status

Activity Logs Table

This table records the administrative actions performed within the system to support accountability and auditing.

Field	Description
id	Primary Key
actor_id	Foreign Key referencing Users
actor_name	Name of the Actor
actor_role	Role of the Actor
action_type	Type of Action Performed
target_type	Affected Record Type
target_id	Affected Record Identifier
description	Description of the Action
created_at	Timestamp of the Action

Daily Reports Table

This table stores the end-of-day operational and financial summary prepared during cashier reconciliation.

Field	Description
id	Primary Key
report_date	Report Date
cash_on_hand	Recorded Cash on Hand
pos_reading	POS Reading
maya_dp	Maya Digital Payment Collections
is_locked	Report Lock Status

The relationships among these tables allow the application to trace customer transactions, monitor appointment activity, compute therapist commissions, track inventory movement, and generate comprehensive operational reports and analytics. Proper indexing, foreign key constraints, and normalization help maintain referential integrity and optimize query performance, ensuring that the system efficiently supports both daily operational transactions and business analytics for Recovery Iloilo Spa.

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CHAPTER V

CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter presents the conclusions of the development and evaluation of the **Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently**. It summarizes the results of the completed project, determines the extent to which the objectives of the study were achieved, and presents recommendations for future improvements and continued implementation of the system at Recovery Iloilo Spa.

The study aimed to design, develop, and implement a centralized web-based system that would address the spa's manual and fragmented operational processes. The developed system integrated appointment scheduling and management, customer account and record management, service and product management, inventory monitoring, Point-of-Sale and payment processing, therapist management, commission computation, voucher and discount management, customer feedback, security and accountability controls, automated notifications, reports and analytics, and formula-based short-term revenue forecasting.

Throughout the development process, the proponents coordinated with Recovery Iloilo Spa to identify its operational requirements, existing procedures, and user needs. The completed system was subjected to system testing and evaluation to determine its functionality, accuracy, usability, reliability, performance efficiency, security, and overall effectiveness. User feedback and evaluation results were used to assess whether the developed system addressed the identified operational challenges and met the requirements established for the study.

This chapter presents the Summary of Findings, Conclusions, and Recommendations based on the completed development, testing, and evaluation activities. The findings are presented according to the specific objectives of the study to establish a direct connection between the identified problems, the developed system features, and the results obtained during evaluation. The conclusions determine whether the objectives were achieved, while the recommendations identify possible improvements and future enhancements that may further support the operational needs of Recovery Iloilo Spa.

5.2 Summary of Findings

The development and evaluation of the **Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently** produced results that corresponded to the specific objectives established in the study. The completed system addressed the major operational requirements identified during the requirements-gathering phase and integrated the identified business processes into a centralized web-based platform. The findings were based on the system development activities, functional testing conducted by the proponents, and the validation and evaluation provided by the client and respondents during the User Acceptance Testing. The results demonstrated that the developed system provided functional support for appointment management, customer account and record management, service and product management, inventory monitoring, payment and transaction processing, therapist management, commission computation, promotional management, feedback, security, notifications, reporting, analytics, and forecasting.

The following findings were established based on the specific objectives of the study:

1. Operational Requirements Gathering and Analysis

The proponents successfully gathered, analyzed, and evaluated the operational requirements of Recovery Iloilo Spa through consultation with the client and relevant users, together with interviews, surveys, observation, and document review. The requirements-gathering activities provided information regarding the spa's existing procedures, operational challenges, user responsibilities, and information needs. The identified requirements covered appointment scheduling, customer management, services and products, inventory, payment processing, therapist management, commissions, discounts and vouchers, feedback, security, notifications, reporting, analytics, and forecasting.

The gathered requirements served as the basis for identifying the system modules and functions included in the developed system. The client and respondents, including the spa owner, management, HR personnel, receptionists, therapists, and other relevant staff, participated in the validation of the identified operational requirements. Their feedback helped establish that the developed functions corresponded to the business processes and operational needs identified during the study.

2. User Interface and System Accessibility

The proponents developed a user-friendly, responsive, and organized system interface that provided separate and appropriate functions according to the roles of administrators, authorized staff, therapists, receptionists, and customers. The interface organized the major system functions into accessible modules and provided users with navigation mechanisms for performing their respective activities.

The developed interface allowed customers to access account-related functions, browse services and products, manage their appointments and transactions, and submit feedback. Authorized personnel were provided with functions for managing operational records and monitoring business activities. During the evaluation, respondents were able to interact with the system and perform the functions relevant to their roles. Their feedback indicated that the interface was understandable, accessible, and manageable for the intended users.

3. Customer Registration and Account Management

The proponents successfully implemented customer registration and account management with email OTP verification through PHPMailer. The registration process provided an additional verification step before an account could be activated, while the account management functions allowed customers to maintain their profile information and access their account-related activities.

The customer module also provided access to service and product browsing, shopping cart functions, appointment booking, appointment history, transaction history, and profile management. The implementation provided customers with a centralized account through which their interactions with the spa could be managed. During evaluation, respondents confirmed that the account verification process operated properly and that the customer account functions supported secure and organized access to customer-related activities.

4. Appointment Scheduling and Management

The appointment scheduling and management module was successfully implemented to address the difficulties associated with manual appointment scheduling. The system supported online, walk-in, home-service, and hotel or partner bookings while maintaining appointment details such as customer information, selected services, schedules, service duration, therapist assignments, and booking status.

The module implemented minute-level scheduling and server-side conflict detection to prevent incompatible appointment schedules. It also supported therapist assignment and reassignment, appointment updates, rescheduling, cancellations, and applicable add-on services. For home-service appointments, the system considered the required travel-time allowance in scheduling. These functions provided authorized users with a centralized means of monitoring available schedules and managing appointments.

The testing and evaluation activities showed that the appointment module operated according to the established requirements. Respondents indicated that the centralized scheduling functions supported the management of appointments and helped address issues associated with duplicate reservations, scheduling conflicts, overbooking, and missed appointments.

5. Centralized Customer Record Management

The proponents successfully implemented a centralized customer record management feature that organized customer information in a single database. The system stored relevant information such as personal details, contact information, appointment history, service history, transaction records, preferences, feedback, and applicable voucher or discount usage.

The centralized records allowed authorized personnel to retrieve customer information without relying on separate manual records. The availability of organized customer information also supported the processing of appointments, transactions, customer inquiries, and service-related activities. During evaluation, respondents indicated that the centralized customer records provided a more organized and accessible method of retrieving and managing customer information.

6. Service and Product Information Management

The proponents implemented a centralized service and product management feature that organized the information required by customers and employees during service and transaction activities. Service records included information such as service names, descriptions, categories, prices, durations, availability, packages, and add-on services. Product records included product names, descriptions, categories, prices, stock quantities, and availability information.

The centralized management of these records provided a consistent source of information for browsing, appointment selection, product selection, shopping cart activities, and Point-of-Sale transactions. Authorized personnel were able to manage and update service and product information through the system. This reduced dependence on separate records and supported more consistent presentation of service and product information to customers.

7. Inventory Management

The inventory management module was successfully implemented to monitor product availability, stock movement, inventory usage, and low-stock items. The system provided authorized personnel with centralized inventory records that supported monitoring of products and supplies used in spa operations.

The module helped organize information regarding available quantities and inventory movements and provided a basis for identifying items requiring replenishment. During evaluation, respondents indicated that the inventory functions supported more organized stock monitoring and made it easier to identify low-stock items. The module therefore addressed the identified difficulty of maintaining inventory information through manual monitoring.

8. Point-of-Sale and Payment Processing

The proponents successfully implemented the Point-of-Sale and payment processing module to support the recording and management of customer transactions. The system digitally recorded transactions, calculated transaction totals, applied applicable discounts and vouchers, generated receipts, maintained transaction histories, and supported refund requests.

The system also supported multiple applicable payment methods, including cash and online payments. For supported online transactions, the system was integrated with the PayMongo Payment Gateway. The system handled the related payment request, received the payment result, updated the corresponding payment status, and recorded the transaction information, while PayMongo handled the online payment transaction.

The implementation provided a centralized method of monitoring payment-related information and transaction records. Testing confirmed that the payment and Point-of-Sale functions operated according to the established requirements. Respondents also indicated that the digital transaction-recording process supported more organized and efficient payment and sales activities compared with the manual procedures identified during requirements gathering.

9. Therapist Management

The therapist management feature was successfully implemented to support therapist assignment, reassignment, attendance monitoring, specialty management, service scheduling, and performance tracking. The system provided authorized personnel with information that supported the assignment of therapists according to their availability, specialties, schedules, and service requirements.

The attendance feature recorded therapist clock-in and clock-out activities, while the management functions supported monitoring of assigned services and therapist activities. Therapist assignment and reassignment functions also provided greater flexibility when scheduling changes or conflicts occurred. During evaluation, respondents indicated that these functions supported improved coordination of therapist schedules and assignments and provided more organized information for managing therapist activities.

10. Therapist Commission Management

The therapist commission management feature was implemented to reduce the need for manual commission computation. The system calculated commissions based on recorded completed services, applicable service categories, commission rates, multi-person sessions, and authorized flat-rate arrangements, including applicable influencer-related arrangements.

By basing commission calculations on recorded transactions and completed services, the system provided a more consistent method of maintaining commission information. The commission records could also support payroll preparation and employee performance monitoring. During evaluation, respondents indicated that the automated computation reduced the administrative effort associated with manual commission calculations and helped maintain more organized commission records.

11. Discount, Voucher, and Promotional Management

The proponents successfully implemented the discount and voucher management feature to organize promotional transactions and customer incentives. The system supported the validation and application of applicable discounts and promotional offers, including authorized senior citizen, PWD, staff, celebration, and gift certificate-related discounts based on the established business rules.

The system also recorded issued and redeemed vouchers and monitored promotional usage. Validation mechanisms helped reduce the possibility of duplicate or unauthorized voucher usage. These functions provided authorized personnel with a centralized method of monitoring promotional activities and supported more consistent application and recording of discounts and vouchers.

12. Digital Feedback and Rating Management

The digital feedback and rating feature was successfully implemented to provide customers with a centralized means of submitting comments, ratings, and suggestions regarding their spa experience. Feedback could be associated with the corresponding service and therapist, allowing management to identify the context of customer evaluations.

The feature provided management with organized customer feedback that could be reviewed in relation to service quality and therapist performance. During evaluation, respondents indicated that the digital feedback mechanism supported the collection of customer opinions and provided useful information for assessing customer satisfaction and identifying areas that required improvement.

13. Security and Employee Accountability

The proponents implemented security and accountability features to protect system access and support monitoring of sensitive activities. Administrative functions were separated from customer-facing functions, and authorized access was controlled according to assigned roles and permissions.

The system also incorporated hidden administrative access, gate code verification, PIN verification for designated sensitive transactions, and activity logging. Activity logs recorded relevant information regarding user actions and timestamps, allowing authorized personnel to trace important activities performed within the system.

Testing confirmed that the implemented security controls operated according to the established requirements. The respondents also indicated that these controls strengthened oversight and provided a clearer means of monitoring critical administrative and financial activities.

14. Reporting, Analytics, and Revenue Forecasting

The reporting and analytics module successfully generated organized information regarding sales, appointments, customer activities, inventory, therapist productivity, attendance, commissions, payment transactions, cash reconciliation, promotional activities, and other operational records.

The system also provided dashboards and analytical views that allowed management to review summarized operational information. Formula-based short-term revenue forecasting was implemented using recorded historical transaction data to provide estimates that could support short-term planning and monitoring. The forecasting feature did not rely on artificial intelligence or machine-learning-based predictive analytics.

Reports could be filtered and viewed according to relevant criteria and exported into Excel and PDF formats. The generated reports provided management with organized information that could be used for documentation, operational monitoring, analysis, and decision-making. During evaluation, respondents indicated that the reporting and analytics functions supported more organized access to business information and assisted management in reviewing operational performance.

15. Automated Notifications

The automated notification feature was successfully implemented to support communication between Recovery Iloilo Spa and its customers. The system delivered email notifications for important activities, including account verification, appointment confirmations, appointment updates, cancellations, and payment confirmations.

The notification feature reduced the need to communicate every system event manually and provided customers with timely information regarding their accounts, appointments, and transactions. During evaluation, respondents indicated that the automated notifications supported clearer and more timely communication between the spa and its customers.

16. Integration of Operational Processes

The completed system successfully integrated the major operational functions of Recovery Iloilo Spa into a centralized platform. Instead of maintaining separate procedures for appointments, customer information, services, products, inventory, payments, therapists, commissions, promotions, feedback, notifications, and reports, the developed system connected these activities through a centralized database and coordinated workflow.

The integration allowed information recorded in one operational process to support related activities in other modules. For example, appointment information was associated with customer records, selected services, therapist assignments, transactions, feedback, and related operational records. Likewise, completed transactions supported payment records, receipts, sales reports, commission computation, and analytics. This integration reduced the need for repeated data entry and provided authorized users with more consistent access to operational information.

17. System Testing and Evaluation

The proponents conducted system testing and User Acceptance Testing to determine whether the developed system operated according to the established functional and operational requirements. The evaluation covered relevant system quality characteristics, including functionality, accuracy, usability, reliability, performance efficiency, and security.

The testing activities verified the operation of the implemented modules and their major functions, while User Acceptance Testing provided feedback from the intended users and stakeholders. The results indicated that the developed system addressed the identified operational requirements and provided the intended functionality for managing the spa's major business processes.

Overall, the findings demonstrated that the specific objectives of the study were substantially achieved. The proponents successfully developed and integrated the major system modules identified during the requirements-gathering phase, while the client and respondents participated in the validation and evaluation of the completed system. The developed platform provided centralized management of appointments, customer information, services and products, inventory, transactions, payments, therapist activities, commissions, promotional activities, feedback, security, notifications, reports, analytics, and forecasting.

The results further demonstrated that the system provided a structured technological solution to the operational challenges identified at Recovery Iloilo Spa. By consolidating the major business processes into one platform, the system reduced dependence on fragmented manual records, supported more organized information management, strengthened transaction and employee accountability, improved access to operational information, and provided management with analytical outputs for monitoring and decision-making.

Therefore, the completed development and evaluation established that the Recovery Iloilo Spa Integrated System provided the intended operational support and substantially fulfilled the objectives established for the study. The findings also provided the basis for the conclusions and recommendations presented in the succeeding sections of this chapter.

5.3 Conclusion

The development and evaluation of the **Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently** established that the system was a viable and effective solution for supporting the daily operations of Recovery Iloilo Spa. The developed system centralized the spa's major operational processes into a single web-based platform, including appointment scheduling and management, customer registration and records, service and product management, inventory monitoring, Point-of-Sale and payment processing, therapist management, attendance and commission monitoring, discount and voucher management, customer feedback and ratings, automated notifications, security and accountability controls, reporting, analytics, and formula-based revenue forecasting. Through this integration, the system addressed the fragmented and manual procedures identified during the requirements-gathering phase and provided users with organized access to operational information and transactions. The centralized database further supported the consistent storage and retrieval of customer, appointment, service, product, transaction, therapist, feedback, notification, and reporting information, providing a more structured approach to managing the spa's business processes.

The developed system addressed the identified operational challenges by providing dedicated functions for the spa's daily activities. The appointment module supported online, walk-in, home-service, hotel, and partner bookings and included appointment details, therapist assignments, schedules, booking status, and additional services. The system also supported minute-level scheduling and conflict detection to assist in managing therapist availability and appointment schedules. Customer account and record management provided customers with registration, email verification, profile management, service and product browsing, shopping cart and checkout functions, appointment and transaction history, payment and refund status, and feedback submission. Service and product management

organized information such as service names, descriptions, categories, prices, durations, availability, and product details, while inventory functions supported monitoring of product availability and stock movement. These integrated functions provided a more organized workflow for managing customer requests, services, products, appointments, and related operational records.

The system also strengthened the spa's transaction, workforce, financial, promotional, communication, and accountability processes. The Point-of-Sale and payment module supported digital transaction recording, multiple payment methods, online payment processing through the PayMongo Payment Gateway, receipt generation, transaction history, and refund requests. Therapist management supported therapist assignment and reassignment, specialties, attendance, scheduling, and performance monitoring, while the commission feature supported the computation and recording of commissions based on completed services and applicable arrangements. The system also organized discounts, vouchers, and gift certificates and provided validation and redemption monitoring. Customer feedback and therapist ratings provided management with organized information regarding customer experiences and service performance. In addition, security controls such as role-based access, gate code verification, PIN verification, and activity logging provided stronger access control and accountability for sensitive activities. Automated email notifications also supported account verification, appointment updates, payment confirmations, refund-related updates, and other important system activities.

Based on the completed system testing and User Acceptance Testing, the proponents concluded that the developed system substantially achieved the objectives established in the study. The testing and evaluation results indicated that the implemented functions addressed the identified operational requirements and provided the intended support for the spa's users and management. The reporting and analytics functions further provided organized information regarding sales, appointments, inventory, therapist productivity, commissions, cash reconciliation, and other operational activities, together with formula-based short-term revenue forecasting and report export capabilities in Excel and PDF formats. Overall, the system improved the organization of operational records, supported more systematic transaction and appointment management, strengthened accountability, facilitated customer communication, and provided management with accessible information for monitoring and decision-making. The completed system therefore provided Recovery Iloilo Spa with an integrated technological platform that supported its current operational requirements and established a practical foundation for continued system use, maintenance, and future enhancement.

5.4 Recommendations

Based on the operational findings, user feedback, system testing, and User Acceptance Testing conducted during the development and implementation of the **Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently**, the proponents recommend several improvements that may further enhance the system's functionality, usability, security, scalability, and long-term sustainability. The recommendations are categorized into System Improvements and Technical Enhancements, Implementation and Operational Strategies, and Future Research and Enterprise Scaling Directions. These recommendations are intended to build upon the existing capabilities of the developed system rather than replace its current functions.

System Improvements and Technical Enhancements

1. Integration of a Wellness Information and Pre-Booking Assistance Chatbot

It is recommended that a conversational chatbot be integrated into the Customer Portal to provide customers with immediate assistance before booking services. The chatbot may provide general information about available treatments, service duration, massage pressure options, preparation guidelines, and common wellness-related questions. It may also assist customers in identifying services based on stated preferences, comfort levels, and general wellness goals.

For health-related concerns, the chatbot should provide general informational guidance accompanied by clear safety disclaimers rather than diagnosing medical conditions or determining whether a treatment is medically safe. When

customers disclose medical conditions, pregnancy, injuries, medications, or other circumstances that may require professional assessment, the chatbot should advise them to consult a qualified healthcare professional or the spa's trained personnel before selecting a treatment. The chatbot may also recommend that customers disclose relevant conditions to the spa before an appointment. This approach would allow the feature to improve customer assistance while maintaining appropriate boundaries for health-related information.

The chatbot may also provide service and package recommendations based on customer preferences, such as preferred massage pressure, treatment duration, relaxation goals, or interest in additional services. This would complement the system's existing service catalog, appointment booking, and add-on functionality.

2. Regular System Maintenance and Updates

The current system integrates multiple modules for appointment scheduling, customer records, inventory management, payment processing, therapist management, commissions, discounts and vouchers, feedback, notifications, security, reporting, and analytics. As the system is continuously used, regular maintenance may be required to address technical issues, correct identified errors, update software dependencies, and apply necessary security patches.

Regular system maintenance may include monitoring system performance, checking database integrity, reviewing implemented functions, updating dependencies, and resolving issues identified by users or administrators. Conducting these activities regularly would help maintain the reliability, security, and functionality of the system and ensure that it continues to support the operational requirements of Recovery Iloilo Spa.

3. SMS Gateway Integration

The current system uses an Email/SMTP service for OTP verification, appointment confirmations, booking updates, payment acknowledgements, refund notifications, password reset instructions, and other system-generated messages. As an enhancement, an SMS gateway may be integrated to complement the existing email notification mechanism.

SMS notifications may be used for important time-sensitive activities such as appointment reminders, booking confirmations, cancellations, payment confirmations, and verification codes. Using both email and SMS would provide customers with additional communication channels, particularly when an email notification is not immediately noticed.

4. Establish Regular Data Backup and Recovery Procedures

The current system stores centralized operational information, including customer records, appointment details, payment and transaction records, inventory information, therapist records, commission records, discounts and vouchers, feedback, notifications, and reports. As these records are essential to the daily operations of the spa, regular data backup procedures should be established to protect the information from accidental deletion, system failure, hardware problems, or other unexpected incidents.

Scheduled backups may be performed regularly and stored in a secure location separate from the primary system environment. A data recovery procedure should also be established and tested periodically to verify that important records can be restored when necessary. Implementing these procedures would help minimize the risk of permanent data loss and support the continuity of Recovery Iloilo Spa's operations.

Future Research Directions

1. Advanced Predictive Analytics and Machine Learning

The current system provides formula-based short-term revenue forecasting using recorded transaction data and generates reports and analytics for sales, appointments, inventory, therapist productivity, commissions, and other operational activities. Future researchers may explore the integration of advanced predictive analytics and machine-learning techniques to further improve the system's forecasting capabilities as more historical data becomes available.

Machine-learning models may be used to analyze historical sales, appointment patterns, service demand, seasonal trends, and other relevant operational data to produce more comprehensive forecasts. The performance of these models may also be compared with the current formula-based forecasting method to determine whether they provide improved forecasting accuracy and more useful insights for management decision-making.

2. Mobile Application Development

The current system is implemented as a web-based platform that can be accessed through supported web browsers on computers and mobile devices. Future researchers may develop a Progressive Web App (PWA) or dedicated Android and iOS applications to provide customers and staff with a more optimized mobile experience.

A mobile application may provide features such as push notifications, appointment reminders, booking updates, payment notifications, appointment monitoring, and easier access to relevant customer or therapist functions. Developing a dedicated mobile application may improve accessibility and convenience while extending the existing functions of the web-based system to a more mobile-oriented platform.

3. Biometric Attendance and Authentication

The current system provides digital therapist attendance monitoring through clock-in and clock-out functions and includes security controls such as role-based access, gate code verification, PIN verification, and activity logging. Future researchers may explore the integration of biometric technologies, such as fingerprint or facial recognition, to provide an additional method of verifying therapist attendance and authorized access.

Biometric authentication may help strengthen attendance verification and reduce the possibility of unauthorized attendance recording. Future studies should evaluate the accuracy, reliability, cost, privacy considerations, security requirements, and user acceptance of biometric technologies before they are incorporated into the system. The implementation should also follow applicable data-protection and privacy requirements.

Implementation and Operational Strategies

1. User Training and System Orientation

The proponents recommend that Recovery Iloilo Spa provide proper training and system orientation to administrators, receptionists, therapists, cashiers, and other authorized users before and during system implementation. Training should cover the functions relevant to each user role, including appointment management, customer records, Point-of-Sale transactions, payment processing, inventory management, therapist attendance and commissions, discounts and vouchers, feedback, notifications, and reporting.

Training sessions may include demonstrations, guided practice, and basic procedures for handling common system activities. Additional orientation should also be provided whenever significant system updates or new functions are introduced. Proper user training would help users understand their responsibilities, minimize incorrect data entry, and ensure that the implemented features are used consistently.

2. Regular Data Management and Backup Procedures

The proponents recommend that Recovery Iloilo Spa establish clear procedures for maintaining and protecting the information stored in the system. Authorized personnel should regularly review customer records, appointment information, transaction records, inventory data, therapist records, commission information, and other operational records to ensure that information remains accurate and up to date.

Regular database backups should also be performed and stored in a secure location. A recovery procedure should be established and tested periodically to ensure that important information can be restored in the event of accidental deletion, system failure, hardware problems, or other unexpected incidents. These procedures would help protect the integrity and availability of the spa’s operational data.

3. Regular Security and Access Review

The proponents recommend that management regularly review user accounts, assigned roles, access permissions, gate code access, PIN verification, and activity logs. Access privileges should be updated whenever an employee's responsibilities change, while inactive or unauthorized accounts should be disabled or removed when no longer required.

Management should also regularly review activity logs and monitor sensitive transactions to ensure that system activities remain traceable and accountable. These practices would strengthen the security of customer, payment, and operational information and help ensure that system functions are accessed only by authorized personnel.

4. Continuous System Evaluation and Maintenance

The proponents recommend that Recovery Iloilo Spa conduct periodic evaluations of the system after implementation to identify technical issues, usability concerns, and additional operational requirements. Feedback from administrators, receptionists, therapists, and customers may be collected to determine whether the system continues to meet the spa's changing business needs.

Regular maintenance should include monitoring system performance, addressing reported errors, reviewing database integrity, updating necessary software components, and applying appropriate security updates. Continuous evaluation and maintenance would help preserve the reliability and effectiveness of the system and provide a basis for future improvements.

5.5 Impact of the Study

The **Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently** contributes to the field of Information Technology by demonstrating the practical application of an integrated web-based information system in a service-oriented business environment. The study demonstrates how multiple operational processes, including appointment scheduling, customer and therapist management, inventory monitoring, Point-of-Sale and payment processing, commissions, discounts and vouchers, feedback, notifications, security, reporting, and analytics, can be organized within a centralized platform. The system also demonstrates the use of operational data for generating reports, dashboards, and formula-based short-term revenue forecasts. Through the integration of these functions, the study provides an example of how information systems can be designed to address actual business requirements, reduce dependence on fragmented manual procedures, improve information organization, and support data-informed operational decision-making.

For Recovery Iloilo Spa management and staff, the study provides a centralized system that supports the management of the spa's daily activities. Management benefits from organized appointment records, customer information, inventory records, transaction histories, therapist attendance and commission records, discount and voucher records, feedback, and operational reports. The reporting and analytics functions provide management with information regarding sales performance, appointment activities, inventory usage, therapist productivity, commissions, cash reconciliation, and other operational indicators, together with formula-based short-term revenue forecasting. The system also provides security and accountability mechanisms, including role-based access, gate code and PIN verification, and activity logging, which support controlled access to administrative and sensitive transactions. These functions provide management and staff with more organized information and support more systematic monitoring of the spa's operations.

For customers and therapists, the study provides benefits through features that support more convenient and organized interactions with the spa. Customers can register and verify their accounts through email OTP, browse available services and products, manage their profiles, use the shopping cart, book appointments, view appointment and transaction histories, monitor payment and refund information, receive automated notifications, and submit feedback and ratings. The system also provides organized information about services and products, including

relevant descriptions, prices, categories, durations, and availability. Therapists benefit from organized appointment assignments, therapist schedules, attendance records, specialty information, completed-service records, commission records, and customer feedback associated with their services. These functions provide customers and therapists with clearer access to relevant information and support more organized service coordination.

For the proponents, Information Technology students, and future researchers, the study provides practical experience and a reference for the application of software development, web programming, database management, systems analysis and design, information security, payment integration, and data analytics in an actual business setting. The development process demonstrates how identified business requirements can be translated into system modules and integrated into a centralized database-driven platform. The study may also serve as a reference for future researchers who intend to develop or evaluate integrated management systems for spas and other service-oriented establishments, particularly in areas involving appointment management, inventory, payment processing, workforce management, customer feedback, security, reporting, and forecasting. The study also identifies areas for future enhancement, such as advanced predictive analytics, mobile application development, and biometric attendance, which may be explored as the operational requirements and available data of the establishment increase.

5.6 Summary

In summary, the **Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics efficiently** demonstrated the practical application of software engineering, database management, systems analysis and design, information security, and business analytics in addressing the operational requirements of Recovery Iloilo Spa. Guided by the developmental research design and Agile methodology, the proponents developed and implemented a centralized web-based platform that integrated the spa's major operational processes into a single system. The completed system provides functions for appointment scheduling and management, customer registration and records, service and product information, inventory monitoring, Point-of-Sale and payment processing, therapist management and attendance, commission computation, discounts and vouchers, customer feedback and ratings, automated notifications, security and accountability, reporting, and analytics.

The development of the system addressed the operational challenges identified during the requirements-gathering process. The appointment management module supported online, walk-in, home-service, and hotel or partner bookings and incorporated minute-level scheduling and server-side conflict detection to help manage therapist availability and prevent scheduling conflicts. The customer management functions centralized customer information, appointment history, transaction records, preferences, feedback, and voucher usage. The system also organized service and product information and supported inventory monitoring, allowing staff to monitor product availability, stock movement, and low-stock items. These integrated functions provided a more organized approach to managing customer requests, appointments, services, products, and operational records. The findings showed that the implemented modules addressed the identified requirements and supported the daily activities of the establishment.

The system also integrated transaction processing, workforce management, communication, security, and business analytics within the same platform. The Point-of-Sale and payment module supported multiple payment methods, online payment processing through PayMongo, digital receipts, transaction history, and refund requests. The therapist management and commission functions supported therapist assignment, reassignment, attendance monitoring, specialty management, performance tracking, and commission computation based on completed services and applicable arrangements. The system also managed discounts and vouchers, collected customer feedback and ratings, and provided automated notifications for important account, appointment, payment, and transaction activities. Security mechanisms, including controlled access, gate code and PIN verification, and activity logging, provided additional support for accountability and the protection of sensitive operations.

The development and evaluation process further established that the system supported the operational requirements identified for Recovery Iloilo Spa. System testing and User Acceptance Testing were conducted to evaluate the

implemented functions, while feedback from the client and respondents was used to assess the system's functionality, usability, reliability, performance efficiency, accuracy, and security. The results indicated that the developed system provided the intended support for the spa's daily operations and that the specific objectives of the study were substantially achieved. The reporting and analytics functions also provided management with organized information regarding sales, appointments, inventory, therapist productivity, commissions, cash reconciliation, and other operational activities, together with formula-based short-term revenue forecasting and report export capabilities. Overall, the completed system provided Recovery Iloilo Spa with a centralized technological solution that improved the organization of operational processes, strengthened record management and accountability, supported customer and employee services, and provided management with information for monitoring and decision-making.

End of Proposal

Note to the instructors handling this subject

CHED BSIT ALIGNMENT STATEMENT

This capstone project complies with CHED BSIT standards by demonstrating competencies in application development, database management, systems analysis and design, and emerging technologies such as machine learning. The project follows outcomes-based education principles and integrates ethical con

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